The Creation and Complexity Analysis of a Corpus of Educational Materials in Irish (EduGA)

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Declaration

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Summary

This thesis sets out to develop a set of complexity metrics for educational materials written in Irish. In order to achieve this, the way language in educational materials becomes more complex from one level of education to the next was objectively analysed (e.g. from primary level to post-primary level, or from one CEFR-GA[1] level to the next). This research was undertaken because of a combination of issues with Irish- language and Irish-medium educational materials that were identified in the literature.

Materials for teaching Irish have been criticized for being overly simple and materials for teaching subjects through the medium of Irish have been criticized for being overly complex; such as Mac Donnacha (2005), Ó Muircheartaigh (2009), An Chomhairle um Oideachas Gaeltachta agus Gaelscolaíochta (2011), de Brún (2007). This would suggest that the contents of materials may be an unnecessary barrier to the acquisition of Irish at primary and post-primary levels, or in mainstream education. The counter-point to this is that the materials may have been unfairly criticized and that other socio- and psycho-linguistic factors are negatively impacting language acquisition and/or the extent to which language learners use the language inside and outside the classroom (Harris, Forde, Archer, Nic Fhearaile, and O'Gorman (2006), Ó Giollagáin, Mac Donnacha, Ní Chualáin, Ní Shéaghdha, and O' Brien (2007), Central Statistics Office (2011), Ó hIfearnáin and Ní Neachtain (2012), Ó hIfearnáin and Ó Murchadha (ND), Pétervári, Ó Curnáin, Ó Giollagáin, and Sheehan (2014)). Regardless of which is true, educational materials are at the core of every classroom and therefore influence the majority of the language used in classrooms.

Educational materials are designed to become more complex from one level to the next. Syllabi and curricula are typically presented as guides, providing examples of which language features are prescribed for each level of education. They do not, however, describe the prescribed language features in terms of their relative complexity. As a result, the syllabi and curricula are used here as the starting point from which the metrics are developed. This is the first attempt, to my knowledge, at a complexity analysis of any Irish-language content. In this research the term complexity refers to the objective ranking of a set of phenomena (typically lexical, morphosyntactic, or semantic language features) and their interrelationships. Complexity is distinct from difficulty, which here refers to learner-centric phenomena (such as cognitive cost, language interference, learner errors). The two phenomena can overlap, but need not.

[1] Irish-language applications of (Council of Europe, 2001)

To develop these metrics, a 7.5 million word corpus of educational materials was compiled.

The corpus is tagged for lemma and part of speech (POS). The corpus contains materials for teaching Irish as well as materials for teaching other subjects through the medium of Irish. The collection, pre-processing, and annotation of this corpus was a major undertaking and as well as being a product of this research the corpus will be a valuable resource for future research. All materials are categorised according to their level of ability and course of learning.

In order to develop the metrics three types of analyses were carried out, each of which uses prescribed language features from curricula and syllabi. The first type contains a comparative frequency analysis, the second type contains an analysis of sentence and word length, and the third type of analysis combines term matching with a statistical method that applies a topicality score to each term.

The first type of analysis compares in two ways the frequencies at which prescribed language features occur in educational materials. The first comparison is done across levels of ability within a single course of learning and the second comparison is done between different courses of learning. The comparative courses of learning are mainstream Irish (primary and post-primary school) and CEFR-GA materials for levels A1 through B2. The contents of the educational materials for these courses are structured in a way that is guided by syllabi, curricula and frameworks. Following the guidelines in these documents the educational materials generally introduce simpler features earlier than more complex features. Therefore this analysis seeks to identify if the frequency of prescribed language features can be used to mark complexity in the corpus of educational materials. The seven features analysed have been prescribed for both mainstream education (primary and post-primary school) and for CEFR-GA levels A1 through B2 in Teastas Eorpach na Gaeilge (TEG, www.teg.ie) syllabi. Each of the features is first prescribed for a different level of ability.

This analysis found that three of the prescribed features showed frequencies consistent with the gradual increase in complexity expected in mainstream materials (Introductions, Greetings and Salutations, and Relative Clauses). Another three of the prescribed features indicated that the sub-corpus of primary-level materials was more complex in this regard than the sub-corpus of materials for junior cycle at post-primary level (Noun-adjective collocations, Conditional verbs, and other Conditional-mood structures). Autonomous verb forms showed varied frequencies, depending on tense or mood, either progressing in complexity as expected or indicating primary-level materials were more complex than those for junior cycle post-primary level. Finally, the frequencies of features in senior-cycle materials (mainstream) were found to be comparable to a mid-point between those found in CEFR-GA materials for levels B1 and B2. The features in CEFR-GA materials, on the other hand, predominantly occurred at frequencies that were consistent with expected changes in complexity. However, half of both the autonomous verb forms and conditional-mood structures analysed indicated CEFR-GA materials for A1 were more complex than materials for A2, in a manner that is similar to those found in materials for mainstream education.

The second type of analysis focusses on sentence length and word length in EduGA. Word length (Flesch (1948), Kincaid, Fishburne, Rogers, and Chissom (1975), McLaughlin (1969)) and sentence length (Klare (1976), Mikk, Uibo, and Eits (2001), Lewis and Frank (2016)) are commonly used measures of complexity. These studies primarily focus on English and to my knowledge no work has been done in this area for Irish. Having the ability to construct and understand longer sentences is referred to in syllabi and curricula for mainstream education as markers of developing language ability (An Roinn Oideachais agus Scileanna (2010a), An Roinn Oideachais agus Scileanna (ND), An Roinn Oideachais agus Scileanna (1999a)). The analyses for Irish were conducted on materials for teaching Irish in mainstream education and in CEFR-GA classes, as well as materials for teaching Business Studies, History, and Science through the medium of Irish. The analysis of average sentence lengths found sentences at lower levels contained fewer words on average than sentences at higher levels, which consistently contained a higher average number of words per sentence. During this analysis the average number of words per sentence was graphed and exceptionally large sentences were manually checked so that, if erroneous, they could be removed from the calculations. This metric was deemed to work for Irish language educational materials in the same way it works for data from other languages. Increases in average word length did not correlate with increases in level, and average word lengths were found to fluctuate significantly across all sub-corpora. It was concluded that word-length is not a suitable metric for the analysis of Irish-language complexity. The language's morphology is believed to be the primary reason for this.

The third type of analysis focusses on terminology use in EduGA. Accurate use of terminology is frequently associated with –and prescribed for– learners at more advanced levels. Terms frequently refer to concepts that are of a higher complexity than those referred to by non-technical or non-terminological words or phrases. Therefore, terminology was selected as the basis for the third type of complexity analysis. It was first found that simply matching terms did not give any indication of a document's complexity because there was little change in the frequency of terms used across levels of ability. Furthermore, there are currently no semantic disambiguation tools for Irish. This resulted in the addition of two further dimensions to the term- based metric. In order to rank the terms, term matching is combined with TF-IDF[2] topicality scoring and each term-score combination is then analysed in the context of the term's frequency in a corpus of general Irish; Nua-Chorpas na hÉireann[3]. The resulting dataset contains topical term-tokens from the chosen educational materials that are filtered according to five NCI-based frequency ranges. The metric successfully distinguished more complex documents from simpler documents in each sub-corpus, and facilitates analysis at both a document level and a term level. Furthermore, the metric found Irish-medium materials to be far more complex than Irish-language materials.

[2] Term Frequency-Inverse Document Frequency
[3] link: http://corpas.focloir.ie/

1. The results of the frequency analyses have indicated that in some cases the inconsistent complexity reported in the literature may be true, particularly in materials for mainstream education, but not in all cases.
2. The analysis of average sentence length was found to be consistent with expected increases in complexity in all materials, however, average word length did not and does not appear to be suitable for analysing Irish-language complexity.
3. The term-based analysis used objective means to identify a significant difference in complexity between Irish-medium materials and Irish-language materials. This difference in complexity aligns with criticisms of these materials in the literature.
4. Metrics are best used in combination in order to provide a more complete picture of the different types of complexity present in educational materials.
5. Additional analysis of the corpus would have been possible if additional NLP tools and resources were available for Irish, specifically, semantic-analysis tools and additional corpus data for general Irish or Irish on other domains.

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List of Abbreviations and Key Concepts

- **BNC.** British National Corpus.
- **CEFR.** Common European Framework of Reference for Languages: Learning, Teaching, Assessment (Council of Europe, 2001).
- **CEFR-GA.** Irish-language adaptations, implementations and interpretations of the CEFR. GA is the two-letter code for Irish, as attributed by ISO 639-1 (International Organization for Standardization, 2002). This includes syllabi and materials published by multiple organisations.
- **Complexity.** The term complexity refers here to the objective ranking of a set of phenomena and their inter-relationships. Complexity is distinct from difficulty. The two phenomena can overlap, but need not. In this research, complexity is being examined in terms of its occurrence from one level of education to the next. Complexity is typically defined as Rescher (1998) defines it; "a matter of the number and variety of an item's constituent elements and of the elaborateness of their inter- relational structure".
- **Difficulty.** Difficulty is used here to refer to a phenomenon that is unique to each interlocutor typically measured in terms of the cognitive cost of acquiring, understanding, or producing a given language feature. Cognitive cost can be influenced by language interference, or errors arising from the learning process, or by the extent of an interlocutor's exposure to a language feature. Therefore, difficulty can intersect with complexity, but need not.
- **EduGA.** The corpus of Irish-language (GA – Gaeilge) educational materials and Irish-medium educational materials (subjects other than Irish).
- **EduGA-Sci.** Irish-medium educational materials for Science
- **EduGA-Bus.** Irish-medium educational materials for Business Studies
- **EduGA-His.** Irish-medium educational materials for History
- **EduGA-Ga.** Educational materials for learning Irish
- **ELT.** English-language Teaching
- **Gaelscoil.** A school in which the language of instruction is either entirely or predominantly Irish that is not located in a designated Irish-speaking community.
- **Irish-language educational materials.** This term refers here to educational materials that have been made **to teach the Irish language**. This includes materials for Irish (or *Gaeilge*) in primary and post- primary education as well as materials created for any CEFR-GA aligned course.

- **Irish-medium educational materials.** This term refers here to educational materials that have been made **to teach subjects other than Irish through the medium of Irish**. For example: Chemistry, Business Studies, Geography, or Music. This applies to materials that were made for primary and post-primary levels of education.
- **Level of Education.** The term *level* refers here to one tier within an education system. For example: Level A1 within the CEFR structure. The term can also refer to Junior Cycle within the post-primary mainstream education system, despite junior cycle including three years of post-primary education. Level is determined here either by the writers of syllabi, curricula, and frameworks or by the creators of educational materials.
- **Mainstream Education.** Educational materials, or syllabi, or examinations for any classes or subjects taught at primary level, at post-primary level, or at university level.
- **Reference Documents.** The term reference documents is used here to refer to syllabi, curricula, and frameworks developed, for example, by The Department of Education and Skills[4] for use on mainstream courses.
- **Scoil Ghaeltachta.** An Irish-medium school that is located in a designated Irish-speaking community.

[4] URL: https://www.education.ie

1. Introduction

This chapter defines key concepts and provides context for the present research. The chapter also includes research aims and questions, followed by a description of the structure of the thesis. Finally, the chapter includes a list of key terms and concepts.

1.1. Context of the Present Research

The Irish language is taught at primary and post-primary levels. Subjects other than Irish are also taught through the medium of Irish at these levels. Irish is a compulsory subject for all students in mainstream education in Ireland[5]. Three categories of students study Irish in school: Those who live in Irish-speaking communities and study all subjects through the medium of Irish[6], those who live in English-speaking communities but study all subjects through the medium of Irish[7], and those who live in English-speaking communities and who study all subjects through the medium of English. Learners in the first two school types will typically have more contact with their respective L2 than most other language learners, however, it is worth acknowledging how this distinction is less unique than it used to be because of how omnipresent English has become globally. Most Irish students receive either 13 or 14 years of Irish-language education; 8 years in primary school and either 5 or 6 years in post-primary school depending on whether the student completes the transition year, which is typically administered in 4th year. Numerous anecdotes continue to surface (Nic Pháidín (2003)) about an inauthentic type of Irish that is learned in Irish-medium school[8]s, a possible reason for which is offered here. Others who elect to learn Irish outside mainstream education typically do so by attending evening classes or by doing immersion courses in Irish-speaking communities; some of these courses are accredited and certified courses of education, e.g. CEFR-GA classes (Irish-language adaptations, implementations and interpretations of the Common European Framework of Reference for Languages: Learning, Teaching, Assessment (Council of Europe, 2001)).

The Gaeltacht schools and Gaelscoileanna are grouped together at primary level, where there is a curriculum for teaching Irish in an Irish-speaking environment and a separate curriculum for teaching Irish in an English- speaking environment. All Leaving Certificate[9] candidates sit the same terminal examinations, divided only by the following levels of ability: Higher, Ordinary, and Foundation level. The Irish exam paper is entirely in Irish. Exams for other languages are predominantly written in the studied language with instructions and select questions presented in parallel Irish and English. Exams for subjects other than a language, such as Physics or Business, are entirely in Irish or entirely in English.

The CEFR has had a considerable influence on Irish-language education. CEFR-GA courses are offered nationwide, some of which have been incorporated into other qualifications or aligned with existing courses and qualifications. The CEFR (Council of Europe, 2001) and CEFR-GA reference documents and materials have been included in this research for comparative purposes. The frameworks, syllabi, and curricula that guide all of the courses of education included in this section are described in more detail in Chapter 2. These documents are collectively referred to as reference documents here, each of which influences and guides the contents of the educational materials used.

The Department of Education and Skills website currently provides access to a database (https://www.education.ie/en/find-a-school) from which a list of schools can be generated according to the language of instructions. A report from this database was generated on 02/07/2018 using the following criteria: include schools for either males or females, as well as mixed-gender schools, schools of any ethos, and schools in all geographic areas. The results report 248 primary schools in Ireland where the language of instruction is all Irish, and 29 primary schools in Ireland where the language of instruction is both Irish and English. The same Department of Education and Skills database reports 48 post-primary schools where all pupils are taught all subjects through the medium of Irish, 14 post-primary schools where some pupils are taught all subjects through Irish, and finally 9 post-primary schools where some pupils are taught some subjects through the medium of Irish. No schools in the "special education" category have Irish as their sole language of instruction, but 6 are categorized as teaching in Irish and in English. In this context "special education" relates to special educational needs that have been professionally assessed. This number of Gaelscoileanna (Irish-immersion schools) has been growing steadily since the early- to mid-80s, reflecting the positive attitude towards Irish as a language and Irish as a language of instruction, particularly in non- Gaeltacht areas.

[5] Except for those that seek an exemption from the subject due to learning difficulties, due to time spent outside of Ireland, due to their residence being temporary, or due to the student having insufficient English.

[6] The literature refers to these schools as "Scoileanna Gaeltachta" or Gaeltacht Schools.

[7] The literature refers to these schools as "Gaelscoileanna" or (Irish-)Immersion Schools.

[8] Examples and a more complete explanation of this reportedly inauthentic type of Irish are provided later in this Section.

[9] The "Leaving Certificate" is the name of the certificate received by students after completing their final post-primary examinations. This certificate details the subjects, levels, and grades of all examinations.

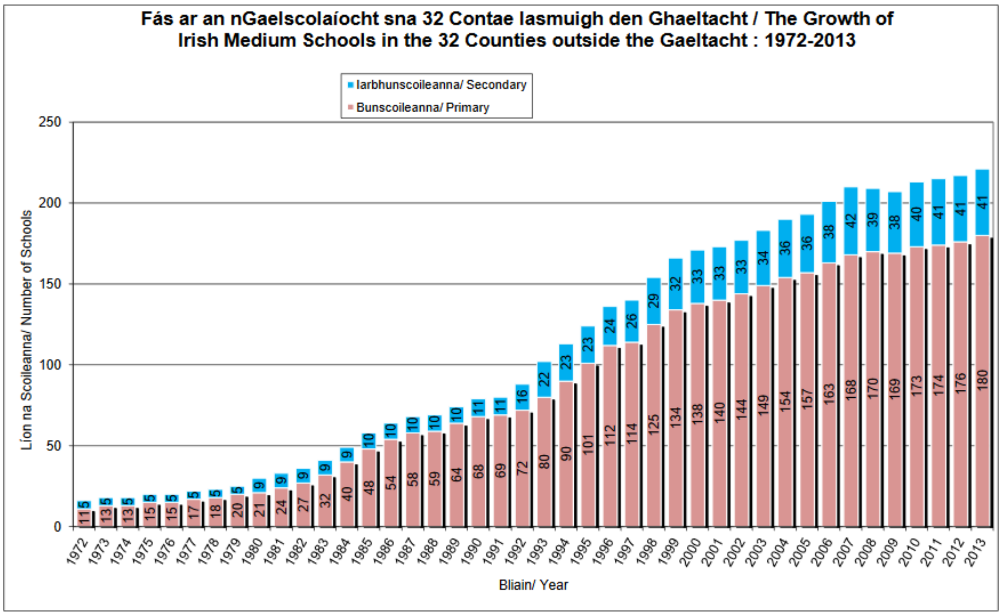


Figure 1: Number of Irish-immersion schools (*Gaelscoileanna*) in the Republic of Ireland

(URL: http://www.gaelscoileanna.ie/files/Growth.F%C3%A1s-gscoil-72-13.pdf)

The majority of universities in the Republic of Ireland, and some institutions of technology, offer courses in Irish at undergraduate level and postgraduate level. The majority of these courses are Bachelors of Arts. Irish-medium tracks for degrees in Business, Journalism, or Computer Science are also available. Universities (including centres within universities) and QQI (Quality and Qualifications Ireland) accredited institutions also host the majority of the accredited evening or extra-curricular classes in Irish. These courses are typically aligned with, or based on, levels of the CEFR (Common European Framework of Reference for Languages: Learning, Teaching, Assessment). Some universities and institutions also provide online language- learning materials; TEG[10], Vifax[11], and the Irish University Syllabus[12], to name but a few. Mainstream media and enthusiast groups also publish a substantial amount of materials, particularly at exam times, the majority of which can be found free of charge online. While these types of materials can frequently be valuable and well-constructed they are not always part of a focussed course of learning because they opt to focus on a small number of features, topics, or useful phrases for a prescribed topic.

[10] www.teg.ie
[11] http://vifax.maynoothuniversity.ie/
[12] http://www.arts.ulster.ac.uk/irish_syllabus/

This section has thus far introduced the structure of the Irish-language education system in terms of school type. I now introduce key points regarding materials, followed by reported issues with the type of Irish learned in the education system.

Regarding the materials selected for students, in some cases, special materials have been created for Gaeltacht schools and Gaelscoileanna, particularly at primary level where there is a specific curriculum for schools that share a language of instruction. At post-primary level Gaeltacht schools and Gaelscoileanna typically use the same Irish-language and Irish-medium materials, provided they are studying for the same exam. Students in English-medium schools who are sitting the same examinations as students in the other two school types can use the same Irish-language materials, or elect to use different materials where appropriate or required. Students in English-medium schools do not study other subjects through the medium of Irish, despite having the option to.

Materials for primary and post-primary levels are published by a number of private commercial groups, limited companies, and state-sponsored publishing companies. For example: a substantial number of materials are published by An Gúm, which is a company of the Irish State founded in 1925. Materials are also published by an *Chomhairle um Oideachas Gaeltachta agus Gaelscolaíochta* (henceforth, COGG) who are also a State sponsored organisation. COGG was founded under the provisions of Article 31 of the Education Act of 1998 to cater for the educational needs of Gaeltacht schools and of Gaelscoileanna in particular.
There are also functions in Article 31 relating to the teaching of Irish in the country's other schools, but in general terms COGG's remit resides with supporting Irish-medium education. The role of COGG relates to both primary and post-primary education and the three main areas of work are the provision of teaching resources, the provision of support services, and the research (primarily focused on Irish in the domain of education). In order to achieve the first of these tasks, COGG are maintaining and developing a database of educational materials via http://www.cogg.ie/aiseanna/.

The contents of educational materials heavily influence the language that is acquired in the classroom. There have been reports, both evidence-based and anecdotal, that the variety and standard of Irish learned in the three school types is different. The increase in the number of Gaelscoileanna, as shown in Figure 1, has coincided with interesting language examination results. Harris et al. (2006) found that there was a significant decline in the performance of children in Gaeltacht schools during the period 1985-2002 in the majority of tests. Gaelscoil students outperformed Gaeltacht students during this period and the Gaeltacht decline was also mirrored in English-medium schools over the same period. An anecdotal point that is worth noting at this point is the reported increase in families electing to speak Irish or both Irish and English in the home in urban areas and outside designated Gaeltacht areas. If true it may be the result of a number of factors:

1. Families with 1st, 2nd, or 3rd generation connections to a Gaeltacht area
2. Families where one or both parents speak Irish fluently as a result of their schooling, employment, an interest in the language, or other reasons.

It should also be noted that English is omnipresent in modern-day Ireland, including in Gaeltacht areas. Pétervάry et al. (2014) concluded that a significant number of the students from Gaeltacht schools who participated in their research performed better in the English portion of their tests than the Irish portion. The improved performance in Gaelscoileanna reported by Harris et al. (2006) occurred in tandem with the increase in the number of Gaelscoileanna. In spite of these performances in ability tests a contradictory phenomenon was reported (Harris et al., 2006) during the same period that a less authentic form of Irish was frequently used by *Gaelscoil* students. These less authentic constructions are most frequently referred to as *Gaelscoilis*. The *–is* suffix on *Gaelscoil* is indicative of a language name in Irish, for example: *An Ghearmáin* translates as *Germany* and *An Ghearmáinis* translates as German. Similarly, *An Fhrainc* translates as France and *An Fhraincis* translates as French. In Irish the language name is not a homonym of the adjectival form nor the demonymic form of countries' and nations' name. "Gaelscoilis" is a term used to refer to incomplete acquisition of Irish that is reportedly spoken in some Irish-speaking schools. The term is pejorative, but is not been precisely defined, and the sociolinguistic motivations for the phenomenon have not been investigated. The following is a brief discussion of the phenomenon, accompanied by some anecdotal examples of *Gaelscoilis* (Nic Pháidín (2003), Walsh (2007), Ní Ghearáin (2011)). Anecdotal examples of "*Gaelscoilis*" typically fall under the following categories:

i) Over-use of proverbs, for example:

(1) Aithníonn ciaróg ciaróg eile

Recognizes beetle beetle other

It takes one to know one.

(2) Níl aon tinteán mar do thinteán féin Is-NEG any hearth like your hearth own *There is no place like home.*

(3) Is i_ndiaidh a_chéile a thógtar na caisleáin
is-COP after each other rel_part build(autonomous) the castles
Rome wasn't built in a day

ii) Over-use of adverbials, for example:

(4) Go tobann!
(Adverbial particle) sudden
Suddenly!

(5) Ar muin na muice
On (upper) back the pig
On the pig's back
(lucky)

iii) mistranslations, learner errors, and/or word-for-word translations of figurative speech and idioms, number of examples of which appear in Nic Pháidín (2003):

Cad a bhfuil sé mar? ('What is it like?')
Tá chomh méid acu ann ('There are so many of them (there)') *Sin cad i gcónaí a dhéanann tusa* ('That is what you do always.') *Cén fáth nach?* ('Why not')
Mar! ('Because!')
A deireann cén duine? ('Says who?')
Mise, má chaitheann tú know! ('Me, if you must know!')
(emphasis and format as original)

iv) Other anecdotal examples are:

(6) Tá mé Twix

Am me Twix

I am a Twix [sic.]
Intended sentence: "Can I have a Twix?"

(7) Ag_tafann suas an

crann mí-cheart Barking

upward the tree

wrong *Barking up the wrong tree*

(This word-for-word translation does not carry the same meaning in Irish)

To the best of my knowledge, no evidence has been found to show that these examples of errors have come into common usage, which indicates they may be errors and part of the learning process. I would speculate that some examples are possibly tongue-in-cheek errors made on purpose by learners in order to elicit a response from a teacher, or for ironic or comedic ends. All examples could be categorised as learner errors of different sorts; L1 interference, insufficient vocabulary or incomplete knowledge of the appropriate grammar, or finally, over-extension of an idiom or of a word's meaning. The burden of proof lies with those who are creating this categorisation and, while the phenomenon may exist, a handful of anecdotes do not constitute evidence for a phenomenon. These learner errors can frequently be a part of the learning process, however, differentiating "school" Irish from "real" Irish is not impossible. For example, reference is made in Römer (2005) to a similar phenomenon in English-language coursebooks used in Germany where one of her core aims was to identify if there was a significant difference between "school" English and what she calls the real English from the reference corpus. This categorisation was empirically defined in Römer (2005), which is possibly the reason it was not used pejoratively, with differences found between the forms of verbs used and certain tenses (ibid, p.244-246). This field is receiving more attention in recent years (Ní Ghloinn, Uí Dhonnchadha, and O'Keefe (2018b), Ní Ghloinn, Uí Dhonnchadha, and O'Keefe (2018a)). When considering this language use it is also worth considering the recent changes in education described in this section, as well as changes in technology and media (Mac Mathúna, 2008) that have occurred during the same period.

In summary, Irish is being learned in a number of different language communities, each of which is subject to a number of issues relating to language change and language use. Whether one issue is causing another or these issues are simply co-occurring has not yet been empirically proven. For example, the test performances reported in Harris et al. (2006) co-occur with issues raised with materials in Mac Donnacha (2005), but there is no evidence of the issues with materials causing this drop in performance. Issues with educational materials have been reported for almost ten years, but examples of overly simple or overly complex language features were not given.

1.2. Issues Impacting Irish-language and Irish-medium educational materials

This section presents the issues from the literature relating to educational materials written in Irish. I first present the issues that relate to materials directly, followed by issues relating to language acquisition in mainstream education and language use in Irish-medium education. The stance adopted here with issues relating to language learning is that they are heavily impacted by the contents of materials.

Mac Donnacha, Ní Chualáin, Ní Shéaghdha, and Ní Mhainín (2004) reported on issues with materials, teacher teaching, and teaching practices in Gaeltacht schools. Among the issues reported was the level of the language used in mainstream education. Some Irish-language materials have been deemed overly simple for the level at which they are used and some Irish-medium materials have been deemed overly complex for the level at which they are used. Educational materials are central to the learning process and dictate much of the language used in the classroom (Burton, 2012), therefore, this research focuses on issues with educational materials alone. References to the other issues with the Irish language and mainstream education are only included to add context or for illustrative purposes.

One of the questions arising out of Mac Donnacha et al. (2004) was why Irish-medium materials were not being used more frequently. It was concluded that one of the biggest issues for Gaeltacht schools was the lack of materials containing level-appropriate content (ibid, p. 135-6). This report was concluded with a set of recommendations. The following bullet points paraphrase the recommendations relating to educational materials:

- An understanding needs to be arrived at between teachers and publishers regarding the appropriate level of the language used in educational materials. The level of fluency in the school and surrounding community ought to be taken into consideration when making recommendations about the level of the language used in educational materials.
- That a professional relationship be developed between teachers and the creators of educational materials such that the contents of materials and the teaching priorities are discussed. An annual conference was recommended as a forum for presenting this information.

Mac Donnacha et al. (2004) did not contain examples of the language that was not level-appropriate but was followed up by Mac Donnacha (2005), which included additional feedback from respondents. The question of appropriate materials was put to teachers, principals, and parents in Mac Donnacha (2005). The following excerpt is indicative of issues with the semantic content of Irish-medium lessons raised in the literature:

> *[N]íl siad sásta go bhfuil an beagán áiseanna atá ar fáil oiriúnach ach oiread. Thuairisc siad go mbíonn an Ghaeilge a bhíonn iontu ródheacair, níl dóthain ábhar ar fáil i gcanúint na bpáistí, bíonn an t-ábhar gearrshaolach [póstaeirí, bileoga le tabhairt amach sa rang &rl.] beagnach ar fad i mBéarla, níl dóthain ábhar 'nua-aimseartha' ar fáil i nGaeilge, agus ní bhíonn siad ar an eolas faoin ábhar atá ar fáil i nGaeilge*. (Mac Donnacha, et al. (2004:9)
>
> **"**[T]hey are not happy that the small number of resources that are available are suitable either. They reported that the Irish in them is too difficult, that there are not enough materials available in the children's dialect, that the ephemeral materials [posters, hand-out sheets for the class, etc.] are almost all in English, there are not enough 'modern' materials available in Irish, and that they do not know about the materials that are available in Irish."

They go on to say that teachers have reported the contents of the current materials result in two lessons; a language lesson that prepares the learners to use the materials, followed by the lesson about the subject matter. Respondents added that the reading materials for Irish lessons were too simple, with the parents from a particular area asking why more of their rich local literature and folklore was not used in school(s). The specific elements of the materials that were considered unsuitable, out of date, or too difficult were not specified by the respondents of Mac Donnacha (2005). While these issues were reported for both primary and post-primary schools in Gaeltacht areas, the aspects of the materials that were considered too simple or too complex were not specified. Finally, Mac Donnacha et al. (2004) found that some parents believed teachers hid behind a reported lack of appropriate textbooks and used it as an excuse not to teach certain unspecified subjects through the medium of Irish.

Continuing with this topic de Brún (2007) reports on what type(s) of reading materials children enjoy and on how few books are bought. De Brún goes on to investigate why this is happening and finds the difficulty of the language used in educational materials is among the reasons. De Brún (2007) found that educational materials initially published in a language other than Irish, that were then translated into Irish, were more likely to contain a level of Irish that was too complex for the level at which they were meant to be used. De Brún (2007) points out that high-frequency English words were selected for the original texts and that the translations of these words are not necessarily high-frequency Irish words. She goes on to add that unfamiliar words or unnatural structures can be included in the sections of a translated book where space is limited by pictures or graphics. One reason for this is because the short simple English words or phrases that fill these areas of the page can have translations that are either more grammatically complex or longer and more syntantically complex. In her conclusions De Brún (ibid, p.32) adds that COGG (An Chomhairle um Oideachas Gaeltachta agus Gaelscolaíochta) have found teachers reporting that translations of textbooks cause particular problems for learners, and that the level of Irish in them is too high.

This topic of research continues in Ó Muircheartaigh (2009), where the opinions of teachers from Gaeltacht areas with regards to materials, among other topics, were collected by survey. The following are anonymised quotes from the survey:

i) "*Mar shampla, bíonn ort gníomhachtaí ó leabhair shaothair agus ó théacsleabhair a athscríobh chun iad á dhéanamh níos simplí ó thaobh teanga de.*"
"For example, you have to rewrite exercises from workbooks and from textbooks in order to simplify the language."
ii) "*Bíonn gan amhras, bíonn ceacht teanga i gceist le gach aon ábhar leis, obair bhreise cinnte dúinn.*" "There is undoubtedly, there is a language lesson within every subject too, certainly more work for us."
iii) "*Tá an-chuid ama caite ag déanamh acmhainní Gaeilge agus ag aistriú acmhainní Béarla.*" "A lot of time is spent creating Irish resources and translating English resources."
iv) "Is minic go mbíonn orainn obair a aistriú go Gaeilge muid féin sul má féidir linn tabhairt faoin rang." "We frequently have to translate work to Irish ourselves before we can go about teaching a class."

The issue of provision is noted here because it has come to light that what is meant frequently relates to a provision of appropriately-levelled materials, rather than there being no materials at all. This distinction is made in the survey conducted for Ó Muircheartaigh (2009).

An Chomhairle um Oideachas Gaeltachta agus Gaelscolaíochta (2011) probes the same recurring topic as the researchers above, but with the aim of developing materials and ensuring best practice in the creation, provision, and development of educational materials. It was found that the findings from Mac Donnacha et al. (2004) continued to be true in COGG (2011:37 of 67) where the highest frequency response was: The Irish textbook is out of date.

The second highest frequency complaint was with the difficulty of the language in textbooks.

Thus far it has been established that issues with educational materials have been reported in the literature for almost ten years. The most persistent of the issues across subjects and levels of education has been the overly simple or overly complex language used in materials. When teachers are translating materials to English on a regular basis because of comprehension difficulties it would appear as though the issues are not localised around a set of learner difficulties, which the teacher would likely address, but are wide-ranging issues that are true for an entire domain. When this has been reported the features of the language that were overly simple or overly complex were not specified. As a result, identifying the features of the language that are either complex or simple must be done before changes can be made. In the context of this research language features facilitate the performance of communicative functions and the communication of meaning. Examples of language features are set out in curricula and syllabi and can include categories and Forms of nouns and verbs, particular phrases, vocabulary, punctuation, and sentence structure. These Language features prescribed for each level of education vary according to the communicate needs and Goals, related subject matter, audience, and medium.

The impact these findings have on learners' acquisition have not been quantified, it is safe to assume that the impact is not null. Maunsell (2009) reminds us of the function of what is taught and examined in Irish- language classes, i.e. to enable people to communicate effectively with each other. The 2011 Census (Central Statistics Office, 2011) reported that only 1.8% of the population aged 3 and above[13] speak Irish on a daily basis outside the education system[14]. This suggests low levels of conversion, or people transitioning, from learning Irish to speaking Irish to/with the greater Irish-speaking community. While the issues impacting education are wide-reaching this research focusses on analysing educational materials in order to find objective means of assessing the complexity of educational materials.

1.3. Research Aim

The primary aim of the present research is to quantitatively analyse the contents of Irish-language educational materials and Irish-medium educational materials with view to developing a set of complexity metrics that can be used to objectively rank educational materials according to their complexity.

Achieving this aim requires the construction of a corpus of educational materials, therefore, the construction of this corpus is the other primary aim of this research. The metrics have been designed so that they provide objective feedback for materials creators. To achieve this the present research aims to combine multiple complexity measures in order to give as detailed a description as possible of language complexity. The topic of complexity is approached from three different perspectives: lexico-grammatical complexity, syntactic complexity, and semantic complexity.

1.4. Research Questions

This section contains the three research questions answered in this thesis. This study seeks to answer the following research questions:

i) How are items prescribed in the curriculum or syllabus reflected in educational materials? By this I mean, do prescribed language features occur at a frequency that is statistically significant at the level of education for which they are first prescribed?
ii) Can complexity measures used on other languages be used to objectively evaluate the complexity of Irish-language and Irish-medium educational materials?
iii) Can terminology be used to measure the semantic complexity of educational materials?

[13] Although 3-year olds would not be attending school, the Central Statistics Office has included children of this age. This is likely the case
[14] All statistics and reports are available via: http://www.cso.ie/en/index.html

1.5. Structure of the Thesis

This thesis has five chapters. This section outlines the content of each chapter in turn. Section 1 of each chapter gives an overview of that specific chapter. The summary findings of each chapter are given at the final section of each chapter.

Chapter 2 is split into five subsections that introduce the different aspects of Irish-language education relevant to the present research. In doing so I introduce the different types of schools in which Irish is taught in the Republic of Ireland, as well contexts in which courses in Irish have been based on the Common European Framework of Reference (Council of Europe, 2001). I also summarise the syllabi, curricula, and frameworks that guide each of the courses of learning.

Chapter 3 is split into three subsections and a set of conclusions. Each subsection introduces and discusses literature relavent to one of the three pieces of analysis done in the present research. The first subsection includes analyses of individual or multiple language features in the domain of education. The second subsection includes analyses of length of word and of length of sentence. The third sebsection includes term extraction and information retrieval research, with reference to the way terminology is viewed in the language-learning context. The conclusions describe how the literature has influenced the methodologies applied in the present research.

Chapter 4 details the corpus compilation process, including its design, collection, pre-processing, annotation and finally the specifications of the corpus in terms of the number of tokens and documents per sub-corpus. The corpus is called EduGA.

Chapter 5 contains three analyses that were done on EduGA. The first is a frequency analysis of 7 prescribed items which that were selected from the syllabi and curricula. The second is an analysis of word length and of sentence length. The final piece of analysis combines term extraction with a statistically-based weighting scheme.

2. Syllabi, Curricula, and Frameworks for Irish-language Education

This chapter contains an overview of the curricula, syllabi, and frameworks that guide language learning in mainstream education and in other accredited courses based on the CEFR (Common European Framework of Reference for Languages: Learning, Teaching, Assessment). As noted in section 1.1 syllabi, curricula, and frameworks are referred to collectively as “reference documents” in this study.

The Department of Education publish the curricula and assessments used in early childhood education, and in primary and post-primary education. The Curriculum and Assessment Policy Unit (CAP) within the Department of Education are responsible for providing the support and development of curricula, assessments, and general policy[15]. CAP provide guidance and assistance to teachers and schools directly and via a number of distinct bodies: State Examinations Commissions (SEC), National Council for Curriculum and Assessment (NCCA), National Council for Guidance in Education (NCGE), An Chomhairle um Oideachas Gaeltachta agus Gaelscolaíochta (COGG), and the Post-primary Languages Initiatives (PPLI). I now include a brief description of the roles and functions of these bodies.

SEC[16] is responsible for the development, assessment, accreditation and certification of the second-level examinations of the Irish state: the Junior Certificate and the Leaving Certificate. SEC is a non-departmental public body under the aegis of the Department of Education and Skills. SEC publish past examinations on their own website (examinations.ie), however, the curricula for each subject at primary and post-primary level are published separately on Curriculum Online[17].

The official role of the National Council for Curriculum and Assessment (NCCA)[18] is to advise the Minister for Education on curriculum and assessment for early childhood education, primary and post-primary education. The advice given is typically the result of consultation processes with the public, with schools, and with other parties who are active in education settings. This consultation is managed through online tools, submissions and focus groups.

[15] https://www.education.ie/en/The-Department/Management-Organisation/Curriculum-and-Assessment-Policy-Unit- CAP-.html
[16] https://www.examinations.ie/
[17] https://www.curriculumonline.ie/Home/
[18] https://ncca.ie/en

The advice and reports given to the Minister for Education are agreed upon by twenty-five council members who have been appointed by the Minster for Education for a three-year term. These council members represent partners from education, industry and relavent trade unions, parents' organisations and other educational interest groups. The council also includes one nominee each from the Minister for Education and Skills and the Minister for Children and Youth Affairs[19]. The NCCA collaborate with academics and subject experts to publish reports[20] [21] on research conducted in and for the domain of education, naturally these reports inform the advice that is past on to the Minister for Education regarding curriculum and assessment, as well as best practice and other related topics. Harris and Ó Duibhir (2011) specifically set out to identify and report on up-to-date research on the acquisition of a second language in the classroom. This is done with a view to informing curriculum development.

PPLI[22] is a dedicated unit providing expertise and support for foreign languages education in Ireland, predominantly at post-primary level where these languages are examined. PPLI's remit currently includes Chinese, French, German, Italian, Japanese, Korean, Lithuanian, Polish, Portuguese, Romanian, Russian, and Spanish.

When changes are made to policy or curricula they are typically announced in a circular of the Department of Education first, followed by the publication of the new or amended reference document. Two examples of which are the ammendments outlined in An Roinn Oideachais agus Scileanna (2010b) and the official implementation notion of these ammendments in An Roinn Oideachais agus Scileanna (2010c). Electronic copies of these reference document are typically found on www.education.ie.

The reference documents are presented here because of the substantial influence they exert over the contents of educational materials. This applies to materials for mainstream education and CEFR-GA courses alike. The aim of discussing the reference documents here is twofold; firstly, to identify prescribed language features that can form the basis of a complexity metric because they ought to be central to the content of educational materials. Secondly, to inform the interpretation of the results of the analytic stage of this research. Both the search for languages and the interpretation requires an overview of how language features are prescribed, and the structure of the content prescribed in each reference document.

[19] https://ncca.ie/en/about/about-ncca/what-we-do
[20] NCCA Research Series: All. https://ncca.ie/en/publications-and-research/research-series
[21] NCCA Research Series: Language. https://ncca.ie/en/search-ncca?q=language
[22] https://ppli.ie/

Regarding the use of reference documents as guiding the development of complexity metrics, this relates to a situation where language feature L is consistently prescribed at a higher level of education, and never for beginners for example. This indicates that syllabus and curriculum designers either consider this feature to be too difficult for beginners or more complex than the features that precede it or they believe certain building blocks or scaffolding are required before L can be acquired.

Some of these features *may* also be considered more complex than others outside the domain of education, but some will not necessarily be considered more complex when encountered outside of an educational environment. For example, relative clauses are considered more complex than single-clause sentences. In a general sense, this complexity is also realised outside of the educational environment. Certain language features simply require building blocks before they can be acquired and other language features are learned before others simply because one or other must come first. For example, learners will likely acquire present tense forms of a verb before they acquire future-tense forms. This order does not make the future tense more complex than the present. As a result, the reference documents are used as a guide and the selection of prescribed features considers factors other than the level at which language features have been prescribed.

Chapter 2 contains five subsections. Sub-sections 2.1 through 2.4 summarise the similarities and differences between the reference documents for Irish-language courses whose educational materials are included in EduGA. While pedagogies and learning aims are not being assessed in this research, they inform the interpretation of results. Finally, Sub-section 2.5 details the three categories of prescribed language features that form the basis of the analyses conducted in Chapter 5.
Irish-medium materials are included in the analysis, however, the syllabi for teaching these subjects (such as Chemistry, History, Business Studies, and so on) only focus on the core principles and learning outcomes of the subjects themselves without specifying any required language skills. As a result, the syllabi for these subjects cannot inform the selection of language features for the analytic stage of this research.

2.1. Primary-level Irish language learning

There are two primary documents that guide Irish language learning at primary level:

- *Curraclam na Bunscoile: Gaeilge. Treoir do mhúinteoirí* ["Primary School Curriculum: Irish. A Guide for Teachers".] (An Roinn Oideachais agus Scileanna, 1999b) (henceforth, CBG-Tr)
- *Curraclam na Bunscoile: Gaeilge* ["Primary School Curriculum: Irish"] (An Roinn Oideachais agus Scileanna, 1999a)(henceforth, CBG).

There is also a third reference document worth acknowledging, A Framework for Junior Cycle (An Roinn Oideachais agus Scileanna, 2015a), which combines the strands of learning from the aforementioned curriculum for Irish with the English-language counterpart that is not discussed here. An Roinn Oideachais agus Scileanna (2015a) provides guidance to teachers regarding transferable skills from one language to the other, on helping the pupil with his/her general understanding of languages, and finally with helping pupils develop their communication skills. These recommendations relate to pedagogy rather than to the subject matter and contents that would be found in educational materials. A communicative approach to language learning features in CBG (ibid, p.8) and throughout CBG-Tr. However, an integrated approach to language learning is also prescribed.

This 2015 curriculum will be important in future, it has not had an influence on the materials included in EduGA since almost all the materials included in EduGA were developed prior to its publication. It is being acknowledged but will not be referred to in this study because it cannot yet be used to develop complexity metrics.

CBG-Tr was designed to help teachers understand and access the contents of CBG and better understand the theoretical basis for its contents. This is done by describing methods of classroom planning and assessment planning, by introducing the core topics (semantic content) for the classroom, by giving some examples of teaching material and content, and by giving some background to the communicative approach to language learning. In general terms, the contents of CBG-Tr are more theoretical than CBG, which includes more examples of prescribed language features. CBG provides two sets of directions: One set of directions is for pupils studying Irish either in a Gaelscoil (e.g. Irish-medium school not in a designated Irish-speaking community) or in a Gaeltacht school (Irish-medium school in an Irish speaking community). The other set of directions is for pupils studying Irish in English-medium schools. Both sets of directions are divided into four class groups: The junior- and senior-infant classes, 1st and 2nd class, 3rd and 4th class, and finally 5th and 6th class. Educational materials are not, however, always divided into these categories.

A set of broad-curriculum learning objectives are set out in CBG[23] (ibid, p.14) that are immediately followed by general objectives that describe language acquisition, enjoyment of the subject, use of appealing materials, development of communication skills through tasks and games, and formally and informally improving abilities in the four communicative modes (or strands) in (vi) above. It is noted that pupils ought to receive language assistance for studying other subjects through the medium of Irish, however, no further elaboration is given. Use of technology through Irish, and the composition and comprehension of short pieces of creative writing are also included, as well as learning to use drafts when writing. The general goals conclude with statements of awareness of national and European identity and culture.

CBG prescribes a range of recommended topics, some of which allow for the teachers' and learners' interests to influence the topics covered in class. Each topic is accompanied by a range of communicative functions relating to that topic. The range of communicative functions accompanying each topic broadens as the learner level increases. The topics for primary school do not broaden. The topics are:

- Myself (*Mé féin*)
- At home (*Sa bhaile*)
- School (*An Scoil*)
- Food (*Bia*)
- Television (*Teilifís*)
- Shopping (*Siopadóireacht*)
- Hobbies (*Caitheamh Aimsire*)
- Clothes (*Éadaí*)
- The Weather (*An Aimsir*)
- Special Events (*Ócáidí Speisialta*).

[23] including personal development, social and cultural awareness, and

These topics are prescribed for all primary-level classes, from Junior Infants to Sixth Class. CBG recommends sub-topics be selected based on these topics depending on the interests of the learners. CBG prescribes a specialized vocabulary for the topics (ibid. p.35, 43, 55, 93) and the development or broadening of this specialized vocabulary is also prescribed (ibid, p.69, 105, 119, 133). Examples of these additional communicative means include a progression from recounting a story or following instructions to expressions of abstract concepts and emotions, debate, and reasoning. Each communicative function is accompanied by one or more examples of how a learner at each given level could perform the communicative function.

Table 1: Prescribed Language: Topic, Communicative Function, Examples

Topic	Communicative Function	Example
Talking about the weekend	Saying you went somewhere	"I went to Cork." "I went to the cinema with my friend, Bob." "I played a match" "I did nothing!"
	Asking if somebody enjoyed something and answering the question.	"Did you like Cork City?" "It was smaller than I thought, but I liked that about it." "Did you enjoy playing the match?" "Absolutely! We lost, but it's always fun to meet my friends and play a match."

The "Communicative Function" tier of the prescribed language taxonomy shown in Table 1 can also include categories of grammar or vocabulary relating to the topic. Examples are included in Table 2.

Table 2: Prescribed Language: Topic, Grammar or Vocabulary, Examples

Topic	Vocabulary and Grammar	Examples
Talking about the weekend	Transport	Car, bus, train, bicycle, public transport, return ticket, road trip, petrol station, motorway.
	Past tense verbs	Drove, bought, went, played, shared, caught, walked, ran, cycled, enjoyed.

The guidance and examples given in both CBG and CBG-Tr are intended for use when speaking, listening, reading, and writing; each of which is referred to in CBG and CBG-Tr as a strand[24]. It is repeatedly stated in both documents that all strands should be learned in parallel complementing one another whenever possible (ibid, p.88-89, 100-101, 114-115, 128-129). From my experience as a teacher, written educational materials are read aloud much more frequently than written content from other domains or registers (such as emails, newspaper articles, or novels). This can happen because a learner is asking or answering a question in the materials, if a section of the materials is being explained by the teacher then it is read aloud, and new words are often both read and heard when they are first encountered. Another key feature of the primary-level curriculum is that learners are expected to learn all the roles involved in an interaction, e.g. both speaker and listener, in parallel.

[24] The medium of communication can also be called a "mode" in the literature.

It is also specified that learners will learn a number of different ways of completing a language function during their time at primary school, each of which is appropriate in different situations. Two illustrative examples are given where a person is asking for chocolate in two different ways:

(8) Seacláid, le do thoil.
Chocolate, with your wish
Chocolate, please.

(9) Ba mhaith liom barra seacláide, le do thoil
[cop.] good with-1SG bar chocolate, with your wish
I would like a bar of chocolate, please.

(An Roinn Oideachais agus Scileanna (1999a), p.18)

The second of these examples is taught at a higher level than the first, but it does not replace the first example. In CBG we learn that the second example is taught second because it allows the user to add a layer of appropriate communication to their repertoire, in this case, for more formal situations. While a considerable amount of repetition can be seen throughout CBG, this example shows one way the language used could change despite the repetition of a very similar communicative goal. The following learning goal is repeated verbatim for 3rd class through 6th in schools where English is the language of instruction and for 1st class through 6th in schools where Irish is the language of instruction:

- ***páirt a ghlacadh i ngníomhaíochtaí éisteachta a léiríonn tuiscint*****:** *éisteacht agus clárú eolais, éisteacht agus breacadh nóta, éisteacht agus leanúint treoracha (cluiche páirce agus clóise, gluaiseacht ceol), aithint agus idirdhealú a dhéanamh ó chur síos taifeadta, scéal a insint ó shraith fuaimeanna.*
  (**taking part in listening exercises in a way that shows comprehension:** listening and recording information, listening and taking notes, listening and following directions (field and yard games, musical movements), recognising and distinguishing based on a recorded description, telling a story based a series

of sounds.)

In this sense, it can be said that the goals and objectives of the primary-level Irish curriculum are relatively broad and inclusive of a wide variety of communicative modes and goals. They are broad and inclusive in order to cater for learner of as many different levels of ability as possible. Some abilities are only prescribed for 5th and 6th class in schools where Irish the language of instruction, for example:

- ***leathnú a dhéanamh ar abairtí simplí****: abairtí a cheangal le chéile; agus, nó, ach, mar.*
- **the extension of simple sentences:** connecting sentences to one another; and, or, but, (such) as

(An Roinn Oideachais agus Scileanna (1999a), p.133)

While a large number of examples would be valuable to a study of this sort in order to differentiate levels of complexity, it is understandable that only a small number of examples are included so that the content covered in classrooms can be tailored to each learner. Equally, the reference documents are designed to guide, therefore, creating or selecting appropriate content for the classroom is largely delegated to teachers and materials creators. Finally, no indication has been given that corpus data was used in the writing of reference documents. The creation of a large number of examples for educational materials could be problematic without corpus data upon because they would likely be subject to the creators' bias or idiosyncrasies.

2.2. Post-primary level Irish language learning

There are two primary reference documents for junior-cycle Irish. I will be referring to the *Framework for Junior Cycle* (An Roinn Oideachais agus Scileanna, 2012a) rather than *Framework for Junior Cycle* (An Roinn Oideachais agus Scileanna, 2015b) because it applies to the contents of the EduGA Corpus. Post-primary junior cycle courses for all subjects are currently being changed dramatically, and Irish is no exception. While extensive changes will occur, with the changes to the Irish courses only coming into effect in September 2017 they will not have any impact on the materials in EduGA. Ní Mhaonaigh (2013) presents issues with the outgoing approach, chief among her points is the shift from separate guidelines for English-medium and Irish-medium classrooms at primary level to syllabi that are focussed solely on level of ability at secondary level. Noteworthy among the changes are separate approaches to language teaching in Irish-medium and English-medium schools. Prescribed reading is in place under the outgoing approach, however, separate prescribed reading lists for Irish-medium and English-medium schools have been compiled for the new approach and use of CEFR's universal level descriptors and self-assessment level descriptors have also been added. Finally, the grading of the course is changing from a terminal examination to continuous assessment. This study remains relevant because the analysis development steps outlined will continue to apply to corpus-linguistic analyses for Irish, and these changes only affect one sub-corpus. More specifically the analyses and results detailed in Section 5.1 are reproducible, the analyses in Section 5.2 are both reproducible and the discussion lays groundwork for further development of analyses of this sort. Finally, the analysis developed and used in Section 5.3 is reproducible. All results can be referenced in future comparative analyses, for example, in a comparison between the outgoing and incoming junior-cycle materials. The reference documents for post-primary Irish education that related to the contents of EduGA are as follows:

Junior Cycle

- The Framework for Junior Cycle (2012a)
- The Junior Certificate: The Irish Syllabus (ND)

Senior Cycle

- A syllabus for higher and ordinary level (2010d)
- A syllabus for foundation level (2010)

The frameworks and syllabi above prescribe a list of communicative topics (semantic content), sample learning statements, methods of assessment, and describe the way all post-primary courses are to be administered. Each subject within the Junior and Senior Cycles has a syllabus which describes the subject- specific details of each course, subject-specific examinations, and grading. A number of clarifications and distinctions regarding courses' goals and the difference between foundation, ordinary, and higher level are also given. Similar to the primary level syllabus (An Roinn Oideachais agus Scileanna (1999a)) the stated goals of the Irish language course include personal development, and cultural and social awareness as well.

The syllabi for Irish (An Roinn Oideachais agus Scileanna (ND) and An Roinn Oideachais agus Scileanna (2010d)) state that pupils undertaking the ordinary-level course ought to be able to do everything prescribed for foundation level, as well as that which has been prescribed for ordinary level. Likewise, pupils undertaking the higher-level examination ought to be able to do everything prescribed for both foundation and ordinary level, as well as that which has been prescribed for their own levels. This is important to the present research because it implies the contents of lower-level materials will be included, rather than replaced, in higher-level materials.

Continuity is the first stated aim of the Junior Cycle syllabus, which purports to consolidate and build upon that which has been learned in primary school —particularly the pupil's knowledge, understanding, skills, and competences. In addition, a stated aim of the Senior Cycle curriculum is to build upon that which has been learned during Junior Cycle. The remaining aims are very similar to those listed for Junior Cycle, including citizenship and national identity, themes for the course, and language functions —all of which are followed by examples (An Roinn Oideachais agus Scileanna (2010a), p 11-26).

The communicative approach to language learning is not explicitly prescribed in either syllabus. Having the ability to communicate with classmates is a goal, as well as with the public and the media, as is completing group work and other communicative tasks. The description and structure of the course contents are similar to primary level and to communicative approaches to language learning. Themes and Topics are specified for each course (foundation, ordinary, and higher) each of which comes with a collection of communicative functions. Table 3 (below) contains the titles of topics for ordinary- and higher-level post-primary education for Irish. Each topic contains a similar list of sub-topics that can be found in the appendices of the curriculum.

Table 3: Topics from Appendices 1 and 2 of *An Ardteistiméireacht, Gaeilge*. (An Roinn Oideachais agus Scileanna, 2010a)

Topic	**Ordinary Level Irish**	**Higher Level Irish**
The learner	√	√
Home and place of living	√	√
School and education	√	√
Friendship	√	√ (and Relationships)
Music	√	√
Sport	√	√
Pastimes	√	√
Irish (language)	√	√
Mediums of communication	√	√
Work life	√	√
Travelling and languages	√	√
The weather	√	√
Particular People	√	√
Religion and ethics	√	√
Culture	√	√
Young people's problems	√	√
Industry	√	√
The environment	√	√
National and international affairs	√	√
The State	√	√
Money	√	√
Fashion	√	√
Food and Health	√	√
Holidays	√	√
Information and communication technologies	√	√
The Third World	X	√
Literature	X	√
Other topics in which the learners show an interest	√	√

It is expected that these topics will be found in educational materials to varying extents at appropriate levels of ability, and depending on the preferences within the classroom. The communicative functions for each topic are almost always followed by brief and isolated illustrative examples. For example, pages 17-30 of An Roinn Oideachais agus Scileanna (ND) contains 29 communicative functions specifically written for materials creators and teachers who create lessons, the 4th of which is "*Coinne a dhéanamh*" [Making an appointment]. This communicative function is followed by two groups of examples; four ordinary-level examples and two higher-level.

Ordinary-level examples

*(10)*Buailfidh mé leat tar_éis lóin
Will_meet I with.2SG after lunch
I will meet you after lunch

*(11)*Feicfidh mé tú ag
a hocht
Will_see I you at
[num.part.] eight
I will meet you at eight

*(12)*Cathain a bheidh tú saor
When [rel part.] be
you free *When will you be*
free?

*(13)*An mbuailfidh tú liom san óstán
[interog. part.] will_meet you with.1SG in.DET hotel
Will you meet me in the hotel?

Higher-level examples

(14) Ba mhaith liom coinne a dhéanamh leis an dochtúir

[cop.past.] like with.1SG appointment [rel. part] doing with.DET doctor

I would like to make an appointment with the doctor

(15) An bhféadfá bualadh liom ag an deireadh seachtaine [sic.][25]

[interog.part.] could.2SG meeting with.1SG at the end week

Could you meet me at the weekend?

These examples, and their use in this study, are discussed more in Section 5.In conclusion, although the approaches to education differ from primary level to post-primary level, up as far as An Roinn Oideachais agus Scileanna (2010a) one of the stated aims is building upon that which was learned at the previous level. An Roinn Oideachais agus Scileanna (2010a) and An Roinn Oideachais agus Scileanna (ND) are similar to An Roinn Oideachais agus Scileanna (1999a) and An Roinn Oideachais agus Scileanna (2015a) in respect to language prescribed in them, and the topics and examples given.

[25] The expected question marks were not present.

2.3. Common European Framework of Reference for Languages: Learning, Teaching and Assessment (CEFR)

The Common European Framework of Reference for Languages: Learning, Teaching, Assessment (CEFR) (Council of Europe (2001)) was published by the European Council to provide a common basis for the elaboration of language syllabuses, curriculum guidelines, examinations, textbooks, and other materials related to language learning (CEFR, 2009:1). This common basis was chosen for two reasons, to inform people working in language learning and to enable course administrators, teachers, and examiners to understand the level of ability attained on courses in other languages or countries via this language-independent framework. The CEFR is the result of decades of work, research and co-operation across multiple institutions and organisations in Europe. The CEFR achieves its common basis in a language- independent way by providing can-do statements across a number of levels. This delegates the language specific features appropriate to different levels to experts from each language. One of the primary reasons the CEFR was designed this way is because language learners frequently pass through a number of educational sectors and institutions, and the provision of common levels was seen to be an effective way of encouraging collaboration between these sectors and institutions (CEFR, 2009: p.2, p.17). The language-independent approach also gives experts from each language the freedom to create a course of learning for Language A that *can* be aligned with a course for Language B; without the course for Language A being based on, or necessarily related to, a course for Language B.

Given that the focus of this research is on teaching materials, this summary primarily focuses on chapters 3 to 5 and appendices A-D of the CEFR, i.e. the chapters and appendices that have the biggest influence on the contents of educational materials. These are Chapter 3 “Common Reference Levels”, chapter 4 “Language use and the language user/learner”, and chapter 5 “The user/learner’s competence”, as well as Appendices A, B, C, and D of Council of Europe (2001).

Chapter 3 defines the common reference levels, which include general competencies that are not associated with any particular situation or mode of communication. The descriptors in Chapter 3 are provided as context-free can-do statements, such as, “ I can understand texts that consist mainly of high-frequency everyday job-related language. I can understand the description of events, feelings and wishes in personal letters.” (Council of Europe (2001), p.26)

Chapter 4 defines the requirements on a learner in order to achieve a certain level of ability. The criteria are presented as a kind of checklist that give learners the opportunity to explore the extent of their abilities. An important feature of Chapter 4 is that the domain of use is included in each topic, for example: Private, Public, Occupational, or Educational domains can be used to filter the learner's ability to communicate effectively on topics like locations, persons, events, or texts (Council of Europe (2001), p.48).

Chapter 5 outlines competencies that enable a learner to use his/her language, such as flexibility, turn- taking, coherence, cohesion, and thematic development. It is important to note that the CEFR describes the communicative proficiency of the learner in each of the modes of communication, i.e speaking, listening, reading, writing, and the general Global Scale.

The CEFR is frequently described as a concertina-like reference tool that provides expanding categories, levels and descriptors for language learning. They have been written in this way so that educational professionals can merge or sub-divide, elaborate or summarise, adopt or adapt according to the needs of their context (Martyniuk, 2010). This means the CEFR can be tailored to the needs of academic, business, or casual language learners. For example the levels of ability defined in the CEFR range from A1 (Breakthrough) through A2, B1, B2, C1, to C2 (Mastery) but a course may require subdivision within a particular level such as B1.1, B1.2, B1.3 —all of which can be done using the CEFR. These levels are collectively referred to as the common reference levels and are based on recommendations made by Wilkins (1978). Each level is described in terms of can-do statement.

The following paragraphs explore what level a person is considered to be at depending on what they *can do* with the language they are learning. There is no validation body for any language, at present, to check the alignment of course materials with levels of the CEFR. Each of these points are key, and will be seen to be important as the materials are analysed and compared to one another.

Semantic content in the CEFR is specified in several different ways, one of which is called Descriptive Categories. These descriptive categories are titled as follows: Locations, Institutions, Persons, Objects, Events, Operations, and Texts. Each of these can be categorized according to one of the following uses: Personal, Public, Occupational, or Educational. What this means is that where the category "Persons" intersects with "Personal" the vocabulary for relations, family members and friends is needed, and where "Persons" intersection with "Occupational" the vocabulary for employer/employee, manager, colleague, subordinate, and so on, is needed.

Up to this I have introduced the reference documents and described their similarities, however there are some differences worth noting. The CEFR was primarily written for adult learners who are learning a second or foreign language and its associated culture, however, these learners are expected to have another mother tongue and culture as a reference or guide (CEFR, 2001:43). Primary and post-primary school pupils will still be developing their general language skills and cultural awareness, be it through Irish or alongside English. Young learners of Irish have not fully developed their mother tongue in the way the CEFR describes learners[26]. This can be reflected in the materials, for example, an adult learner may be given a translation for a new word, within the context of a lesson, whereas a child may not be given a translation and may rely on a picture or the context. The CEFR has undoubtedly had an effect on language education in Ireland[27]. Another key difference between the CEFR and all other reference documents included here is that it is a language-independent framework, therefore, the CEFR does not prescribe the pairs of communicative functions and examples or grammatical features and examples that other reference documents do prescribe. Topics are not included but are referred to descriptors. The closest equivalent to the topics or themes included in the CEFR are the descriptive categories (ibid, p.48-49). The descriptive categories describe four domains; personal, public, occupational, and educational. Each of the four domains is associated with the following categories; locations, institutions, persons, objects, events, operations, and texts. When topic is referred to in descriptors it typically refers initially to topic identification, context-based comprehension of the language used, and moving on to a range of familiarity in different settings. This familiarity frequently includes reference to the extent of a learner's topic-specific vocabulary.

The CEFR has undoubtedly had an influence on the other reference documents included here therefore leading to a number of similarities, however, the differences arising from its context-free and language- independent approach are considerable.

[26] The CEFR (Council of Europe (2001)) had adult language learners in mind when it was initially written, despite it now influencing a wider cohort of language learning courses.

[27] The CEFR has been explicitly referenced in An Roinn Oideachais agus Scileanna (2017)

2.4. Documented applications of the CEFR for Irish: CEFR-GA

A number of institutions, universities, and organisations offer Irish courses that use the CEFR to varying extents. A number of these bodies have contributed materials to EduGA; the corpus created for this research. In this section I summarise applications of the CEFR in the Irish-language context (CEFR-GA). These are situations where the majority of the course of learning is CEFR-based, as opposed to some courses that only apply elements of the CEFR to certain elements of their language-learning courses. As was the case with other reference documents, the aim here is to identify features that can be used to categorize educational materials and identify markers of complexity.

Teastas Eorpach na Gaeilge (TEG)[28] [*European Certificate of Irish*] provides a series of general Irish language proficiency examinations and qualifications for adult learners of Irish. TEG courses are also used as the language element of under-graduate and graduate qualifications, such as teacher training. TEG exams give candidates an opportunity to show their ability in speaking, listening, reading and writing Irish at different levels, from absolute beginner (A1-A2) through intermediate (B1-B2) and advanced levels (C1-C2). TEG courses are currently offered at levels A1 through C1 to adults. TEG provide a syllabus for each of these, as well as sample materials, and examinations twice per year. A number of organisations nationwide offer courses that are based on the TEG CEFR-GA syllabi (http://www.teg.ie/eolas-molta%C3%AD/ionaid- teagaisc.260.html). The TEG syllabi for CEFR-GA levels A1, A2, B1, B2, and C1 are all available online[29]. The syllabi prescribe topics, language functions or grammar-based competencies and examples, as well as some topic-specific vocabulary. The following topics are presented in the TEG syllabi.

Table 4: Topics Prescribed in TEG Syllabi

Topics	A1	A2	B1	B2
Meeting People	√			
Background and Place of Residence	√		√	√
Household	√	√	√	√
House and Accommodation	√			
Hobbies	√	√	√	√ (and Free Time)

28 www.teg.ie
29 http://www.teg.ie/leibh%C3%A9il-teg.171.html

Daily Life	√		√	
Abilities and Skills	√			
Work	√	√	√	√ (and Emploment)
Food and Drink	√	√	√	√ (and Health)
Illness and Injury	√	√		
Clothes and Shopping	√	√	√	
Making Arrangements	√	√	√	
Social Occasions		√		
Daily Travel		√	√	
Talking about the Weekend		√		
Describing People		√	√	√
Planning Holidays		√	√	
Directions, Locations and Movement		√		
Favours and Permission		√		
Learning Languages			√	
Talking about the Future			√	
Organising an Irish Course in the Gaeltacht				√
Attending an Irish course in the Gaeltacht				√
Personal Details				√
Education				√
The Media				√
Technology				√
Social and Legal Issues				√
National and International Politics				√
What's in the News?				√

The topics in the TEG syllabus for CEFR-GA C1 are presented in a different way to the previous four levels, as are the examples of language for this level. Appendix 1 details the topics that are prescribed for CEFR-GA level C1, giving a heading for general topics followed by a list of sub-topics. The following headings are given: Personal information and personal relationships, House and Home, the Environment, Travel and Transport, Travel for Holidays, Food and Drink, Shopping, Services, The Body and Health, Work and Career, Education and Studying, Hobbies and Entertainment, Society and Politics. The topics and sub-topics are inclusive of previous levels as well as adding to that which was previously prescribed.

The examples given with communicative functions are presented similarly to those given for post-primary level in Section 2.2, for example. Under Topic 2 in Ionad na dTeangacha (ND-e), communicative functions are prescribed with a handful of examples. The following examples are given with communicative function 2.4, "Describing a place"

(19) Ceantar tuaithe atá ann, tá sé
timpeall fiche míle siar ó chathair
na Gaillimhe.
Community rural is.REL.PRES in.3SG, is he
around twenty mile west from city
the Galway
It is a rural community, it is around twenty miles west from Galway city.

(20) Ceantar tionsclaíochta a bhí ann ach
dúnadh a lán de
na monarchana le blianta beaga anuas.
District industrial [rel part] was in_it but
[auton.]closed [phr. part.] lot of
the_plural factories with years small downwards.
It was an industrial district but a lot of the factories were closed in recent years.

(21) Tá sé iargúlta go maith.
Is he rural [adv part.] good.
It is pretty rural.

(22) Tá an baile is cóngaraí deich míle uainn.
Is the home [part. sup.] near ten mile from.1PL.
The nearest town is ten miles from us.

Gaelchultúr[30] offer a range of language courses and qualifications for adult learners of Irish, among which are their evening classes that are offered at levels analogous to the CEFR. Evening courses are offered at the following levels: Beginner (e.g. A1), Elementary (e.g. A2), Lower-intermediate (e.g. B1), Upper-intermediate (e.g. B2), Advanced 1 (e.g C1), and Advanced 2 (e.g. C2)[31]. É. Ó Dónaill (2013) confirms this, "Language classes for the general public at six levels (A1–C2)".

Gaelchultúr also offer a certificate in applied Irish, a post-graduate diploma in translation, as well as preparatory courses for people entering teacher training courses and other third-level courses. The materials for each of these courses draw on a CEFR-like approach to varying degrees, with the evening courses and preparatory courses most closely aligned with the CEFR. No Gaelchultúr syllabi were available, therefore, the materials used in EduGA were the CEFR-based courses and the accredited third-level courses they offer.
These accredited QQI-NFQ: HETAC courses (Quality and Qualifications Ireland - National Framework of Qualifications: Higher-education and Training Awards Council) fall under the same accreditation standards as third-level courses taught in any university or institution.

Siollabas Nua Ollscoile na Gaeilge [*A New University Syllabus of Irish*] (aka: (an) Siollabas Ollscoile) [Translation: (*the*) *University Syllabus*] is a third-level syllabus for teaching Irish that is hosted by Ulster University and was developed by Ulster University; St. Patrick's College, Drumcondra; NUI, Galway; and Foras na Gaeilge. This project delivered syllabi and sample materials for the language element of each of the three years typically undertaken during an undergraduate degree. These syllabi and materials are made available via http://www.teagascnagaeilge.ie/.

Separate syllabi have been developed for first year, second year, and third year undergraduate Irish courses, all of which are based on bringing all the learners' language abilities (writing, reading, listening, speaking) to CEFR level B2 at least. The introduction to the syllabus for each of these years describes a situation where university students enter first year at very different levels of language ability, the example of a native speaker who speaks at level C1 but writes at level B1 is given as part of the introduction. The syllabi state that the decision to aim for level B2 in all abilities was not taken lightly, but that it was thought that level B2 would best serve the undergraduate group as a whole.
10 topics are provided at the beginning on page 10, most of which can be aligned with topics from the aforementioned reference documents, and each of which is accompanied by a short description (pages 51- 60). A number of examples are given in the chapter "**Na**

Scileanna Teanga sa Seomra Ranga (An Fheasacht Teanga)" [Translation: *Language Skills in the Classroom* (*Language Awareness*)], [ibid. p. 89]. The examples are not laid out as topics, language functions and examples like they are in the other reference documents, but they will inform this research and the examples provided will be marked as CEFR-GA: B2 in the corpus.

CLILstore (Content and Language Integrated Learning Store) is an online repository used by teaching professionals that contains "a store of copyleft content and language integrated teaching materials."[32]

At the time of collection, CLILstore-GA hosted 140 educational materials for teaching and learning Irish, which are presented in audio-visual and written formats. Administrators and language-learning practitioners have independently aligned these materials with levels of the CEFR ranging from A1 to C1. No references are made to any Irish-specific syllabus for the uploaded materials. Which element(s) of the chosen CEFR levels are the focus of the uploaded materials is never documented, furthermore, the materials are rarely accompanied by usage notes or directions.

[30] www.gaelchultur.com

[31] http://www.gaelchultur.com/en/courses.aspx or http://www.gaelchultur.com/ga/cursai.aspx (Last accessed: 13:00, 13/11/2016)

2.5. From Prescribed Language Features to Candidate Complexity Metrics

The language features selected must have been consistently prescribed in all reference documents if they are to guide the direction of the Corpus Analysis section, where the analysis conducted informs the development of complexity metrics.

Reference documents for mainstream education describe the language learning process in broadly similar terms to the CEFR (Council of Europe (2001)), the reference documents also use relatively uniform structures and terminology throughout when prescribing language features for each of their respective levels of ability. Figure 2 was created to show the similarities between the terms and structures used in reference documents, and the way each layer of the taxonomy feeds into the next. These taxonomic structure can be found in reference documents for primary level, post-primary level, CEFR-GA syllabi:

[32] http://multidict.net/clilstore/about.html (last accessed: 14:30, 14/11/2016)

Figure 2: Illustration of the Taxonomic Structure of Language in Reference Documents

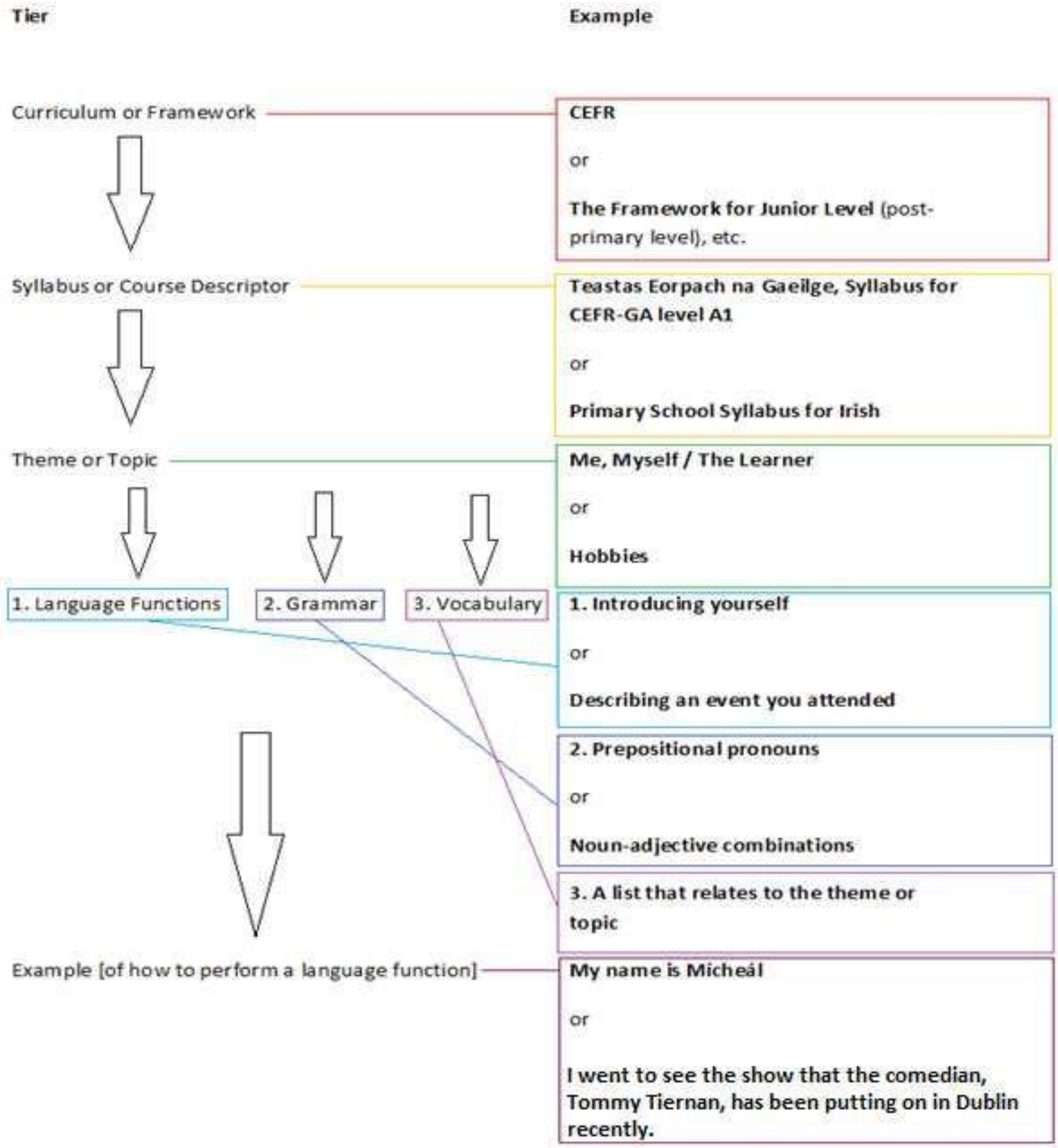


Figure 2 shows each of the elements taken into consideration when selecting the language features that are analysed in this study. The type of course (e.g. CEFR) and level (e.g. CEFR-GA Level A1) dictate the boundaries of each sub-corpora. Topics are not being analysed in this study but each prescribed topic (e.g. Hobbies) dictates the type of language functions and examples associated with it. Therefore, while the analysis focuses on language features the context in which the language features are prescribed remains important.

Each level of ability after absolute beginners is defined as a progression or development from the previous level. Each level is inclusive of the contents of the previous level of ability. Both mainstream-education and CEFR-based reference documents adopt a building-block approach where the first blocks laid are as much a part of the structure as the last blocks laid. Repetition has long been a key part of the learning process, and whether an empirical analysis of this repetition can be used to reflect the complexity of educational materials will be tested in Chapter 5.

When seeking similarities and difference between reference documents the following type of language features were identified.

A number of lexico-grammatical features are prescribed for materials in all EduGA sub-corpora. For example: Greetings and Salutations, Introductions, Noun-adjective collocations, Conditional-mood verbs, Conditional- mood particles, Autonomous verbs, and Relative clauses. These lexico-grammatical features represent different stages in the building-block process described in the reference documents. The analysis aims to identify whether or not they can be used as complexity measures. Further details regarding the lexico- grammatical features that form the basis of this types of analysis are included in Section 5.1. The suitability of these language features and the methodology used when analysing these language features is discussed in the context of complexity research in Chapter 3.

The reference documents discuss degrees of language complexity is in terms of a connection between the length of the construction and the simplicity of the message being communicated. Longer constructions appear to be considered more complex because they require the ability to chain, or connect, constructions both grammatically and conceptually. Longer examples such as these are seldom included in the reference documents, therefore, the ranking of constructions based on length will. The methodology used to rank construction length is discussed in Chapter 3 and the analysis is detailed in Section 5.2.

The reference documents for mainstream education (An Roinn Oideachais agus Scileanna (1999a), An Roinn Oideachais agus Scileanna (ND), An Roinn Oideachais agus Scileanna (2010a)), the CEFR (Council of Europe (2001)), and the TEG syllabus for CEFR-GA level C1 (Ionad na dTeangacha) all explain the importance of domain-specific vocabulary required of learners who wish to improve their language ability, for example, the domain-specific vocabulary needed to discuss politics in CEFR-GA syllabus for level C1 (Ionad na dTeangacha (ND-b), p.7). Terminology and domain-specific vocabulary is also specified in the CEFR as a competency that denotes upper-intermediate or advanced ability (p. 67, 70, 231, 234). This is a competency that is relatively frequently referred to, compared to other high-complexity features, but for which examples are seldom given because of their dependence on topic. The expansion of a learner's topic-specific vocabulary from one level of education to the next is also prescribed across numerous levels of ability at primary and post-primary levels. The CEFR also prescribed terminology. Terminology and specialised language/words are only referred to at higher levels of ability (e.g. primarily at C1 or C2, occasionally at B2 when consulting a dictionary) throughout the CEFR (Council of Europe (2001)); see descriptors and definitions on pages 67, 70, 231, 234 as examples. The following references to technical or domain specific language are made in the CEFR (Council of Europe (2001)):

> "**Reading** [...]
>
> C1: Your test result suggests that you are at level C1 in reading on the Council of Europe scale. At this level people can understand **long and complex factual and literary texts** as well as differences in style. They can understand **"specialised" language in articles and technical instructions**, even if these are not in their field.
>
> C2: Your test result suggests that you are at or above level C2 in reading on the Council of Europe scale. At this level people can read, without any problems, almost all forms of text, including texts which are abstract and contain **difficult words** and grammar. For example: manuals, articles on special subjects, and literary texts" (ibid, p. 235) (**emphasis added**)

DIALANG is a self-assessment system aimed at adult learners who want to get feedback on the strengths and weaknesses of their language abilities, which can be accessed via: https://dialangweb.lancaster.ac.uk/. I make reference to it here because of the CEFR's reliance on it. It is an application for diagnostic purposes of the Common European Framework. In Document C3 (in ibid, p.231-243) we can read the elaborated descriptive scales that are used in the advisory feedback section of DIALANG. We can also find reference to specialised language or words and clear evidence that they are considered complex or difficult in the educational context. For example:

> "Reading [...] B2 [...] Range and types of text only a minor limitation – can read different types of text at different speeds and in different ways according to purpose and type. **Dictionary required for more specialised or unfamiliar texts**. (ibid, p. 239) (**emphasis added**)

> "Writing [...] C2 [...] No need to consult a dictionary, except for occasional **specialist terms in an unfamiliar area**." (ibid, p. 241) (**emphasis added**)

Terminology therefore forms the basis of the analysis conducted in Section 5.3. The methodologies used to extract and analyse terms are introduced in Section 3.3, and discussed in Section 3.4.3.

2.6. Conclusions and Summary

In this chapter reference documents have been used to identify three types of language features that have been prescribed for both mainstream education and CEFR-GA courses. More specifically: seven prescribed lexico-grammatical language features from different levels of education, the length of the construction was also found to be associated with complexity in the reference documents, and finally terminology and domain-specific vocabulary. The contents of reference documents have not been designed for complexity studies, therefore, an appropriate methodology for analysing the use of prescribed language features in educational materials is now sought out and discussed in Chapter 3.

3. Empirical Methodologies used in Language-learning Studies

The focus of this chapter is on identifying methodologies that have been used to analyse the categories of language features found in reference documents in Chapter 2; specifically, prescribed lexico-grammatical features, length of construction, and term-based analyses.

Section 3.1 includes research on lexico-grammatical language features in the domain of education. The research is grouped according to the data type; Section 3.1.1 and Section 3.1.2 present analyses of educational materials, with the first set in English and the second in languages other than English. Section
3.1.3 presents research into Learner Language, in both Irish and English. Section 3.2 introduces research on length of construction, predominantly focussing on length of length of word and length of sentence. The research presented here has its foundations in English, therefore, the transferability of the methodologies requires additional consideration. Finally, Section 3.3 presents research on term identification and extraction. The methodologies adopted in this research, and reasoning for adoption, are included in Section 3.4.

The transfer of methodologies from research for other languages is necessary because of how under- developed this field is with respect to the Irish-language research, especially when compared to English research in the same field. McEnerey and Xiao (2010) introduce an upsurge in corpus-based approaches to linguistic research and language-learning research that has occurred since the mid-80s. McEnerey and Xiao (2010) go on to summarise the depth and breadth of this research, which they categorise into two groups. The first group includes ways in which corpus-linguistic research has indirectly influenced language learning, by informing syllabus and materials design for example. The second category includes ways in which corpus- linguistic research has directly influenced language learning, by creating reference corpora for students of syntax. Learner-generated corpora are also included in the second categorisation because they directly influence language pedagogy. Unfortunately the Irish language is several decades behind in both categories.

3.1. Analyses of Language Features in The Domain of Education

This section presents an overview of empirical analyses conducted on educational materials. No empirical research has been done on Irish-language materials, therefore, the focus is on transferable methodologies from languages other than Irish.

3.1.1. Educational Materials for Learning English

In this section I present research where the contents of educational materials have been analysed. The aim continues to be the identification of methodologies that can be applied to the language features identified in Chapter 2. The focus here is on methodology therefore only the most basic details of the language features and/or linguistic features analysed are provided here. Further reason for this stems from the fact that the majority of the research included in this section has languages other than Irish as its subject matter.

Different approaches to feature selection became clear during the course of this review with some researchers focussing on a single language feature, while others focussed on a selection of language features. The goal(s) of the research usually dictate(s) whether a single feature is, or multiple features are, the focus of the research rather than one approach being "better" than the other. I begin by introducing methodologies used in studies that have focussed on a single language feature, after which I introduce methodologies used by researchers who are analysing a wider variety of language features.

Römer (2005) analyses differences in the use of progressives between English-language educational materials that are used in Germany and a reference corpus of spoken British English. Progressives are defined as the combination of "TO BE" and the present participle -*ing* form of a verb. Römer (2005) begins by stating that progressives are considered "troublemakers" in the foreign language learning context, especially for learners from language backgrounds where the progressive does not exist as a grammatical category (e.g. German, Norwegian, Polish, Swedish, etc.). In addition to this, Römer has found that the difficulty with teaching them may be related to the definitions of progressives being both inconsistent and not empirically based. Römer therefore aims to take a corpus-driven approach to defining progressives empirically, while comparing their use. Similar to the present research, a corpus of educational materials was compiled for Römer (2005). That corpus contains 494 text files in 12 directories, and 108,424 tokens/words collected from six volumes each for two series of educational materials. By adopting this corpus-driven approach to analysing progressives, Römer (2005) observes all examples of the phenomenon rather than filtering the examples observed by applying a pre-existing theory. This approach was believed to be more inclusive than any theory-driven approach that could have been applied. The observed progressives were analysed according to the verbs' form, function, aspect, and frequency. Differences between the language used in materials and that used in the reference corpus typically begins with frequency, be it a count or a percentage, but the discussion of the forms and functions require more detailed discussion. A feature of the results from the two series of educational materials were the differences in the forms used, for example, which may have occurred because they were developed without corpus data.

Gabrielatos (2003) and Gabrielatos (2006) present different perspectives on the same research topic, the use of if-conditionals in educational materials for teaching English. Similar to Römer, Gabrielatos focuses on a single language feature and compares its frequency and function in educational materials with its frequency and function in a sample taken from a corpus of general language. More specifically, the if- conditionals in a collection of ELT coursebooks is compared to a sample of 1,000 if-conditionals from the BNC (British National Corpus[33]). After filtering idiomatic uses of "if" out, he discovered the coursebooks only provided 44% coverage of the range of if-conditionals found in the BNC sample. This finding was attributed to the syntactically straightforward nature of the examples in the course materials. Gabrielatos also discovers examples that are treated as special cases in the course materials despite them having a similar frequency to the if-conditionals that are included in the ELT (English Language Teaching) framework for his course materials. The point being, higher frequency irregulars ought not to be treated the same as lower frequency irregulars. Gabrielatos (2003) goes on to recommend coursebooks be written in the following way: coursebook corpora are created, the contents of which are compared with L1 and learner corpora, and the results are then used to write (what he calls) syllabi coursebooks and associated examinations. This research is published at a similar time to Gabrielatos (2005) which highlighted the usefulness of corpus data and methods in language learning, as well as its limitations. He goes on to described the relationship between corpora and language teaching as "a blind date" because of how little understanding there is in language education regarding the value of corpus data. He notes the difference between native intuitions and actual usage found in a corpus, or a move "from over-generalised and exception-ridden rules towards flexible and context-specific patterns" (ibid, p.20). Burton (2012) echoes this point, suggesting little progress had been made in this area.

A number of other researchers in the domain of language learning have also based their analyses on multi- word structures. Uçar and Yükselir (2015) sought to reveal the impacts of corpus-based activities in the classroom on verb-noun collocation learning in EFL (English as a Foreign Language) classes. Two groups of 15 learners were used for the experiment, one experimental group and one control group. The main finding was that learners who were taught a lesson on noun use using noun collocations that were extracted using corpus-linguistic techniques, rather than traditional lessons, performed better when examined on the subject than the control group who were taught using a traditional lesson. Analysis of multi-word structures, specifically collocation analyses, have also been found to be effective

in corpus-linguistic research for the domain of education. Chan and Liou (2005) and Jafarpour, Hashemian, and Alipour (2013) found that using verb-noun collocations extracted from a reference corpus lead to higher test scores than a control group.
Research conducted by Sun and Wang (2003) focuses on language acquisition, however, they make a strong case for collocation accuracy causing difficulties for learners. They go on to say that correctly collocating words cannot be learned implicitly, and that using concordance software to study collocations can be beneficial to learners.

Similar to multi-word structures of different sorts, vocabulary can also be analysed as a language feature in this domain; particularly if the vocabulary range has been prescribed. A corpus of Taiwanese English- language coursebooks used at university level was created for Hsu (2009) in order to compare the vocabulary coverage in the materials with the British National Corpus. The comparative analyses were done using 1st–14th of the 1,000 high-frequency word families from the BNC, as well as the 570 word families in Coxhead's Academic Word List (AWL)(Coxhead, 2000)[34], and the curriculum-prescribed wordlists for grades 1-9 in Taiwanese high schools. When preparing for the analysis, the corpus was lemmatized and proper nouns were removed from the word list because Hsu found the meanings attributed to proper nouns are normally either well known or can be inferred more easily than common nouns from the contexts in which they are used. Further reason not include them in the study was that no proper nouns featured in the 2000- word list prescribed in the Taiwanese curriculum. Hsu (2009) concluded that for dependable reading comprehension 95-98% of the vocabulary ought to be known. In other words, there can be between 5 and 2 new words per hundred. It was noted that a range of 1.30% to 6.54% of the vocabulary in the Taiwanese coursebooks for learning general English were tokens from Coxhead's Academic Word List (AWL). The distribution of AWL word families in Hsu's corpus was mixed and unpredictable, and an increase in distribution was not found to be indicative of an increase in the level of ability for which the materials were created. It was concluded that a learner ought to know roughly 12,000 words in order to understand 95% of the selected texts. The following pieces of research have sought to achieve their aims by analysing multiple language features, rather than focussing on an individual language feature. The criterial or categorical features analysed need not be prescribed features. Regardless of their source, all are frequently used to distinguish between levels of ability in this domain.

[33] British National Corpus. URL: http://www.natcorp.ox.ac.uk/ (last accessed: 12/01/2019)

Wordlists or vocabulary also play a part in the following studies, structural and grammatical features form the basis the research as well, for example, in Murakami (2009) or Takahashi and Tono (2015). Murakami (2009) aimed to identify “textual characteristics and differences between English textbooks in Japan and other major Asian countries (China, Korea, and Taiwan).” and found that the literature showed gaps in the analysis of “genre/text type” in English textbooks. Murakami further explains the need for research of this sort by declaring that textbooks have a strong influence over what learners acquire in the classroom, as did Burton (2012), and that the language used in classrooms is important for both their personal and professional development. Murakami (2009) also notes that the use of wordlists in ELT in certain East Asian countries is not frequently mirrored in other regions of the world. While Murakami says this bias towards wordlists can be detrimental to the students’ ability to use correct syntax, specific issues or examples of this phenomenon were not given. The textbook corpus used for Murakami (2009) was 1,367,112 words in size. To focus the study, Murakami selected the most widely adopted textbooks rather than including every possible ELT textbook. The 69 textbooks that were included in this corpus were divided into 662 texts, with spoken sections separated from reading sections, and with a mean length of 2,065 words per subsection. This sampling was done to follow the example given in Biber (1988) and (2001). The final preparation done by Murakami before analysing the corpus was as follows: Each text was tagged for parts of speech and lemma using *TreeTagger* (Schmid, 1994). Perl programs were developed and used to count the 56 linguistic features and frequencies were normalized to the instances per 1000 words (Murakami, 2009:5). The position taken by Murakami (2009) was that factors[35] from advanced levels did not need to be searched at lower levels, because it was felt that these searches may interfere with the factor analysis. Murakami goes on to say there’s an absence of “advanced features”, such as the passive voice and relative clauses, at lower levels of education. Resulting from this, Murakami says that this is why they were not searched for at lower levels. The factor values are not shown here because they relate to English, however, the language features analysed are listed below in order to illustrate the variety of analyses conducted:

[34] The Academic Word List was generated using a collection of texts and does not include the 2000 most common words in the English language according to West (1953)’s General Service List.

"**Factorial Structure 1 (beginner level)**: prepositions, mean word length, Guiraud's index of lexical richness, attributive adjectives, *to* clauses, contractions, *wh*-questions, *be* main verb, first person pronouns, second person pronouns, pronoun *it*, proverb *do*, noun (1), noun (2), phrasal conditions.

"**Factorial Structure 2 (intermediate level)**: nominalizations, passives, *that* relatives, *wh*-relatives, split auxiliaries, conjunctions, emphatics, down-toners, demonstrative pronouns, predictive modals, place adverbials. (ibid, p.6)

Murakami (2009) found that Japanese texts for spoken lessons were more likely to contain a wider variety of linguistic features than written texts, and a wider variety than other equivalent texts from the region as well. It was also discovered that certain countries were more similar to others, resulting in a China-Taiwan group and a Japan-Korea group. Dimension 4 (persuasive language in a text) and Dimension 5 (whether a text is abstract or technical) of the Biber (1988) analysis yielded no results as far as Murakami was concerned, leading to the conclusion that English textbooks included in his corpus were non-technical. Murakami concludes that the approach successfully distinguished textbooks from one another in a way that had not been previously possible.

3.1.2. Educational Materials for Learning Other Languages

A similar approach is adopted in Berendes and Vajjala (2018), whereby a collection of language features are analysed. They first define what they call the "systematic complexification assumption" as a process that yields systematic differences in the complexity of the language that is used in textbooks across grade levels and schools tracks (in their context; vocational or academic tracks in the German school system). That is to say, the complexity of the language used at different levels is assumed to be different. Their aim is to systematically assess whether the complexity level of textbooks systematically increases with grade level and the academic orientation of the school track (ibid, p.2). Before beginning their analysis they go on to discuss various approaches to measuring or predicting the complexity of texts, of which they say there are many. It is also noted that a relatively small amount of work has been done on the application of these methods to the contents of textbook.

[35] In this context, a factor is a numerical value that is the result of a statistical analysis – or a factor analysis.

They then go on to introduce the mixed methods approach they take to analysing this complexification in a corpus of 35 Geography textbooks, for grades 5 through 10. They state that their interest was in the linguistic features of the information presented in the body of texts. They therefore excluded the following from their analyses: instructions, summaries, interviews, exercises, primary sources, definitions, picture captions, and miscellaneous material such as the table of contents and publisher information. Each individual reading unit was then defined and considered a text unto itself, of which there were 2,928 of varying sizes in the 35 textbooks. The texts were converted to raw text using optical-character recognition (OCR) software, the accuracy of which was manually checked. In order to complete their analyses, after pre- processing was completed, the following types of annotation were completed: lemma and part of speech, sentence segmentation, phrase-structure trees, morphology and dependency parsing, and compound-word splitting. The grade- and track-level classification provided by the publishers were accepted and used when processing the results.

Their analysis of the texts they collected was based on 165 features that they say represented lexical, syntactic, and morphological characteristics of language and discourse cohesion. The 165 features are detailed in *Appendix B* of Berendes and Vajjala (2018), which are summarized as follows:

- Surface measures: average sentence length, average number of syllables per word.
- Lexical measures: lexical diversity (type-token ratio), variation (verb variation), and lexical density
- Frequency-based word-usage features are analysed, using the dlexDB project[36] as a reference tool
- Semantic relatedness measure: using the GermaNet (Hamp & Feldweg, 1997) lexical semantic network for German.
- Syntactic measures: based on phrase structure and dependency representations of sentences.
- Morphological measures: based on verbal and nominal inflection (e.g., passive participle, genitive nouns, etc.) as well as the use of German suffixes.
- Cohesion measures: In the case of sentences, the overlap of 27 features between sentences were measured.

[36] URL: http://www.dlexdb.de/

The analysis was split into two steps. For the first step, all 165 features were used in a computational process known as supervised machine-learning in order to classify documents. The second step learns from the first, with 10 of the 165 features chosen in order to compute different training models. The results of the second step are analysed in terms of frequency and then more complex statistically-based measurements, such as linear-regression models. The classification accuracy was improved by 0-2% when the two corpora were split according to tracks, i.e. vocational or academic. These improvements were not considered statistically significant. Ultimately the classification was most accurate when distinguishing grades 5-6 from grades 9-10, giving scores of roughly 75% accuracy. They take this to imply that grades 7-8 are indeed situated between grades 5-6 and grades 9-10. Grades were then analysed by publisher, where materials from one publisher were used as training data and the others as test data. The results show that the grade-level classification accuracy across publishers was lower *between* publishers (accuracy rates between 37.8% and 46.1%) than *within* publishers (accuracy rates between 52.3% and 56.7%). Ultimately, they conclude that these results provide only limited support for the systematic complexification hypothesis because of the variety shown by each publisher.

The findings of the regression models that were based on 10 of the 165 features were similar, with regard to the variation exhibited by publishers in particular. Equally, they find that their results imply a certain complexification from one grade to the next but that it did not appear to be systematic. This results applies to both grade-classification measurements and track-classification measurements to a very similar degree. They therefore conclude that their overall results provide only partial support for the systematic complexification hypothesis. They go on to say that additional supports are required for materials creators so that they can decide *when* (i.e., for which age groups or grade levels) to give *what* (i.e., the kind and complexity of linguistic features) to *whom* (i.e., which school track or for good vs. poor readers).

3.1.3. Learner Language

There is an intrinsic link between the language presented to learners in educational materials and the language acquired by learners. This is the primary function of the educational materials. The research included in this section was completed on learner-generated data, rather than data from educational materials. As noted in Section 1.1 Irish is acquired in three different types of school; Gaeltacht schools, *Gaelscoileanna*, and English-medium schools. The research presented here is divided into groupings that reflect the distinction between Gaeltacht schools and Irish-immersion schools given in Section 1.1. This section begins with a study of learner language in the *Gaelscoil* context and learners studying in the L2 context, followed by a study of learners attending a *Gaeltacht* school and a study of language acquisition in the L1 context.

Walsh (2007) collected essays for the subject of Irish (henceforth: Irish essays) and essays for other subjects that were written in Irish (henceforth: Irish-medium essays). Both were collected from groups of senior-cycle pupils in seven Irish-immersion schools not in a Gaeltacht. 60 essays were collected in total, some of which were written by transition-year students (typically 4th year post-primary) and some of which were written by 6th year students (final year post-primary). These essays were collected at a similar time to the reporting of issues in Mac Donnacha et al. (2004), Mac Donnacha (2005), de Brún (2007) and Ó Muircheartaigh (2009) therefore representing examples of errors being made by learners at this time. The strength and usefulness of Walsh (2007) to the present research lies with its presentation of quantitative results and examples of errors made by learners. As well as this, Walsh (2007) is an example of research where a broad spectrum of language features are analysed empirically. These errors represent the language features that learners had not fully acquired, therefore implying the learners attribute a degree of difficulty to these erroneously-used language features.

The distribution of verbal errors and nominal errors was found to be almost even, as was the distribution of errors between Irish essays and Irish-medium essays. 369 errors were identified in Irish essays and 383 errors were identified in essays for subjects other than Irish (ibid, p.34-35). Illustrative examples for each of the categories of errors are provided as well as explanations of the communicative function the learner intended to perform. In some cases the examples included were full sentences, and in others only the clause or individual word is given as an example.

The percentage of each category of errors was recorded. The category "General verbal errors" accounted for 4.25% of all errors, which is considered to be quite a high percentage in Walsh (2007). Walsh described some of the errors in this category as quite basic and goes on to say it's surprising that higher-level Leaving Cert (senior cycle) students would have difficulty with regular verbs in the past and present tenses. Other notable results in relation to the present research were the verbal-noun errors that accounted for 2.8% of errors, which were also considered relatively basic errors, and the relative-clause errors that accounted for 1.6% of errors. While the number of relative-clause errors was considered low by Walsh (2007), she goes on to say that this result may not be representative of the learners' ability because she believes some learners may have avoided using a relative-clause structure because it was too difficult (ibid, p.62). It is considered to be both difficult and complex, and Walsh's belief is confirmed in the grammar textbook, Mac Murchaidh (2002): "*Tá an Clásal Coibhneasta Indíreach níos casta agus tógann sé am ar mhic léinn dul i dtaithí air*", "The Indirect Relative Clause is more complex and it takes time for students to get used to it." (ibid, p.188). The examples included with the verbal errors were typically a sentence or clause. On the other hand, the examples of terminology that feature in Walsh (2007) are presented as individual words. The following list of erroneously used terms are listed in Walsh (2007), these examples were identified in essays of 4th year post- primary level students (ibid, pages 101-102).

Technical Terms

Marxist, Syndicalist, Abdication, Exchequer, Anti-comintern pact, Corruption, Decrees, Direct action, Mafia, Na Allies, Opposition, Pact of steel, Pact, Rule by decree, Syndicat, Treaty, Vatican, Charcoal, Halo, Tonal, Cartoonist, Genre, Texture, Modernism, Racketeering, Suicide bombers, Brush strokes, Impressionism, Lyrical, Post impression, Yugoslavia, Czechoslovakia, Hectare, Pasta, Balfour declaration, Electoral amendment act, Great depression, Popular sovereignty, Statute, Statute of Westminster, Ultimate financial agreement, Public safety Act.

Common Terms

Battle, incident, marches, mockery, now, peacemaker, puppet, river, skiing.

Teenager Speak

Oh nó! (i. Oh no!), Stoned, Adrenalin, Ecstasy, Fix, Hash, Drug, Yob.

Not all terms are recognised as terminology by the term databases Téarma[37] (The National Terminology Database for Irish), IATE[38] or Termium PLUS[39]. For example: *Pact of Steel, Popular sovereignty[40], Ultimate financial agreement* and *Electoral amendment act*. *Pasta* and *Mafia* are international words that are seldom translated or transliterated. Another issue is the definition of term categories; differentiating Technical Terminology and Common Terminology can be problematic where all uses are taken from school essays rather than, for example, technical terminology being found in a technical manual and common terminology being found in newspaper articles. On the other hand Teenager Speak is problematic because of the variety of examples included in the list. The interjection *Oh nó!* typically isn't considered a term, nor are the examples of colloquial slang, *yob* and *fix*. The remaining items on the list could be found in media articles, suggesting they may align with the common terms presented here. Perhaps presenting erroneous uses of terms in full sentences, in the same way verb errors were presented in Walsh (2007), would have clarified these errors. The inclusion of examples of errors in Walsh (2007) clarified the errors being discussed, and the presentation of percentage frequencies for each category contextualises their occurrences. Walsh concludes that the level of ability, even among the better pupils, was inconsistent. She goes on to say that the basic structures[41] of the language had not been learned by some, and that it appeared as though grammar needed to be taught using the communicative approach. However, Walsh does also note that this can only be implemented in Irish classes and that the approach taken in Irish-medium classes presents a challenge to teachers and learners. The majority of pupils had difficulties with terminology, but Walsh found a big leap in ability between fourth and sixth year. The following categories of errors are the most significant, despite not being the highest in frequency in all cases. "[*B]hí líon ró-ard earráidí sna catagóirí seo a leanas: fadhbanna le tuisil agus le comhaireamh, úsáid an bhriathair i gcoitinne, (an t-ainm briathartha ach go háirithe), agus séimhiú agus urú a bhaineann le réamhfhocail*." (ibid, p.72), [There were too many errors in the following categories: problems with casing and with counting, verb usage in general, (the verbal noun in particular), and lenition and eclipsis pertaining to prepositions.]

The research presented thus far in Section 3 have approaches their analyses in a positivistic manner. By this I mean, it was the quality and quantity of the language features that were present that were analysed. Indeed, where fault has been found or where researchers aim to improve educational materials then it may also be wise to search for what is not present.

[37] National Terminology Database for Irish, available from https://www.tearma.ie
[38] Interactive Terminology for Europe, available from https://iate.europa.eu/home
[39] Termium PLUS, available from https://www.btb.termiumplus.gc.ca/tpv2alpha/alpha-eng.html?lang=eng
[40] This term features in IATE with translations in Estonian and Danish, but not in the three procedural langues: English, French, or German. This suggests the term was not yet needed by any of the European Union's or European Parliament's translators.
[41] primarily referring to "syntax"

Hawkins and Buttery (2010) achieve this by setting out to identify criterial features by analysing learner- generated data from the Cambridge Learner Corpus (CLC), which was reported as being 45-million words in size at the time of writing. The learners' abilities are considered in association with features of their L1. For example: if the learner's L1 does not have a definite article, their definite-article errors will be viewed in this context and particular attention will be paid to these errors. It follows that if the learners' L1 is known then their materials could also be tailored to compensate for these L1-L2 differences.

Four categories of criterial features are described in terms that are not language or feature specific:

1. Positive linguistic properties, which are acquired at a CEFR level and their presence denotes a CEFR level of L2 ability
2. Negative linguistic properties, which are errors that denote a CEFR level of L2 ability
3. Positive usage that matches the distribution of the feature in L1 native speech
4. Negative usage that does not match the distribution of the feature in L1 native speech

If these categorisations were to be adapted to the present research, numbers 1 and 3 are the most easily transferable and category 4 may apply where language features have been prescribed but where their distribution was significantly low. The categorisation of criterial features was done as follows, regarding positive-type properties are given for L2 English levels (2010:6-7).

4. **Verb constructions**: For example, new verbal constructions that arise at level B1, such as a ditransitive NP-V-NP-NP (*she asked him his name)* which are criterial at levels [B1, B2, C1, C2]. The object-control structure NP-V-NP-AdjP (*He painted the car red*) which are criterial for levels [B2, C1, C2].
5. **Relative clauses**: A certain distribution of relative clauses formed on indirect object/oblique positions (*e.g. the professor that I gave the book to*) to relativizations on other clausal positions (subjects and direct objects) was found to approximate CEFR level C, or the type of language a native speaker could use and understand.
6. **Lexical skewing**: Certain common English verbs (*know, see, think*) were found to be over-represented at lower levels in the learner corpus when compared with a corpus of native language.

Filipović and Hawkins (2013) also approach positive-type properties as measurable in L1 English (p.8-9). Examples of relative clauses in L1 texts were extracted from a sub-corpus of the BNC which included texts from The Guardian (newspaper), Mills and Boon (novel), New

Scientist (Scientific Journal/Magazine), The Law Report (Legal Journal/Magazine). Each title showed 2 to 4 times more subject relative clauses than object relative clauses, with indirect or oblique relative clauses appearing least frequently of all[42]. I have omitted error-tagged elements of Hawkins and Buttery (2010) because the results are specific to English and also because the present research contains no learner-generated content. It will not be within the scope of this research project to analyse the language features that are not in learners' L1.

This expresses a difficulty that examiners, teachers, and learners can encounter when trying to interpret reference documents in order to examine, teach, or learn a language. Hawkins and Buttery remind us that there can be multiple ways to perform a language function. The approach outlined appears to have been successful, although English Profile Programme has ceased activity. The present research tackles this issue by bridging the gap between reference documents and educational materials, and by showing what is taught at what level. The research presented thus far has focussed on language learning in an L2 setting. The learners in the following research are learning the language in question both in the classroom and in the community or at home.

Péterváry et al. (2014) collected a spoken corpus, that was transcribed, from pupils attending two schools in two different Gaeltacht areas. The spoken corpora were collected in a manner that allowed for spontaneous speech, whereby all pupils were asked to describe the events of one picture in English and a different picture in Irish. Some control stimuli were used in an attempt to balance the semantic content of the corpus, more specifically, once the pupil had completed his/her description additional questions could be asked by the researchers. Péterváry, Ó Cunáin et al. (2014) state their three research questions as follows:

1. Are Irish–English bilinguals balanced or unbalanced in their linguistic ability?
2. Is ability in one language dominant over the ability in the other?
3. Is there evidence of bilingual acquisition on the linguistic achievement in both languages, especially with regard to Irish, and if so, in which grammatical and linguistic domains?[43]

(Péterváry, Ó Curnáin, et al. 2014:21-22)

[42] The three types of relative clause are defined as follows: (i) **Subject Relatives:** The student who/that wrote the paper, The film which/that shocked the world. (ii) **Direct Object Relatives:** The student who(m)/that I taught, The student [oblique] I taught, The film which/that I saw. (iii) **Indirect/Oblique Relatives:** The student to whom I gave the book, The student who/that I gave the book to, The student [oblique] I gave the book to. (Hawkins and Buttery, 2012:13)

In order to answer these questions the researchers analysed a relatively broad spectrum of language features. Firstly vocabulary was analysed according to token and lemma. The following parts of speech were analysed as a group, as well as the subcategories of the part of speech that are specified here between brackets:

- nouns (all numbers and declensions)
- verbs (regular, irregular, impersonal verb)
- adjectives (comparative degrees, plural forms)
- verbal nouns (native verbal nouns, borrowed forms)
- prepositions (individual forms, preposition with the article, prepositional pronouns, and borrowed prepositions)

Elements of the following were also analysed:

- possessive pronouns
- verbal particles (particularly conditionals)
- compound words terminating with *rud*[44]

The final type of analysis that can be found in this report is labelled as syntactic, focussing on particular non- traditional (Irish language) syntax and borrowed discourse markers.

Pétarváry, et al. (2014, p.32) report that 21.5% of the lemmas used in the Irish interviews were English, and they provide sentence-long examples where production was halted due to a missing term (ibid. p.65). It is not known how many of these English words were technical terms, however, the examples given contained words that are included in Téarma. While adults *may* not find it difficult to recall the Irish words for "wing" or "cliff", the subjects of this research are in primary school and may have had limited experience with the Irish for these terms. In the following example the subject could not recall the Irish word for "cliff" and used the Irish word for "hill" in an attempt to explain him/herself.

[43] When searching the entire document, it appears as though the author is referring to linguistic features that belong to a particular domain of knowledge when using the term "linguistic domains".
[44] This word translates as "thing" and was used when pupils could not think of the appropriate word.

> *"Ní cnocáin iad ach cén t-ainm atá orthub ar chor ar bith? Ní cnocáin iad ach tá siad cineál suas díreach [*
>
> *i. 'alltrachaí']"*
>
> (They are not hills but what are they called at all? They are not hills but they are sort of going up straight [i. 'cliffs'])

In order to verify the word in question is a term it was sought in a terminology databases. According to Téarma[45] the term "cliff", in this sense, is from the domain of ***Geography > Topography***. The same concept can be found in the Termium PLUS[46] termbase, and associated with the domains "***Toponymy, Geomorphology and Geomorphogeny***". This shows the concept to be relatively specific.

> "Em... sin, em... síoch-, em... síochán-, em... em... em... suí-, tá a fhios a'm é just ní ... síochán em... suíocháin... sciatháin, sciatháin!"
>
> (**Explanation:** in this example, the participant is searching for the word "sciatháin", which translates as "wings". The participant searches for the word by saying similar-sounding words that begin with S, before remembering the target word.)

In order to verify the word in question is a term it was sought in a terminology databases. According to www.téarma.ie the term "wing", in this sense, is from the domain ***Medicine > Anatomy***. The same concept can be found in the Termium PLUS termbase, and associated with the domains "***Birds, Mammals, Botany, Insects, Centipedes, Spiders, and Scorpions***". This shows the concept to be more general than cliff, while also being relatively specific.

Parts of speech were analysed by their frequency relative to one another, for example: verbs were found to account for 17.2% of all words used. The frequency of mistakes or non-standard constructions used were also analysed on a part-of-speech basis. A considerable number of errors appear to be the result of the over- extension of known rules or patterns, which are typically corrected over time and with continued exposure to the language and lessons. Irish has a number of conjugation patterns, however, there are two primary conjugation patterns. The first is generally used to conjugate verbs whose root form is monosyllabic, and the second is generally used to conjugate verbs whose root form is multisyllabic. 68.4% of the erroneously conjugated verbs simply applied a pattern from the first conjugation to a verb that ought to have been conjugated using another of the standard conjugation patterns. Furthermore, the conflation of dependent and independent verb forms was reported; with particular attention to irregular verbs. 24% of pupils attempted to use the autonomous form of a verb (ibid, p.39), a percentage that was considered to be both low and a sign of this category of verb fading out of use in Péterváry et al. (2014). The other parts of speech listed above are analysed in the same manner; with the number of uses for each type presented alongside the % of errors, and specification of error types.

Péterváry, et al. concluded that pupils who live in Irish-speaking communities all performed better in English than they did in Irish. The English advantage(s), as they define it, in pupils' bilingual ability is described as being: in vocabulary, in functional codeswitching, in grammatical accuracy, in morphology, and in syntax (with the exception of tense in indirect speech), in phonetic accuracy, in semantic and pragmatic usage, and in disfluency pauses (ibid, p. 236-7). Conversely, all of the above are listed as Irish disadvantages to pupils' bilingual ability, with the addition of "number of words and syllables of hesitation in fluency" as a relative weakness identified in the pupils' Irish ability. It is noted in Péterváry et al. (2014) that a distinction is needed between Irish materials for native speakers and Irish materials for learners and they join Kavanagh (2013), Ní Laoire (2013), Ní Shéaghdha (2010), Ó Muircheartaigh (2009), Nic Cionnaith (2008), Mac Donnacha (2005) in calling for additional community supports so that native speakers can function completely through the medium of Irish where it is believed they currently cannot.

Learner-centric features cannot be measured in a corpus of educational materials; i.e. errors, pronunciation, compensation tactics, code-switching, or types of disfluency. The frequencies of parts of speech and terminology can however be measured in the present research, and results can be presented in a manner that facilitates comparisons between educational materials and the emerging language abilities of young learners.

The recorded code-switching results give reason for additional research, while other results require additional specification before they can be extended or re-analysed. For example: the errors that resulted in "inappropriate" or "divergent" categorisations are based on the forms not concurring with "Traditional Irish (e.g. the norms of previous generations of native speakers of Irish) or with a version of Standard Irish"(ibid, p.20). These potentially distinct varieties of Irish are yet to be defined. Furthermore, no evidence or definition is given with statements such as, "Both 'inappropriate' forms and 'divergent' forms are common in the vernacular, especially among those born after approximately 1980" (ibid, p.20). Finally, although a number of the results are based on the frequency of lemmas no lemma tagging technologies are cited.

[45] https://www.tearma.ie/q/cliff/

[46] http://www.btb.termiumplus.gc.ca/tpv2alpha/alpha-eng.html?lang=eng

Similar to the range of language features selected for analysis in the present research, the research introduced in this section has analysed a range of language features. The aim of the research presented in this section was to identify features of the language used by learners, the features were then manually categorised by part of speech or lemma, and the results of this were contextualized according to frequency. Terminology was found to be problematic in both pieces of research, highlighting its topicality in research for this domain, however, no ontologies were used in either piece of research to crosscheck terminology.

In Diessel (2004), *the Acquisition of Complex Sentences*, observational data collected from five English- speaking children aged from 1 year 8 months to 5 years 1 month was analysed. The data were recorded as 357 transcripts of spontaneous parent-child speech. Diessel (2004) sets out to investigate the development of complex sentences in children's speech. Complex sentences are defined as grammatical constructions consisting of multiple clauses, which are most commonly divided into two distinct types: sentences including co-ordinate clauses, and sentences including a matrix clause and a subordinate clause. Diessel (2004) provides a language-independent clause definition, describing them as *situations*. His *situations* have temporal and relational features and contain *things*, where (most) *things* do not carry temporal or relational features[47]. This approach is closely aligned to a number of schools of thought, including scholars from 20th century philosophy of language; Wittgenstein (1922), Austin (1975), and Searle (1969), among others. I will now give a summary of Diessel (2004)'s findings, followed by a closer look at some examples.

Initial clause expansion was found to occur frequently with lexically specific constructions, with semantically restricted uses (such as presentational constructions or descriptions of concrete situations) and, overall, the development is not particularly limited to any one type of multi-clause construction. Diessel (2004) makes two points that clarify this in a crucial way.

- Firstly, that it appears as though there is little grammatical understanding of the constructions that are being used (There is evidence of a dependence on lexically specific constructions, for example). Therefore, the function of the constructions appears to be learned before an understanding of the grammar is developed.
- Secondly, Diessel notes the central roles that both ambient language and language use play in the language acquisition process. In other words; where the majority of the interactions are parent-child interactions, there were biases and use limitations that were conditioned by this ambient language and language use.

The development of complex sentences is described throughout Diessel (2004) as an incremental process. Complex sentences appear to first emerge when a number of linguistic factors have been acquired and begin to overlap, as well as the development of acquired constructions. What follows is a closer look at some of these factors as analysed in Diessel (2004).

Diessel (2004) reports a high frequency of presentational copular constructions in the early multi-clause constructions. While these constructions are bi-clausal by definition they only designate a single state of affairs and frequently contain an intransitive verb. The relative clause and matrix clause merge, in these examples, into a single syntactic unit[48]. This type of matrix clause is grammatical when standing alone, but it can be merged into a relative clause as well. This observation is pertinent to the research at hand, as presentational copular sentences occur at high frequencies in early stages of Irish language learning. Early complex sentences were also found to include adverbial and co-ordinating clauses where two independent sentences were integrated to form a single grammatical unit —a similar or related finding to the previous one.

Other examples of children's early complex sentences in English are lexically specific constructions that have been extended. For example: "I think" or "I know" were found to prefix complement clauses that were either unmarked or introduced by a wh-adverb or a wh-pronoun. While these sentences do contain two finite clauses, they do, according to Diessel, appear to function like simple formulaic sentences that have been affixed to one another. Diessel, at one point, says the early constructions are rarely abstract and frequently relate to concrete situations.

The extent to which this approach is transferable remains to be seen, but I believe shorter chunks and formulaic language also play a part in learning Irish. From my experience, a substantial number of presentational copular constructions are taught at beginner level with the equivalent Irish forms of "I think" or "I know" preceding them at subsequent levels. His findings relate to English, therefore, the specific grammar and syntactic structures do not necessarily relate to a study of Irish. This use of set phrases and constructions may also occur during the acquisition of Irish. It is this analysis of set phrases and constructions that may be transferable from the learner perspective in Diessel (2004) to the materials perspective of the present study.

If this is the case, the acquisition of complex sentences has been presented in Diessel (2004) as a process that would be realised in Irish with a steady increase in sentence length until a certain average length is achieved, as well as phrases and constructions include a broader vocabulary. This average sentence length will need to be identified in a corpus of general Irish as well as being identified in materials from a range of levels.

[47] There are some exceptions, as always. For example: Driving, destruction, learning, explosion, etc.

[48] For example: “John knows”, becoming “John knows what your name is”

3.2. Analyses of Length of Word and Length of Sentence

The second feature selected from the reference documents for analysis in the present research was length of construction. As noted in Chapter 2, the length of a construction a learner can both create and understand correlates with their level of ability. Constructions, in the reference documents, referred to the collection of units of language required to communicate a communicative function. While this most frequently refers to a sentence, it appeared as though the term “constructions” also referred to a single communicative function that spans multiple sentences at more advanced levels (e.g. argumentation, reasoning, technical concepts). Research into length of construction typically takes the form of an average length of sentence analysis. This metric features most frequently in complexity or readability analyses (not only in language-learning research) but in medicine, and in domains where technical writing is the norm; such as Law. Length of sentence was frequently accompanied by length-of-word as a metric in the literature. As a result of both being treated as relatively fundamental metrics in the complexity or readability analyses covered here, the possibility of including a length-of-word analysis is also explored in this chapter. Increases in a learner’s vocabulary frequently feature the reference documents, therefore, should length-of-word provide an opportunity to measure this or add to the analysis of vocabulary then its addition would be worthwhile on a number of fronts.

Length of word has been used as a metric in readability, language difficulty, and language complexity research since the mid-20th century; for example Flesch (1948) and Kincaid et al. (1975) which have been combined to create the Flesch-Kincaid (F-K) Readability Test. In the Flesch-Kincaid Tests length of word refers to the number of syllables in the word, rather than the number of letters in a word, and number of words per sentences represents what you would expect it to represent. The following Steps illustrate which features are used to calculate F-K scores:

> “**Step 1.** Count the words.
>
> Count the words in your piece of writing. Count as single words contractions, hyphenated words, abbreviations, figures, symbols and their combinations, e.g., wouldn't, full-length, TV, 17, &, $15, 7%.
>
> **Step 2.** Count the syllables.
>
> Count the syllables in your piece of writing. Count the syllables in words as they are pronounced. Count abbreviations, figures, symbols and their combinations as one-

syllable words. If a word has two accepted pronunciations, use the one with fewer syllables. If in doubt, check a dictionary.

Step 3. Count the sentences.

Count the sentences in your piece of writing. Count as a sentence each full unit of speech marked off by a period, colon, semicolon, dash, question mark or exclamation point. Disregard paragraph breaks, colons, semicolons, dashes or initial capitals within a sentence. For instance, count the following as a single sentence:

You qualify if-

You are at least 58 years old; and

Your total household income is under $5,000.

Step 4. Figure the average number of syllables per word. Divide the number of syllables by the number of words. **Step 5.** Figure the average number of words per sentence. Divide the number of words by the number of sentences. **Step 6.** Find your readability score.

Flesch (1981)

The lower the score the more difficult the text was to read. The scores could be decoded using the following reference table.

Table 5: Flesch-Kincaid Reference Table

Score	**School Level**
90-100	5th grade
80-90	6th grade
70-80	7th grade
60-70	8th and 9th grade
50-60	10th to 12th grade (high school)
30-50	college
0-30	college graduate

Another tool that uses the same features in order to achieve a similar goal is known as *Coh-metrix*, which was developed in Graesser, McNamara, Louwerse, and Cai (2004) with a view to analysing the readability of a document via the cohesion of text. Cohesion, in this context, refers to relationships between the surface forms of the language used in a text and how they may help a reader mentally connect the concepts expressed. The results this tool generates are based on over 200 features including lexicons, part-of-speech classifiers, syntactic parsers, templates, corpora, latent semantic analysis, among other NLP tools. By including a wider variety of metrics the authors aim to make the tool sensitive to a variety cohesion relations, world knowledge, and discourse characteristics.

This tool was tested on English texts only and can either be used as a web-tool via the Coh-Metrix website (http://www.cohmetrix.com/) or the calculations can be done locally using the subsequently developed Python software. It is noteworthy that the original tool was built with English in mind and that the Python software only supports English, German, and Dutch[49]. A number of other tools were built on relatively similar principles and crucially, to the present research, were built with English in mind or were tested on English texts. For example: Gunning FOG Index[50] ; Fry Readability Formula (Fry (1963) and Fry (1968)); SMOG (Simple Measure of Gobbledygook) formula McLaughlin (1969) all use the number of syllables in words and sentence length in their formulas. A great number of other algorithms or formulas have been created and used, all of which indicate to varying extents how readable or complex a text is, as seen in Klare (1976) which reviews 36 readability formulas. He points out a major flaw with a number of the readability measures he analysed, namely, that they do not consider the order of the words that have been scored by the formula. Klare concludes that the readability formulas he has reviewed can sometimes make a difference and improve the readability of a text, but that more work needs to be done in this area if they are to be considered reliable.

He suggests in Klare (1976) that the number of different formulas invented and in use are indicative of this conclusion and of the lack of certainty surrounding this topic. Another key finding of Klare (1976) in relation to the 36 studies were the different motivations and purposes for research at the micro level. At the macro level some researchers assessed readability with a view to simplifying the texts, while others assessed readability with a view to aligning the text with one level among a number of scales of ability. It appears to me that both approaches continue to be applied.

The results of these studies aim to provide indications of complexity or difficulty rather than absolute results (cf. Mikk et al. (2001) and Lewis and Frank (2016)). One of the major difficulties in complexity studies is the way language is constantly evolving in a number of ways as well as what I call the reader variable being constantly at play. Every reader brings both different amounts and different types of experience with a text type to the table when reading a text. The reader variable either influences the style of writing adopted and/or the reception of the written text. Upon considering these factors it appears to me as though striving for relativistic results is the most sensible approach for the present study, therefore, analysed texts are ordered relative to one another within a domain rather than being given an absolute complexity score of some sort. This approach is similar in many respects to the Flesch-Kincaid Test's reference scores presented in Table 5.

Lexical complexity has been defined as a correlation between word length and word frequency in Saggion, Bott, Rello, and Drndarevic (2012), with reference to the findings of Rayner and Duffy (1986), who found that shorter and more frequently-used words were considered simpler than their longer and more infrequently- used synonyms when being read. Saggion et al. (2012) remind us of how the majority of text simplification research has been done for English texts, motivating them to develop the lexically-based text simplification tool for Spanish that has been called LexSiS. "LexSiS uses (i) a word vector model to find possible substitutes for a target word and (ii) a simplicity computation procedure grounded on a corpus study and implemented as a function of word length and word frequency. LexSiS uses available resources such as the free thesaurus OpenThesaurus and a corpus of Spanish documents from the Web" (ibid, p.2 of 18). Rayner and Duffy (1986) conducted two fixation experiments in order to identify more complex words. Fixation in this context refers to a reader's eyes pausing on a certain word in order to process the meaning of a word or extract information. This fixation methodology cannot be applied in the present study, therefore, it has been introduced here in order to contextualise Saggion et al. (2012) rather than informing the present study. Two points from Rayner and Duffy (1986) are worth noting. Firstly, it appears as though cognitive cost is being related to a form of complexity. Secondly, Rayner and Duffy (1986) found that, when durations were aligned with word frequency and length, their results adhered to a Zipfian curve (Zipf, 1936[5])[1]. The relevance of these points to the present study are clarified in the context of other elements of the literature in Section 3.4, the conclusions to this chapter.

[49] URL: https://pypi.org/project/readability/
[50] URL: http://gunning-fog-index.com/

LexSiS was developed with the Simplext Corpus as its basis. The Simplext Corpus is described as a small corpus that contains 200 news articles, 40 of which have been manually simplified and the corresponding sentences aligned. The analysis is then conducted on word length and on frequency in the analysed corpus. Word length, in their study, considered multi-word expressions to be a single lexical unit and therefore treated them as "words". This is primarily because of their aim, which was to substitute difficult or complex words with simpler synonyms.

They acknowledge that not all words will have a synonym and another approach may be needed in those cases. Saggion, Bott et al. (2012) also find that shorter words are typically more frequent in their reference corpus, as well as being function words. While it was not stated that this had skewed the data, it could be inferred that a stoplist was required in order to flag or remove high-frequency function words that are shorter than most other words. Their results were nevertheless positive with 70% of the replacement synonyms found to be shorter than the words they had replaced.
Finally, Klare (1976) provides a number of tables that summarise the variables, features or methods used in the research he analysed. Tables 9 and 10 are of particular interest to the present research, given the prescription of vocabulary, length of construction, and so on.

[51] The reference made to Zipf (1936) relates to the seminal work's frequency curve (known as a Zipfian curve) and, in relation to language studies, to the distribution of word frequencies.

“*Table 9*

Some Word Difficulty Variables

1. Proportion of content (vs. function) words
2. Content word qualities
    a. Frequency
    b. Familiarity
    c. Length
3. Concreteness vs. abstractness
4. Association-value
5. Active vs. nominalized verb constructions

Table 10

Some Sentence Difficulty Variables

1. Length (esp. clause length)
2. Active vs. passive
3. Affirmative vs. negative
4. Embedded vs. non-embedded
5. Low depth vs. high depth

(Klare (1976), p.20 of 24)

It is also worth noting that readability metrics are not without their critics. For example, Bailin and Grafstein (2001) highlight a number of assumptions that limit the accuracy of readability metrics. Namely, that vocabularies can shift with the invention of technologies, rendering one generation more familiar with a word than another generation. They go on to say the problem extends to the different varieties of language that different socio-cultural groups can use. They also say it is unintuitive to say the word “aardvark” is less complex or easier to understand than the word “unemployment”, and that “curr” would then be the least difficult of the three. A ranking they dispute. This desire to move away from a dependence on surface forms is echoed by others who have tested reader comprehension and have found no correlation between longer words and an increased difficulty; such as Randall (1988), Schriver (2000). Benjamin (2011) does this differently by recommending a methodology for analysing readability that does not rely on the previously used algorithms. Despite the criticisms readability measures that depend on surface forms continue to be used, and Benjamin (2011) does note that readability formulas “are useful for measuring the factors reflecting a text’s readability”. The conclusions drawn from this section’s literature can be found in Section 3.4.2.

3.3. Analyses of Term Usage

Term extraction can be considered a branch of corpus linguistics. As a practice, term extraction typically refers to one of two practices: either known terms are extracted or new terms, essentially neologisms, are sought. The practice of analysing the implementation, or use, of known terms is sometimes called “terminometry”. One of the primary challenges in this field of research is the semantic disambiguation of homonyms. Some tokens, or lexical items, can refer to a general non-technical concept in one context, while its homonym refers to a specific technical concept in domain-specific contexts. In this research the aim is to extract known terminology and analyse its use in educational materials. This is being done with a view to terminology usage becoming a measure of the complexity of educational materials. Term extraction is typically conducted using semi-automatic[52] methodologies that take advantage of a combination of statistical and NLP technologies. This usually includes word-sense disambiguation, part-of-speech tagging, dependency parsing, and other NLP technologies that rely on these tools in order to generate concordances and syntactic clusters. Once again, a number of these tools are available for Irish-language NLP with the exception of word-sense disambiguation tools. This task is typically completed semi-automatically because one of the primary challenges in this field can be the semantic disambiguation of homonyms.

Like most types of corpus-linguistic analysis, best results are achieved when technologies are combined. Vivaldi and Rodríguez (2001) investigate this topic and find combining several technologies leads to an improvement in the accuracy of a term extractor used on Spanish medical articles. They cite the main issues in this field as either being noise, too many candidate terms extracted, or silence, too few candidate terms extracted. Unfortunately for the present research, the majority of their research focussed on the identification of new terms in corpora rather than on analysing term usage. Vivaldi and Rodríguez (2007) evaluate term extraction technologies from a different perspective. With regards to term extraction they cite a general purpose lexico-semantic ontology as a primary resource, in this case it is the EuropeanWordNet (EWN)[53] that began with Vossen (1998).

A number of researchers have compiled specialized corpora and used term lists in combination with other NLP technologies to analyse either term use or, more specifically, term implantation. For example, Vila i Moreno and Nogué Pich (2007) seek to analyse the implantation of sports terminology for Catalan. They compile a specialized corpus of texts from the domain of sports, as well as a list of sports terms. A similar study was conducted for Quirion (2003), the focus of which was on the implantation of terminology for the domain of transport in Quebecois. Saint (2013) uses a list of known terms from the domain of I.T. for Quebecois and synonyms were added where available. Other than these similarities the studies introduced adopt slightly different methodologies, each of which is dictated by research aims that are different to those of the present research. For example, each of these pieces of research analyses subsets of term databases in specialized corpora from a particular domain. On the other hand, while the corpus compiled for the present research is a specialized corpus it does contain a wider range of topics than the sports, I.T., and transport corpora.

In Vivaldi, Cabrera-Diego, Sierra, and Pozzi (2012) a list of scientific term candidates was extracted using YATE (Yet Another Term Extractor[54]), the term candidates in its output are then manually verified. The aim of this research was to first identify if Wikipedia could be used to attach the semantic knowledge needed for this list of manually verified terms. Semantic knowledge, in this sense, refers to the statistically significant colligations in which each term is used. Once the semantic-knowledge was added the terms could be sought in a corpus of educational materials for teaching physics, chemistry, biology, mathematics, health education, and ecology that are used in Mexico. It was hoped that the addition of semantic knowledge from Wikipedia would improve the accuracy of future term extraction projects by adding accurate term-hood values to extracted terms, leaving fewer non-terms in the data presented to expert terminologists for verification.

[52] "Semi-automatic" refers to the generation of a list of term candidates, which must then be manually checked.

Termhood refers to a calculated value that determines how likely it is that a lexical unit is a term. It was discovered during the course of the research that some crucial elements of Wikipedia's metadata frequently contained errors, particularly among the hierarchy of categories upon which this research depended. During the course of their research they address certain issues that are required for the present research, namely, lemmatization of all tokens so that items from a term list can be matched in texts without the morphological changes arising from gender agreement interfering with this process. e.g. *grian* (sun), and *an ghrian* (the sun). Other issues that arose were related terms being incorrectly tagged as domain-specific terms. For example, the term "pocket calculator" was tagged with the "Mathematics" domain because of the context in which it was used despite the concept not being a mathematical concept. Vivaldi et al. (2012) considered the outcome of the project a success in some ways, despite finding that Wikipedia was not yet fit for this purpose. Results were positive, therefore they conclude that they will re-run their project with a new list of terms once the Wikipedia metadata has been improved.

SketchEngine's (SketchEngine is a corpus building and text analysis tool) term extractor is known as *OneClickTerm*[55] [56] and is a tool that appears to combine an analysis of part-of-speech clustering and lexical matching from ontologies. To the best of my knowledge, the exact methodology employed has not been published. The range of part-of-speech taggers and lemmatization technologies available to SketchEngine users is substantial, and has been posted on their blog[57]. These are invaluable to the term extraction process. The term candidates that are extracted by *OneClickTerm* are selected using a number of criteria.

One of the primary criteria is the keyness score that is based on the *Simple Maths* keyness algorithm:

$$\frac{fpm_{rmfocus} + N}{fpm_{rmref} + N}$$

Kilgarriff (2009) [58]

[53] URL: http://projects.illc.uva.nl/EuroWordNet/
[54] URL: http://eines.iula.upf.edu/cgi-bin/Yate-on-the-Web/yotwMain.pl?lang=EN (Last accessed: 01/03/2019)

where N, in this case, is 1 and the frequency found in the focus corpus is normalized per million words; as are the number of hits per million words in the reference corpus.

$$Score = \frac{fpm_{rmfocus} + N}{fpm_{rmref} + N} = \frac{0.3117 + 1}{0.1686 + 1} = 1.1224$$

(Kilgarriff, 2009)

A high score is therefore given to common words, and a lower score to lower-frequency or rarer words. This score is coupled with other criteria, such as a set of syntactic rules; "For example, a term in English will most likely take the form of (noun+)noun+noun or adjective+noun while in Spanish, most likely, noun+adjective(+adjective) or noun+det+noun"[59] . These rules can then be extended in language-specific ways, where appropriate. SketchEngine go on to say that this approach does not require blacklists or stoplists. This is because the keyness score essentially removes function words and, for example, high- frequency non-terminological words from the list of extracted terms by giving them such a high keyness score. The conclusions that were drawn from this are detailed in Section 3.4.3.

TF-IDF is a similar, and frequently-used, statistical method that facilitates analysis and/or categorization of text documents containing language data. It is a language-independent statistical method that applies a relativistic score to each of the unique words, or types, in each of the documents of a corpus. TF-IDF as a scoring algorithm is used for both information retrieval and term extraction. Its application in term- extraction research leverages the way terms are among the most informative or meaning-laden words in a text.

The TF-IDF scores are frequently used in conjunction with additional frequency analyses or comparative analyses[60]. For example, TF-IDF scoring has been used for weighting (or scoring) in Spark and Yahoo search results and reportedly by Google searches as well[61]. It is often used because raw frequency can yield very misleading results for ranking or weighting analyses. As pointed out in Jurafsky and Martin (2018), at times we want the frequency of the most important words rather than the frequency of every word. Simply put, the TF (Term Frequency) refers to the frequency of a word in an individual document.

[55] URL: https://www.sketchengine.eu/user-guide/user-manual/term-extraction/ . (Last accessed: 24/05/2018)
[56] URL: https://www.sketchengine.eu/the-best-term-extraction/ . (Last accessed: 24/05/2018)
[57] URL: https://www.sketchengine.eu/user-guide/user-manual/corpora/by-language/ (Last accessed: 13/05/2019)
[58] URL: https://www.sketchengine.eu/documentation/simple-maths/ (Last accessed: 16/05/2019)
[59] URL: https://terms.sketchengine.eu/ (Last accessed: 16/05/2019)

The second factor, IDF (Inverse Document Frequency), is used to add weight to words that only occur in a small number of documents because these words are typically important, or topical, in the documents they've been used in. For example, the word "photosynthesis" compared to the word "light". Both may have the same frequency in a post-primary junior-cycle textbook, but by achieving a higher IDF score we know that "photosynthesis" is a more important word in the documents it has been used in. IDF is calculated by dividing the total number of documents in the corpus by the number of documents in which the word occurs. Each unique word, or type, in each document receives a **TF** (term frequency) score according to its frequency in a single document relative to the number of words in that document. Therefore, if the word "sun" occurs 5 times in a 100-word document, its TF score is 0.05. The **IDF** (inverse-document-frequency) part of the score refers to measurement of how significant that term is in the overall corpus. This is done by dampening the scores of the types that appear very frequently in **all** documents in the corpus. For example: the word "the" appears very frequently across all documents in the corpus, therefore it receives an extremely low IDF score. The word "sun" appears occasionally in a handful of documents, therefore it receives a moderate IDF score. The word "Photoluminescence", on the other hand, appears just three times in the entire corpus, therefore it receives a relatively high IDF score. The IDF is calculated as follows:

> if "sun" appears in 300,000 documents in this 10,000,000-million document corpus, the IDF of "sun" in this case is the natural log of 10,000,000 / 300,000; which is 1.52.
>
> TF-IDF is calculated as TD * IDF, which 0.05 * 1.52 = 0.076[62]. Written as: TF-IDF(*sun*) = 0.076.

The algorithm is relatively similar to the Simple Maths algorithm created in Kilgarriff (2009) in its use of word frequency relative to corpus size; the main difference being the absence of a second corpus for reference

[60] Kaggle a forum-type website where a community of data scientists and programmers compete in friendly programming competitions. As stated, tf-idf is frequently used as part of the processing. For example:
(a) https://www.kaggle.com/hsrobo/tf-idf-svm-baseline or
(b) https://www.kaggle.com/reiinakano/basic-nlp-bag-of-words-tf-idf-word2vec-lstm

[61] Reference for Spark: https://events.static.linuxfound.org/sites/events/files/slides/2016%20-%20Seville%20-%20ranking%20the%20web%20with%20Spark.pdf , and Yahoo: https://www.oath.com/press/open-sourcing-vespa- yahoo-s-big-data-processing-and-serving-eng/, report regarding Google: https://www.searchenginejournal.com/understanding-tfidf-one-googles-earliest-ranking-factors/166856/

[62] See an example at: http://www.tfidf.com/

purposes in the calculation of the TF-IDF algorithm. The TF-IDF algorithm also avoids the addition of numeric value, usually 1, in order to avoid a zero-score in the results (cf. Kilgarriff (2009), p. 2-3). Zero-scores are frequently attributed to words in the results of the TF-IDF process, however, a zero-score merely indicates the lowest possible level of topicality rather than a null value. In order to make the scores more accessible as a set of scores I have converted them to the *natural log* of each score using the pre-made scientific programming library *NumPy* (aka: Numeric Python) which has a built in natural log function[63]. This is one of a default type of log for TF-IDF. The TF-IDF score given will still always be greater than or equal to 0, however, scores for function words like “the” or “and” are effectively 0 because they can be smaller than the 10th decimal place while important words are greater than 1. The higher the score, the more important the word is deemed to be to the document within the context of the associated corpus.

One noteworthy issue arose when analysing the results of the TF-IDF scoring, and was confirmed in the literature. There is no agreed cut-off point for TF-IDF scores, other than the 0 TF-IDF scores being discounted in some cases. Function words and words with extremely high frequencies usually receive this tf-idf score of 0.

Other considerations regarding this cut-off and the accuracy of TF-IDF have been discussed by Kilgarrif (2001), who expresses concerns with the 0 score that is received by extremely common words because of the way IDF is calculated. Kilgarrif (2001) suggests normalizing TFs as well as IDFs may improve performance by avoiding 0 scores. He concludes that the approach has been found to be successful overall, but that it can also be tuned in order to improve performance citing Robertson and Spack Jones (1994) as an example. Robertson and Spack Jones (1994) recommend using stemmed words or roots when analysing texts, so that matches are not missed through trivial word variations. The use of lexicons and/or stoplists is also included as a step that improves the output. They also suggest tuning the TF value in order to modify the extent of the influence TF has on the weighting. They also suggest that limiting the length of included documents or the number of documents included could help avoid skewed results arising from the document side of the algorithm. They add that the extent or type of tuning would be arrived at after systematic testing. Liu and Kit (2008) set out to test the accuracy of process when seeking to retrieve terms. This accuracy was assessed by manually appraising term candidates with the top 10% scores in each sub-corpus. The TF-IDF score is considered a proxy for the termhood of extracted term candidate, this refers to how likely it is that the term candidate is in fact a term. They categorise their research as corpus comparison, an aim that is similar in some respects to aspects of the present research, via their focus on the different statistically-based outputs from a corpus of legal texts. They find the accuracy of TF-IDF approach when attributing termhood, or weighting, scores (a synonym in Liu and Kit (2008) for the most topical words in domain-specific corpus) to term candidates does not correlate with corpus size. Furthermore, they find that accuracy is improved by using stoplists and reference corpora. Other processing steps have been found in the literature that have also improved the accuracy of the scoring, such as stemming, lemmatization, suffix removal, and, as noted here, the use of additional lexicons – such as stoplists, or data from a reference corpus (cf. Salton (1986)). The majority of the literature has focused on using TF-IDF to extract terms or topical words from domain-specific material. Manning, Raghavan, and Schütze (2008) recommend lemmatization over stemming, referring to stemming as a crude method and to lemmatization as "doing it properly" (ibid, p.32). They go on to explore a number of applications of weighting schemes and the outcomes arising from them, one of which is TF-IDF. A useful example is given from a corpus of automotive materials, from which the word "car" occurs so frequently in so many documents that it was not found to be topical.

[63] "This mathematical function helps users to calculate the natural logarithm of x where x belongs to all the input array elements. Natural logarithm log is the inverse of the exp(), so that log(exp(x)) = x. The natural logarithm is log in base e." (Source: https://docs.scipy.org/doc/numpy/reference/generated/numpy.log.html, Last accessed: 06/06/2018)

3.4. Conclusions and Application of the Methodologies to the Present Research

In this section I first present general conclusions regarding methodologies and their applications to the present research. This is followed by three sub-sections that relate directly to the three distinct types of analysis being conducted in this research. The research presented has shown that comparisons with reference corpora, or with referential datasets, are frequently central to the methodologies. In the context of the present research this will be done between sub-corpora for different levels of education or courses in EduGA as well as comparisons with NCI (The New Corpus for Ireland, Kilgarrif, Rundell, and Uí Dhonnchadha (2006)).

It is clear that each of the distinct methodologies is far more developed in languages other than Irish, particularly English. A relatively small amount of Irish-language corpus analyses have been published when compared to English, with particular focus on the domain of education. This point extends to the technologies or tools used to perform the research; specific examples of the technologies or tools that are or are not available for Irish-language research are provided in sub-sections 3.4.1, 3.4.2, and 3.4.3. Naturally, where a tool has not yet been developed for Irish-language research the most transferable methodologies have been prioritised.

The difference between surface forms for Irish and English resulting from morphological differences are substantial, and this will impact almost all of the methodologies discussed here. Irish morphology includes lenition and eclipsis, a wider variety of ways to pluralise nouns than English, among other differences arising from synthetic verb forms. As a result, studies of Irish may be more reliant on lemmatization than studies of English.

3.4.1. Language Features

Some researchers focus on an individual language feature where a “troublemaker”, or a problematic language feature, has been identified (Römer (2005) and Gabrielatos (2006)). In other cases a range of features are chosen when there is a broader issue or research aim (Berendes and Vajjala (2018), Pétervàry et al. (2014), Murakami (2009), and Hawkins and Buttery (2010)). Corpus linguistic studies *of* and *for* education materials frequently include analyses of a wide variety of language features. In the present research, the issues identified are more aligned with the research that focusses on multiple language features. Where possible researchers have typically analysed these language features from both a statistical point of view (frequency, average frequency, linear regression, etc) and a linguistic point of view (form, aspect, declension, vocabulary range, function, etc.).

In terms of methodologies, ontologies, and technologies' more of the English-language research in Section

3.1 has benefitted from previous research than the Irish-language research. Researchers for languages other than Irish have based their studies on a number of statistical models' including mean, average, and regression models that are language independent as well as language-specific models like the Biber (1988) factor analyses. Great strides have however been made in Irish NLP research in the last 10 years with some powerful new technologies available for Irish, for example: part-of-speech tagging in Uí Dhonnchadha (2008) and Scannell, Lynn, and Maguire (2015), lemmatization (Uí Dhonnchadha, 2008), dependency treebank parsing (Lynn, 2016), a spellchecker (Scannell, 2000b) and the technology used to collect the monitor corpus, *Corpas Gaeilge* (Scannell, 2000a), as well as an Irish-language Semantic Network (Uí Mhuirneáin, 2010) that was created with a view to facilitating the creation of an Irish-language mono-lingual dictionary and thesaurus. The Irish-language research included in Sections 1.2 and 3.1 have not used these technologies, therefore I describe them as under-used in this domain if not in general corpus analyses of Irish. Using these technologies with the methodologies, when compared to similar research of English, will therefore be somewhat exploratory. The analysis conducted in Section 5.1 follows these steps:

- Part-of-speech tagging is utilised in order to accurately identify the prescribed language features selected in Chapter 3. This reflects the approach taken in almost all the studies presented in Section 3.1.
- The frequency of these features is then collected in materials for all levels of ability in both mainstream education (e.g. primary and post-primary) and CEFR-GA courses. Using frequency to contextualize a large number of examples from a dataset again reflects the majority of the studies introduced in Section 3.1.
- The final step is to conduct a comparative frequency analysis of the selected prescribed features, across levels of ability and between the two different types of Irish-language teaching. This chapter has thus far presented corpus-linguistic research from the domain of education where structures based around a part-of-speech or lexical unit have been the focus of the analysis.

3.4.2. *Length of Word and Sentence*

I now discuss how the methodologies in Section 3.2 have been used in the present research. Before discussing any specifics, it is important to note that the foundations of these measurements have been built on studies of English texts. Despite this, and the criticisms noted in Section 3.2, these metrics are now applied in research on a number of other languages.

In appraising the transferability of these metrics to an analysis of Irish texts I have considered potential issues arising from Irish-language syntax, grammar, spelling, and so on. The average lengths of sentences is typically either analysed comparatively as part of a ranking scheme between sub-corpora, or the value is used in an algorithm in order to generate a score that aligns the analysed document with a ranking scheme. The methodologies explored in section 3.2 found that either all sentences, or a representative sample, are selected and the calculations are based on data that has been annotated in a way that segments the text.

When selecting sentences for analysis it is imperative they are annotated accurately in the corpus so that they can be segmented. This is not a simple task and the sentence tokenizer used here needs to cater for a number of structural considerations, for example:

- Full stops that do not mark the end of a sentence (e.g. abbreviated titles)
- Sentences that do not begin with a capital letter (e.g. fragment sentences)
- Capital letters that occur in the middle of a sentence (e.g. named entities)
- Brackets
- Multi-line sentences

When the annotation has been completed correctly, the counting of words in each sentence is a relatively trivial task that can be completely quickly and accurately with a small computer program. The counts are then either compared to a reference corpus or used in an algorithm in order to calculate a score. These algorithms' calculations are reliant on the accurate annotation of a number of elements including word segmentation which is known as tokenization, syllable segmentation which is known as syllabification, and the aforementioned sentence segmentation which is also known as sentence tokenization.

While the literature frequently refers to length of word, the majority of the research actually measured the average number of syllables in each word rather than the average number of letters. Generally speaking, the number of letters in a word will correlate with the number of syllables in a word. Length-of-word analyses and number-of-syllable analyses may be more problematic for an analysis of Irish texts than they are for an analysis of an English text, because they have all been developed on English texts and Irish morphology is significantly different from English morphology. If this metric is unreliable in the Irish context, this undermines the transferability of the majority of tests that use this metric. While it remains to be seen whether or not word length is a reliable marker of complexity in the Irish-language context, this will be tested in Section 5.2.1. No issues with length of sentence have been foreseen because grammatical features that would add or subtract words to Irish sentences do so in a manner that is similar to the other languages these technologies have been used on (Phrasal verbs, negative verb forms, relative clauses, etc.). A sentence tokenizer has been identified to facilitate this analysis, and the sentence-length analysis is conducted in Section 5.2.

3.4.3. Analysis of Term Usage

This section presents how the methodologies discussed in Section 3.3 have been used in the present research, while introducing considerations relating to Irish that would not have been included in the literature. Lexical matching and the importance of lemmatization to the current research are introduced first, this is followed by the source and specification of the terminology list used in the matching process. The final element of the methodology adopted here is a statistically-based scoring tool that indicates how important each lemma-type is to each text. Other exploratory analyses were conducted during the development of this methodology, the results of which are presented in Section 5.3.

Lexical matching refers to a process of matching the form (sometimes including casing[64], depending on the matching system) of one word with another word. Therefore, this is typically an exact 1:1 match. As a result, inflected forms in a text will not be matched with headwords or base-forms in ontologies. In this context the first word could refer to a word found in a school book and the second to a term from Téarma. Lexical matching is not without its issues in any language; homonyms frequently cause false-positives in matching. This is referred to here as over-matching. Lexical matching is particularly problematic for Irish because of Irish-language morphology, which means lemmatization is an essential step.

For example, the word *grian*, meaning "sun", appears in science textbooks in the EduGA corpus:

Table 6: Inflected noun forms in Irish and English

Irish wordforms	English wordforms	Grammatical Categories
grian	**sun**	nominative singular form
(an) **ghrian**	(the) **sun**	definite nominative singular form
(solas) **gréine**	(light of a) **sun**	indefinite genitive singular form
(lár na) **gréine**	(middle of the) **sun**	definite genitive singular form
grianta	suns	indefinite nominative plural form
na grianta	the suns	definite nominative plural form
lár grianta	middle of suns	indefinite genitive plural form
lár na ngrianta	middle of the suns	definite genitive plural form

Here we see 5 different inflected forms for 8 grammatical categories, as opposed to the 2 different inflected forms in English for the equivalent 8 grammatical categories. Adding all lexical forms to the matching schema is not a problem as such, however, when we likely want to group all forms of a headword again for our analysis then we are adding unnecessary processing steps that could be avoided if our school book is lemmatized. The educational materials have been lemmatized using technology developed in Uí Dhonnchadha (2008). The output can be manipulated to take a number of forms, Table 2 contains an example of the output for sentence (1) from a science textbook:

1. Sraontar gathanna na gréine

disperse-auto rays the sun

The sun's rays are dispersed

[64] Some approaches to lexical matching are programmed to match "page" with "Page", this can be beneficial because sentence initial words do not need to be changed to lowercase. However, this can also lead to conflation of the nationalaity "Polish" with the cleaning agent "polish". Knowing which approach your system has been programmed to use is recommeded.

Table 7: Sample output from the Irish part-of-speech tagger and lemmatizer

1 Token	2 Lemma	3 Grammar and POS
<sraontar>	"sraon"	[Verb VTI PresInd Auto]
<gathanna>	"gath"	[Noun Masc Com Pl]
<na>	"na"	[Art Gen Sg Def Fem]
<gréine>	"grian"	[Noun Fem Gen Sg DefArt]
[**Translation:** *"the sun rays are dispersed"*]		

Once the textbook is lemmatized, one can effectively access a **L**emma-**O**nly **V**ersion of the **E**ducational materials. This version of the materials will be referred to as LOVEly educational materials. By matching terms against the LOVEly educational materials, the issue of morphology has been overcome. Once the corpus has been prepared, then the terms that will be sought in the corpus need to be identified.

Ratified terminology is provided by Téarma (URL: www.tearma.ie), and this resource was used as a source of terminology for the present research. In order to collect the necessary terminology for the lexical matching phase, data from Téarma was downloaded from http://www.tearma.ie/Liostai.aspx in .TBX format. A Python program was written to parse the entire file as an .XML and extract information. The output can be formatted in any number of ways; Table 8 shows one way of visualising the extracted information.

Table 8: A single example of the data extracted from Téarma

Irish Term	Domain(s) of Knowledge	POS	English Translation
Grian	Tíreolaíocht / Geography	Bain2[65] / Noun	Sun

180,693 Irish-language terms were extracted from the aforementioned TBX file and terms were matched against the subset of educational materials. Once the matching has been completed, the number of homonyms with non-terminological matches can be identified. It is at this stage that a separate process is required, one that is aligned with the goals of the present research and also facilitates the handling of non- terminological matches in the results.

The additional NLP step that selected for this task was **"Term frequency-inverse document frequency**[66] (henceforth: TF-IDF). As noted in Section 3.3 stoplists and checklists are frequently used with TF-IDF processing in order to improve the accuracy of the output. The term database presented in this section is used to filter results, and the TF-IDF score of each lemma-type is contextualized according to its frequency in NCI, a corpus of general Irish. This point of comparison has been chosen in order to relate the contents of EduGA to general Irish. This point of comparison has also been chosen because the frequency of a lemma in a corpus of general Irish will correlate with the likelihood a learner has encountered the lemma, or that the learner has encountered the lemma a sufficient number of times in order to acquire the ability to use the lemma appropriately.

The full workings of six examples of TF-IDF scoring are included in Section 5.3.4. Before proceeding, I note that no numeric tuning values that would benefit the general purposes of the present research were found in the literature, however, the lemmatization (cf. Manning et al. (2008)) and term matching (cf. Liu and Kit (2008)) that are being done in order to achieve the present research aim will have the added benefit of improving the accuracy of the results. The values that form the results are calculated as a log of the values. As a result a relatively standard version of the TF-IDF package is used. The tailoring that is done for the present research simply relates to the use of a lexicon (similar to a stoplist), which is almost always recommended in the literature as a means of improving the accuracy of the process. The use of TF-IDF in the present study takes advantage of its language-independent reliability and its continued use in term- extraction studies and corpus-comparison studies. The order in which these steps are applied is described in detail in before the results of the process are presented in Section 5.3.

[65] “Bain” is a frequently used abbreviation of “baininscneach”, meaning Feminine. The number 2 denotes a noun of the second declension.

[66] A point of clarification before I proceed, this acronym contains a hyphen and not a minus.

4. The EduGA Corpus: Design, Collection and Annotation

This chapter introduces the Corpus of Irish Educational Materials that is known as EduGA. The corpus contains almost three thousand documents, which collectively contain approximately 7.5 million words. EduGA includes materials that were published between the years of 1995 and 2015. The name EduGA describes the domain and the primary language of the data. The corpus data is overwhelmingly Irish, however, a very small number of Irish-language educational materials can include complete English sentences as examples or as translation tasks, for example. The corpus contains documents from the domain of education; both Irish-language educational materials and Irish-medium educational materials. In this research the term "Irish-language materials" refers to materials for teaching the Irish language. EduGA includes Irish-language materials for mainstream Irish courses (primary, post-primary, and university level) and for CEFR-GA (CEFR-based courses in Irish) courses, both of which are similar in many respects but different in application and pedagogy. In this research the term "Irish-medium materials" denotes materials for teaching subjects other than Irish through the medium of Irish (e.g. Science, History, Business Studies, or Mathematics). The details of both types of materials are included in Table 9.

Table 9: Subjects included in EduGA

Subject	Short Name prefix in EduGA	Number of Individual Documents	Total Number of Tokens per Subject
Irish	EduGA-Irish	1,112	3,721,312
Science	EduGA-Sci	28	337,728
History	EduGA-Hist	18	184,724
Business Studies	EduGA-BusS	5	191,062
Mathematics	EduGA-Math	6	544,646
Physical Education	EduGA-Corp	2	66,082
Art	EduGA-Art	8	32,936
Social, Personal, and Health Education	EduGA-SPHE	11	119,705
CEFR-GA	EduGA-CEFR	1,818	2,235,294
TOTAL	**EduGA**	**2,840**	**7,433,489**

The structure of this chapter is as follows. The steps undertaken when collecting this corpus data are included in Section 4.1. This is followed by a description in Section 4.2 of all pre-processing steps taken in order to prepare the data for the annotation phases of this research that are detailed in Section 4.3. Finally, Section 4.4 includes concluding comments about the corpus compilation process as a whole.

4.1. Corpus Design and Collection

When compiling a corpus the aim is to represent the domain in as representative a manner as possible. Corpus compilation projects are essentially sampling projects (Biber, 1993), and the contents of the sample depend on the aims of the research: "The appropriateness of the sample depends on factors including the purpose of the research and the domain within which the data is being collected" (ibid, p.243). The aim of the present research is to develop a set of metrics that can be used to identify the way(s) educational materials become more complex from one level of ability to the next. Therefore, EduGA includes as much data as possible from each of the respective levels of ability for each subject. The aim was for each of the sub-corpora (representing different levels of ability) of each subject to contain a comparable number of tokens. Far more materials for teaching Irish are available when compared to materials for teaching Business Studies, Maths, and Physical Education. Collecting the same number of tokens or the same number of documents for both subjects is unachievable; therefore, when analysing sub-corpora the results are normalized in a manner that is appropriate to the size of the sub-corpora in question. Collecting the same number of tokens for each level of Irish-language education required a larger number of documents be collected for lower levels of ability because they were found to be consistently short. It was found that more materials had been published for Irish-language classes at primary-level.

The collection began opportunistically. By this I mean, as much data as possible was collected from as many sources as possible. This was followed by targeted collection, with a view to balancing each of the sub- corpora in the corpus. This resulted in a large number of materials for Irish-language lessons and relatively few for other subjects taught through the medium of Irish. In order for materials to be suitable for the corpus they had to satisfy certain criteria that were related to the aims of this research:

- The materials had to be published and available to all teachers and learners, rather than only being available in limited numbers or only being available to a restricted group.
- The materials had to be aligned with a single level of ability, such as Senior-Cycle Irish or CEFR-GA B1, either by the publishers or by teacher consensus.
- Finally, It was a requirement that the materials had undergone some form of editorial process, thus disqualifying materials that are uploaded teaching blogs for example. This condition was included in an attempt to lessen the impact an individual's biases might have on results, especially where so many of the other

materials had undergone an editorial process.

A considerable amount of materials were collected directly from publishers, authors, and materials creators. The vast majority of the materials that were collected in person were either DOC(X) format or PDF format.

Both file types had to be converted to TXT format for pre-processing and annotation. It was clear that people had always put a lot of thought, time, and effort into developing their materials and were understandably protective of them. The majority of materials creators therefore required that I agree not to redistribute their materials. Some were satisfied with either a verbal or unsigned emailed agreement that their materials would not be redistributed by me to any third party. Others requested a signed written agreement in order to ensure the copyrighted of their materials would be protected. The standard agreement used during this research has been included in Appendix 1

The question of how much of the materials ought to be included in the corpus was also considered while collecting data. Sampling has been used in this domain by Murakami (2009) on educational materials, and by Gabrielatos (2006) when extracting data from BNC for an analysis for this domain. Upon inspecting the contents of educational materials it was discovered that some textbooks divide grammatical language features into separate chapters. The result of this is, for example, a focus on the present tense in chapter 3 and on the future tense in chapter 4, a focus on modifier nouns in chapter 19 and a focus on the conditional mood in chapter 21. In other Irish-language materials the focus of each chapter is based on topics and communicative functions rather than grammatical language features. The view was therefore taken that different materials would need to be sampled in different ways, and that sampling all Irish-language educational materials in the same way whole would result in non-representative samples. This stance also applies to the inclusion of all lessons and content as long as it was was designed for learners, with some exceptions. Where a glossary was included at the end of a chapter or at the end of a textbook, this glossary was removed as it was not part of a lesson. However, where lessons provide a short list of words (typically 10 or fewer) that ought to be added to sentences or conjugated, or another task be completed using them, these were not removed. The instructions and examples that precede a lesson were preserved in documents, because they appear to be part of the learning process. The reference documents for primary and post-primary Irish repeatedly state that the four modes of communication ought to be taught in parallel, as noted in Sections 2.1 and 2.2. Therefore all lesson types were included in the corpus, regardless of the task type. Some materials included guidance for teachers, in the form of an explanation of the goals, or a version of the lesson for teachers that includes answers and helpful information. This type of content was not included in EduGA because it was not created for language learners and would likely not be written in a way that would reflect the level of ability. Sections containing answers were also frequently included in the Appendices to textbooks or at the ends of chapters. These were also omitted from EduGA because, in some cases, they were simply a list of words and in other cases they mirrored the sentences that were included in the tasks. This was done for all 2,840 documents in Table 10.

Where materials were collected from websites, they were considered to be in the public domain and free to be used for research purposes without contacting the creator(s). These were downloaded manually rather than the websites being scraped[67] and/or locally mirrored. I decided to take this approach for two main reasons:

1. In order to ensure all downloaded documents were educational materials that were aligned with a particular course of learning or level of ability.
2. The number of educational materials available on webpages were relatively limited when compared to the number that were collected from publishers and educational organizations. I discovered it was more time-efficient to manually download the appropriate data, rather than sifting through an entire website after scraping it.

Materials were saved as TXT files if they were part of the webpage. A considerable number of websites provide materials as downloadable PDF files, which required conversion to TXT. Materials for teaching Irish at junior and senior cycles in post-primary schools were not as forthcoming as materials for other subsets. A number of publishers did not respond to written requests for materials for EduGA. In order to balance these subsets and ensure all materials were included in the corpus I bought second-hand copies of textbooks. These books were then scanned, the default output format of which is PDF. Optical-character recognition (OCR) was conducted on all PDFs in order to convert them to TXT. Both Abbyy FineReader[68] and the Tesseract[69] tool were used, the latter of which has been incorporated into Google Drive as an OCR feature. Both pieces of software output either DOC(X) or TXT files. As stated, TXT files were chosen because it is compatible with the annotation software and the pre-processing tools used in the present research, the details of which are included in Section 4.2.

[67] "Scraping" is a form of copying that refers to an automated programming process, in which specific data are gathered and copied from the Internet into a local database or directory for later retrieval or analysis.

The entire corpus has been encoded as UTF-8 TXT files. Of the two OCR tools, the Tesseract tool provided more accurate output as well as being quicker to set up and run on multiple batches of scans. The optimum number of scanned pages per OCR process was found to be between 4 and 6 pages, depending on the amount of text on each page. If the scanned pages contained a particularly large amount of text, the process could fail. Equally, between 4 and 6 pages was preferable to 2 pages because of the amount of time it took to set up each batch of scans for the OCR process. Table 10 details the number of documents collected for each level of education of each subject.

[68] URL: https://www.abbyy.com/en-gb/finereader/ (Last visited: 20/05/2019)
[69] URL: “Google's OCR is probably using dependencies of Tesseract, an OCR engine released as free software” https://opensource.com/life/15/9/open-source-extract-text-images and https://github.com/tesseract-ocr/tesseract (Last visited: 20/05/2019)

Table 10: Collected Documents per Subject-specific Level of Ability

Subject	Long Name of Sub-corpus	Short Name of Sub-corpus	Number of Individual Documents
Irish (Mainstream Education)	Irish for primary level	**EduGA-Irish-Prim**	168
	Irish for Junior Cycle Post-primary level	**EduGA-Irish-PPJC**	18
	Irish for Senior Cycle Post-primary level	**EduGA-Irish-PPSC**	77
	Irish at University Undergraduate level	**EduGA-Irish-Uni**	158
	Irish in other Accredited Courses	**EduGA-Irish-Cert**	15
	General-purpose, edited, and published Irish materials aligned with (a) level(s) of ability[70]	**EduGA-Irish-Gnrl**	676
Irish (CEFR-GA)	CEFR-GA materials for levels A1	**EduGA-CEFR-A1**	369
	CEFR-GA materials for levels A2	**EduGA-CEFR-A2**	281
	CEFR-GA materials for levels B1	**EduGA-CEFR-B1**	313
	CEFR-GA materials for levels B2	**EduGA-CEFR-B2**	247
	CEFR-GA materials for levels C1	**EduGA-CEFR-C1**	441
	CEFR-GA materials for levels C2	**EduGA-CEFR-C2**	167
Science	Science for primary level	**EduGA-Sci-Prim**	18
	Science for Junior Cycle Post-primary level	**EduGA-Sci-PPJC**	8
	Science for Senior Cycle post-primary level	**EduGA-Sci-PPSC**	2
History	History for primary level	**EduGA-Hist-Prim**	7
	History for Junior Cycle Post-primary level	**EduGA-Hist-PPJC**	8
	History for Senior Cycle Post-primary level	**EduGA-Hist-PPSC**	3
Business Studies	Business Studies for Junior Cycle post-primary level	**EduGA-BusS-PPJC**	3
	Business Studies for Senior Cycle post-primary level	**EduGA-BusS-PPSC**	2
Mathematics	Mathematics for Primary level	**EduGA-Mata-Prim**	3
	Mathematics for Post-primary level Junior Cycle	**EduGA-Mata-PPJC**	2
	Mathematics for Post-primary level Senior Cycle	**EduGA-Mata-PPSC**	1

[70] These materials are designated as lower, intermediate, or higher (or similar three-tier structure).

Physical Education	Physical Education for Post-primary level Junior Cycle	**EduGA-Corp-PPJC**	2
Art	Art for Post-primary level[71]	**EduGA-Art-PPxx**	8
SPHE (*OSSP*)	Social, Personal, and Health Education for Post-primary level Junior Cycle	**EduGA-SPHE-PPJC**	11
	TOTALS	**n/a**	**2,840 Documents**

4.2. Corpus Pre-processing

The purpose of pre-processing is ensuring the data collected can be read correctly by the computer so that the annotation processes can be completed. The data must be annotated correctly if the analytic steps are to be accurate and representative, and if it is to be annotated correctly then a computer must be able to read the raw text correctly. This step would also be important for a token-based analyses that does not require annotation. A number of pre-processing steps were required, all of which are detailed in this section.

Some of the PDFs collected were generated in a way that preserved their text-based contents. These were easily read into a TXT file. However, a substantial number of the PDFs did not preserve the data as text and stored the contents of the materials as images. These materials needed additional pre-processing steps. OCR is an extremely useful automatic process, but it is rarely perfect and additional pre-processing steps are frequently required in order to clean the text for the annotation phase of the corpus compilation. Typical errors such as "ó" read as "6", or the letters "ni" read as "m", or vice versa, are still encountered. Errors were particularly evident in scans containing multiple font types, skewed text, cursive-like fonts[72]. An example of this is in Figure 3. These can be found and replaced in a number of ways and do not cause much difficulty, however, care must be taken that legitimate instances of "6" or "m" are not accidentally removed. The process is most accurately completed either manually or semi-automatically, e.g. with human input. Where the collected documents were not encoded correctly or where a PDF was generated to contain image data rather than text data, a greater number of problems were found. Some more complex problems were also found. More specifically, in order to make books attractive, a substantial number of pages contained at least two different types of font and at least two different sizes of font.

[71] Junior or Senior cycle not specified
[72] This was typically used to simulate a written letter or note within a text.

Figure 3: Example of Skewed and Cursive Text

Fonn 1

Leabhrán Scrúdaithe → Aonad a Ceathair

Léigh an fógra agus freagair na ceisteanna thíos.

8 marc

Amharclann
na Cille,
Loch Garman

Romeo agus Juliet

1 Meán Fómhair–16 Meán Fómhair
Luan–Aoine: 8.00
Satharn–Domhnach: 3.00
Ticéid ar fáil ó oifig na hamharclainne
nó ar an suíomh idirlín
www.amharclann.ie
Praghas na dticéad: Déagóirí: €15
Daoine fásta: €20

Romeo & Juliet

1. Cén amharclann ina mbeidh an dráma *Romeo agus Juliet* á léiriú?
2. Cén t-am a thosóidh an dráma ar an Satharn?
3. Conas is féidir ticéid a fháil?
4. Cé mhéad atá ar thicéid do dhaoine fásta?

Some of the pictures in educational materials also include text, especially series of pictures for a role-play lesson.

Figure 4: Example of Text within a Roleplay Picture

Of course these graphic design features are essentially necessary in materials for younger learners, they are only an issue when conducting OCR on these materials (for example: Figure 4). With the same aim in mind, creators sometimes use a cursive text on an image of a piece of notepaper (for example: Figure 5) and these texts run at an angle that is not parallel to the sides of the physical page.

Figure 5: Example of a Letter and Multiple Fonts


Léigh an cárta poist agus freagair na ceisteanna thíos.

10 marc

A Thomáis, a chara,
Beannachtaí ó Chontae na Gaillimhe. Tá mé féin agus Breandán sa Ghaeltacht i dTír an Fhia ar feadh trí seachtaine. Is aoibhinn liom an áit.

Gach maidin téimid ar scoil agus bíonn an-chraic againn sa rang. Canaimid amhráin agus bímid ag damhsa agus ag déanamh drámaí beaga sa rang.

Tá an aimsir go hálainn an tseachtain seo agus téimid ag snámh san fharraige gach tráthnóna.

San oíche bailíonn na daltaí go léir le chéile sa halla agus bíonn dioscó nó céilí againn. Beidh brón orm ag fágáil na háite.

Beidh mé ag filleadh abhaile ar an traein ar a haon a chlog an Satharn seo chugainn.

Do chara,
Stiofán

Tomás Ó Nualláin
23 Páirc na Coille Móire
Dún Garbhán
Co. Phort Láirge

1 Cá bhfuil Stiofán agus Breandán?
2 Ainmnigh rud amháin a dhéanann siad sa rang.
3 Cá dteann siad gach tráthnóna?
4 Céard a bhíonn ar siúl sa halla gach oíche?
5 Cén t-am Dé Sathairn a fhillfidh Stiofán abhaile?

Some extreme examples of this were read as Arabic by Tesseract, meaning they had to be manually corrected because none of the source text was recognised. An example of the text that could be included in these pictures is a poster for an event, or a speech bubble (for example: Figure 4). In this type of lesson, learners could be asked to complete a role-play exercise based on the pictures or they could be asked to write a description of what they see in each picture. On occasion inappropriate spacing was introduced into words during the OCR process; as a result of the graphic design “s p a c i n g” is read by the OCR software, this happened when letters like those in Figure 5 were processed. Another feature of the data that needed to be cleaned was where symbol A resembled symbol B from another alphabet, and the OCR software placed the symbol B in the TXT document that was outputted. For example, punctuation was read as a mathematical symbol rather than a dash. This impacts the annotation and analysis because of how computers read these symbols, where the conjoining dash was replaced by a minus symbol. This happened most frequently for scanned or mis-encoded documents, but was most difficult to clean when the text contained scientific or mathematic symbols or annotation.

Table 11: Mis-encoded punctuation and bullet points alongside the characters they replaced

types of apostrophe:	ʼ ʻ ' ʼ
types of comma:	, ʻ ˎ ˎ
types of dash:	– – – — — – -
types of ellipsis:	… … . . .
hyphen: -	minus: -
quotation marks:	‘ ’ “ ” ' ' " "
slash, stroke, solidus:	/ ⁄
other symbols:	• ■ ➢ ❑ √ | ⊥ ½ ↓ • ⍰ ⍰ ⌶ ⪊

In some cases non-Roman letters or glyphs took the place of Roman letters in documents. Mis-encoded documents (PDFs in particular) were converted to TXT; this also caused letters to be read as a homoglyph of the intended Roman letter, resulting in Cyrillic letters being used in the TXT rather than its Roman homoglyph. For example:

Table 12: Examples of OCR or encoding errors identified during pre-processing alongside the letters they replaced

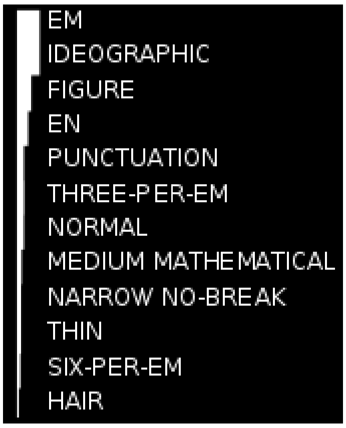
р (not p)	Р (not R)
г (not r)	у (not y)
ά (not á)	о (not o)
о (not o)	Ս (not U)
ύ (not ú)	о (not ó)

The mistakes in Table 12 cannot always be identified by the human eye, by reading the text, because the letters can look very similar to how we might expect them to look. Therefore, because a computer reads the letters differently to humans. In order to reliably identify these letters lists of words from the GaelSpel[7]l[3] spellchecker were used in conjunction with a Python program to check the spelling in the OCR output and flag it. The majority of flagged words needed to be corrected, but some words were purposely published as misspellings or were simply a series of jumbled letters from a word-search lesson. It would not have been appropriate to automatically replace these words, because then the corpus would not represent the original materials. The characters in Table 12 were always replaced with the same incorrect character. In order to clean this, a black list of offending characters was maintained and automated find-replace scripts were written using Python. Caution must be exercised when globally replacing characters, to avoid accidently

[73] The active repository for GaelSpell can be found at: https://github.com/kscanne/gaelspell

replacing “clean” data. A similar homoglyph issue was found, where a space had been replaced with a whitespace character during the OCR process[74]. These caused the annotation tools to crash when trying to process these symbols. The whitespace characters that have been inserted erroneously in the data look like the contents of Figure 6.

Figure 6: Examples of Whitespace Characters on a Black Background from Wikipedia: “Whitespace Character”



When trying to find these symbols the approach that was most successful involved creating a copy of the file and removing all expected characters, thus, only leaving the problematic characters. I refer to this process as: “If you can’t find the needle in the haystack, burn the hay”. Once the offending character had been found, then an appropriate replacement could be selected and inserted in its place in the original document.

Once all the documents have been cleaned, they are ready to be tokenized and annotated. In this study, if a set of documents is processed without error (in the context of this study, tokenized and then tagged for lemma and part of speech) then it was accepted that the pre-processing had been completed for this set of documents. This was accepted because the annotation tools would crash if they received data containing the symbols and characters described in this section. Sentence tagging was then conducted separately.
Annotation processes are detailed in the following section, Section 4.3. When documents have been processed without error, the word-count for these documents can be recorded.

4.3. Corpus Annotation

In this section I describe the types of annotation used in this research, including an overview of the installation and running of the annotation processes. By annotation we mean that the corpus is input to a piece of software, which then processes every single word in every single document, creates a new file or files, and writes the new annotated version of the data into the new file(s). Examples are given in this section. Annotation, or tagging, is always done with a specific analytic task in mind. The processes completed are described in this section, with their use explored in the analyses detailed in Section 5. The entire corpus is tagged for parts of speech (POS) and lemmas, as detailed in Section 4.3.1, and sentence-tagging software has been used on the corpus, as detailed in Section 4.3.2.

4.3.1. Part-of-speech tagging

I have installed the POS and lemma tagger and tagged the entire corpus, as part of the current study. The part of speech tagger developed in Uí Dhonnchadha (2008) is being used. I will henceforth refer to this tagger as the *EUD-tagger*. EUD-tagger needs to be installed on a Linux-based computer, such as Ubuntu. The installation process involved the following steps on a Windows 10 computer:

1. Install Oracle's (Virtual Machine) Virtual Box (URL: https://www.virtualbox.org/). This creates a virtualized machine within your computer, within which a Windows, Linux or Mac operating system can be installed. As a result, a single physical machine can contain multiple virtual machines. Multiple operating systems can be run in parallel, on multiple virtual machines, on one physical computer.
2. Once a virtual environment has been installed, then the latest version of Ubuntu can be installed on it (URL: https://tutorials.ubuntu.com/tutorial/tutorial-install-ubuntu-desktop#0) because it supports the EUD-tagger.
3. The EUD-tagger is available on GitHub URL: https://github.com/uidhonne/irishfst (last accessed: 27/06/2019). The transducer setup upon which the EUD-tagger depends requires Hulden's software called "Foma" be installed first[75], which is currently available from URL: https://bitbucket.org/mhulden/foma/downloads/ (last accessed:27/06/2019). It should be noted that Foma also has a number of dependencies that will need to be installed before Foma. Which dependencies are required can vary depending on the Ubuntu installation selected (For examples: Bison, M4, Flex, and CG3 were all required for my setup; as well as some other minor encoding packages).

The tokenization done for this tagger refers to splitting the document into individual words. A small number of multi-word constructions have been specified in the tagger in order to optimize the tagging process. For example: *go dtí* ["to" (*preposition*)] is tokenized as "go_dtí" so that both of these words are tagged correctly and together. The inputted text is tokenized and then tagged by the EUD-tagger, and the current default output is a vertical version of the input. For example, the output for *Ritheann siad na rásanna* (**"**They run the race") is:

"<Ritheann>"

"rith" Verb VTI

PresInd "<siad>"

"siad" Pron Pers 3P PL Sbj

"<na>"

"an" Art Pl Def

"<rásanna>"

"rás" Noun Masc Com Pl DefArt

The token from the input file is surrounded by "<these>" and is alone on a line, the lemma is between "these" quotation marks indented on the following line. Each of these lemmas is followed by its corresponding part-of-speech tag. If a token cannot be fully disambiguated by the EUD-tagger, but the lemma has been recognised by the software, then the possible lemmas and their corresponding parts-of- speech are given as choices. While this output can be tailored, this was the setting used in the present research. Each lemma-POS-tag combination is split onto its own line. For example, the token "is" can read as a present-tense copula or as a variant form of the conjunction *agus,* which translates as "and". This tagging is done using a lexicon and an extensive set of rules. Where a word cannot be found in the EUD-tagger's lexicon, a "guessed" lemma and POS-tag combination is applied. The guesses are also based on a set of rules, and very frequently guessed the lemma, and tense or declension, correctly.

4.3.2. Sentence Tokenization

The original, untagged, corpus data has also been tokenized on a sentence level. The sentence tokenizer used is called the Punkt Sentence Tokenizer[76]. This is defined in the documentation for the tokenizer itself as: "A sentence tokenizer which uses an unsupervised algorithm to build a model for abbreviation words, collocations, and words that start sentences; and then uses that model to find sentence boundaries. This approach has been shown to work well for many European languages". What is noteworthy here, is that no human input is required. In this context, list refers to lists as a data structure rather than the general meaning. In Python, a list of sentences would take the following form:

```
List_of_sentences = [ "Rith siad an rás." , "The quickbrown fox jumps over the lazy dog. " , " Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua." ]
```

Punkt must be trained on a large collection of plaintext in the target language before it can be used. The NLTK data package includes a pre-trained Punkt tokenizer for English. This pre-trained English tokenizer was used for the present research in order to cut down on training time and because it performed very well in manually-checked tests on Irish-language documents from the corpus.

(Source code: https://www.nltk.org/_modules/nltk/tokenize/punkt.html).

As noted in the documentation for Punkt, sentence segmentation relies on a system that understands that the full stop used in a title like "*Dr.*" does not mark the end of a sentence, equally, sentences do not always begin with a capital letter. Punkt does not include a library for tokenizing Irish sentences, however, Irish shares a number of abbreviations with English. For example: The abbreviations "*mm.*" (millimetre or *milliméadar*), "*cm.*" (centimetre or *ceintiméadar*), "*m.*" (metre or *méadar*), and "*Dr.*" (doctor or *dochtúir*) are used in both languages because the words include the same letters in both Irish and English. On the other hand, "*km*[7].[7]" (kilometre or *ciliméadar*) is used to measure distance in Irish because natural abbreviation for the Irish word *ciliméadar* is "cm." and this abbreviation has already been assigned to another unit of measurement. Roman numerals are sometimes punctuated, a style that can be adopted in both Irish and English. This final point is particularly important in the context of educational materials. As is the aforementioned point about not all sentences begin with a capital letter. Lessons frequently include fragments or begin with a word that is between brackets, all of which the Punkt sentence tokenizer has means of addressing. The success of the sentence tagger is also likely due to the input data being edited and published text that is relatively clearly structured, especially in comparison to web data or other potentially less-structured text.

Rather than keeping the data in the default list format above, each sentence was enclosed in XML tags. XML was chosen to structure the sentences because it is commonly used when structuring data and, as such, a number of out-of-the-box tools are available for parsing and analysing your data. In this research, the LXML version of *Element Tree*[78] was used to parse each XML file. The structure is therefore:

<BigFile>

<s>Rith siad an rás.<s>

<s>The quick brown fox jumps over the lazy dog.</s>

<s>Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.</s>

</BigFile>

Spot-checks were done on the entire corpus in preparation for the sentence-length analysis. This spot checking was done following an initial measurement of sentence lengths. A relatively small computer program was written to output the length of every sentence and to number every sentence. Where a document erroneously contained sentences that contain 30+ words, then these sentence were manually checked. In some situations the same pattern (punctuation, or lack thereof, paragraphing, and so on) has caused the majority of the errors, therefore, correcting the tagging was a pattern-matching exercise that was simply completed.

[76] URL: https://www.nltk.org/_modules/nltk/tokenize/punkt.html (last accessed, 26/10/2018)

[77] URL: https://www.tearma.ie/q/km/ (Last accessed: 16/05/2019)

[78] URL: https://lxml.de/tutorial.html (Last accessed: 14/06/2019)

Equally, if a considerable portion of a document was tagged as 1-word or 2-word sentences then these examples were manually checked. Once again, in some situations the same pattern had also caused the majority of the errors, therefore, correcting them was also done by pattern matching. In some cases, no pattern was apparent and the changes had to be made manually. Take the three sentences presented here as examples of correctly tagged sentences, the number of words between pair of XML tags (or in each XML element) is counted and the following two-column output is generated.

4 words Rith siad an rás.

9 words The quick brown fox jumps over the lazy dog.

19 words Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

The number denotes the number of words in the sentence, and the sentence itself is included alongside the length for reference. This data was then sorted in ascending order, with the largest sentences at the top.

Virtually all sentence tagging errors occurred when a sentence either was not punctuated, or where the boundary of the sentence was not recognized by the software. This resulted in erroneously-tagged sentences of over 100 words in some cases, and with a smaller number of examples of erroneously-tagged sentences in the 30s, 40s and 50s as well. Corrections to the tagging were added manually. The final token count for each sub-corpus is presented in Table 13.

4.3.3. Specification of the Contents of EduGA and Sub-corpora

This subsection includes the size of each sub-corpus in terms of tokens and documents in Table 13, followed by a list of word lists included in Appendix 1: Extended Results and Referential Datasets. Columns 1-4 of Table 13 include subject-specific information divided according to levels of education. Column 5, *Total Number of Tokens per Subject*, summarises the number of tokens for each subject across all levels. The analyses conducted in this research draw on and compare sub-corpora on a level-of-education basis. Where conclusions and summaries refer to the subject as a whole, this is explicitly stated.

Table 13: Specifications for Constituent Sub-corpora of EduGA 2018

Long Name of Sub-corpus	Short Name of Sub-corpus	Individual Docs	Total Number of Tokens per Level of Education Sub-corpus	Total Number of Tokens per Subject
Irish for junior classes primary level	**EduGA-Irish-PrJC**	150	126,386	3,721,312
Irish for senior classes primary level	**EduGA-Irish-PrSC**	18	93,434	
Irish for Junior Cycle Post-primary level	**EduGA-Irish-PPJC**	18	256,760	
Irish for Senior Cycle Post-Primary level	**EduGA-Irish-PPSC**	77	326,869	
Irish at Undergraduate level	**EduGA-Irish-UniU**	158	1,231,550	
Irish in other Accredited Courses	**EduGA-Irish-Cert**	15	384,435	
General-purpose, edited, and published Irish materials not specifically aligned with (a) level(s) of education[79]	**EduGA-Irish-Gnrl**	676	1,301,878	
CEFR-GA materials for levels A1	**EduGA-CEFR-A1**	369	349,702	

[79] These materials are marked as lower, intermediate, higher (or similar three-tier structure) but contain no reference to a guiding framework, syllabus, or course of education.

CEFR-GA materials for levels A2	**EduGA-CEFR-A2**	281	302,660	2,235,294
CEFR-GA materials for levels B1	**EduGA-CEFR-B1**	313	352,863	
CEFR-GA materials for levels B2	**EduGA-CEFR-B2**	247	356,863	
CEFR-GA materials for levels C1	**EduGA-CEFR-C1**	441	656,127	
CEFR-GA materials for levels C2	**EduGA-CEFR-C2**	167	217,079	
Science for primary level	**EduGA-Sci-Prim**	18	134,629	337,728
Science for Junior Cycle Post-primary level	**EduGA-Sci-PPJC**	8	181,956	
Science for Senior Cycle post-primary level	**EduGA-Sci-PPSC**	2	21,143	
History for primary level	**EduGA-Hist-Prim**	7	91,889	184,724
History for Junior Cycle Post-primary level	**EduGA-Hist-PPJC**	8	66,270	
History for Senior Cycle Post-primary level	**EduGA-Hist-PPSC**	3	26,565	
Business Studies for Junior Cycle post-primary level	**EduGA-BusS-PPJC**	3	57,736	191,062
Business Studies for Senior Cycle post-primary level	**EduGA-BusS-PPSC**	2	133,326	

Mathematics for Primary level	**EduGA-Mata-Prim**	3	12,282	544,646
Mathematics for Post-primary level Junior Cycle	**EduGA-Mata-PPJC**	2	435,051	
Mathematics for Post-primary level Senior Cycle	**EduGA-Mata-PPSC**	1	97,313	
Physical Education for Post-primary level Junior Cycle	**EduGA-Corp-PPJC**	2	66,082	66,082
Art for Post-primary level[80]	**EduGA-Art-PPxx**	8	32,936	32,936
Social, Personal, and Health Education for Post- primary level Junior Cycle	**EduGA-SPHE-PPJC**	11	119,705	119,705
TOTALS	**n/a**	**2,898 Documents**	**7,433,489 Tokens**	**7,433,489 Tokens**

Wordlists have been generated from all sub-corpora of EduGA listed in Table 13, each of which includes the 100 highest-frequency tokens for the given sub-corpus, and have been included in Appendix 1: Extended Results and Referential Datasets.

4.4. Conclusions and Comments

This section contains summaries, comments, and conclusions drawn and arising from the various stages of compiling the corpus of Irish-language educational materials.

It had been reported in the literature (Such as in Mac Donnacha et al. (2004) and Ó Muircheartaigh (2009)) that were was a shortage of materials for Irish-medium classes in particular, and a shortage of Irish-language materials at an appropriate level. The variety of materials available for teaching Irish-language classes was vast in comparison to the variety of materials available for teaching Irish-medium classes, however, materials were available for these subjects. Where Irish-medium materials for primary and post-primary levels have not been included in EduGA this is because of the time constraints of the present research or because it was difficult to contact certain publishers. Despite not being quantified here, it appeared as though publications for the domain of education certainly ranked in the top 5 publication types for Irish in terms of the quantity of publications.

The pre-processing of corpus data was the most time-consuming activity of the corpus compilation phase of the present research. This is primarily due to the large number of files received in PDF format, or scanned to PDF, and the fact that OCR technologies used to convert the PDFs to TXT are very powerful but prone to mistakes. While using small computer programs to find or change particular strings of text was hugely advantageous in hundreds of files at a time, the majority of the cleaning phase of pre-processing did require careful manual attention so that the processing was completed accurately. Without Python script (relatively small computer programs), pre-made programming libraries for NLP-purposes, and reference lists, such as Scannell (2000b), the pre-processing of almost 3,000 documents would not have been possible during the course of this research. It appears as though programming skills are becoming more and more necessary for researchers in the field of corpus linguistics and applied linguistics.

Some patterns were identified in the EUD-tagger's guesses, resulting in the following recommendations for improvement. The lexicon used in the EUD-tagger was initally based on the contents of the N. Ó Dónaill (1977)[81] dictionary, and subsequently expanded. The improvements recommended here are predominantly based on datasets and resources that were constructed and/or made available to the public after the EUD- tagger was created.

[80] Junior or Senior cycle not specified

[81] URL: https://www.teanglann.ie/ga/fgb/ (las accessed, 26/10/2018)

Additional modern words & Frequently-used Compounds: A substantial number of modern words are included in the tagger's lexicon. When words were not included in the tagger's lexicon a substantial number of these were scientific terms that were tagged by the Guess function of the tagger. The Guesser's results were very satisfactory, but the tagger would undoubtedly benefit from the systematic addition of entries from the Terminology Database for Irish (URL: www.tearma.ie) and the New English-Irish Dictionary (URL: www.focloir.ie). These additions were not pursued during the course of this research because it is believed the task would require a great deal of editorial work and testing. For example: duplicates should not be added.

Resource Development: A core part of the present research was the development of a resource (namely EduGA) that can be used to develop the complexity measures as well as being used referentially in future research. As such, Appendix 1 includes word lists from each sub-corpus in EduGA.

5. Corpus Analysis

This chapter describes the three types of analysis conducted on EduGA. These analyses have been conducted with a view to developing objective complexity metrics for educational materials written in Irish. In order to develop these metrics, the analyses focus on identifying a number of characteristics in the materials that constitute each of the sub-corpora in Table 13.

The first analysis presented in Section 5.1 looks at the frequency of seven lexico-grammatical features that have been prescribed in reference documents for both mainstream education and CEFR-GA courses. The second piece of analysis in Section 5.2 presents the results of a word-length analysis and the results of a sentence-length analysis, both of which were conducted on 18 subsets of EduGA 2018. The third piece of analysis is presented in Section 5.3. This analysis begins by matching terms in EduGA sub-corpora with the terms from Téarma (The National Terminology Database for Irish, URL: www.tearma.ie) and analysing the frequency of matched terms. A statistical method called TF-IDF (term frequency–inverse document frequency) is used to apply a topicality score to matched terms and these scored terms are then analysed in the context of their frequency in a corpus of general Irish.

5.1. Frequency analysis of prescribed lexico-grammatical items in Irish-language educational materials

This analysis approaches the topic of complexity in the domain of education by working under the assumption that educational materials become more complex from one level of ability to the next. This is based on the fact that language and education experts prescribe language feature A for classes of lower ability, upon which they aim to build and expand the learners' abilities. Where language feature A is more consistently prescribed for lower levels of ability than language feature B, this is being taken to indicate that language feature A is in some way less complex than language feature B. The research question being answered by this analysis is as follows:

> When a language feature is prescribed for a particular level of education for the first time, is the frequency of that language feature statistically significant in educational materials at that level of education over other levels of education?

Where the frequencies of prescribed language features are found to occur at statistically significant frequencies at the level for which they have been prescribed, then the frequency of the language feature in question is taken here to be an indicator of relative complexity within the domain of education. Statistical significance is measured using feature frequency and standard deviation. Feature frequency is being used in order to quantify the extent of spikes in frequency at particular level(s). Standard deviation is being used because it would be incorrect to simply say the highest frequency is the most significant frequency. Baayen (2008) discusses how standard deviation can be used to identify deviations from the normal distribution of frequencies, with which the standard deviation usage applied here agrees.

Relative complexity refers here to one level of education being more complex than another. If all frequencies fall within one standard deviation from the mean, they are not significant and these frequencies will not be taken to indicate any type of relative complexity. This research question addresses the issues identified in the literature; namely, the lack of established methodologies for analysing Irish-language complexity as well as the reports about some materials being overly complex or difficult while others are considered overly simple (cf. Mac Donnacha et al. (2004), de Brún (2007), Ó Muircheartaigh (2009)).

By selecting prescribed items from reference documents as a basis for this analysis I can be certain that the items analysed ought to be both present and relatively important to the level of education for which they were prescribed. In Section 2.5 it was noted that the examples of prescribed language features in reference documents were, for the most part, short isolated chunks of language. These short isolated examples include single words, phrases, clauses, and sentences. The *examples* found in many of the reference documents are very similar – particularly at beginner levels– despite the differences in learning aims, pedagogy, or learner type (e.g. adult learner or child). In order to use these language features in educational materials, creators of educational materials must elaborate and expand upon that which has been prescribed in the reference documents. This elaboration relies on the experience and expertise of the materials creators, and EduGA aims to capture this experience and expertise. This analysis therefore benefits from the experience and expertise of materials creators and their interpretations of the short isolated examples in the reference documents when selecting suitably complex content for learners across a number of levels. This frequency analysis takes into consideration the level of ability for which the lexico-grammatical feature was prescribed, and the level of ability for which the materials were created. One of the most consistently used methodologies for analysing multiple or individual language features in either educational materials or learner language was frequency analyses (Walsh (2007), Pétervåry et al. (2014), Römer (2005), Gabrielatos (2006), among others). These frequencies were often, but not always, combined with other analyses, such as the forms or functions of the feature, and the frequencies themselves compared to the frequencies of the same language feature(s) in a reference corpus.

Table 14 presents the seven lexico-grammatical language features that have been selected for analyses and the level at which they were first prescribed for Irish-language learners in both Mainstream Education and CEFR-based courses. Two of the selected language features, *Greetings and Salutations* and *Introductions*, are prescribed as specific communicative functions that are associated with a limited number of examples. The remainder are either prescribed as grammatical competencies, such as modifying a noun with an adjective, or are given as examples with of how a communicative function might be performed by a learner at a particular level.

The seven analysed language features (cf. Table 14) have been identified in reference documents for Mainstream Irish courses at primary and post-primary levels, and for CEFR-GA courses at levels A1 through B2. Materials for Irish-language university courses were not included in this analysis primarily because syllabi were rarely available for undergraduate courses. The learning aims of degree courses are expressed in a different manner to those of primary and post-primary levels, typically relating to a secondary topic in the degree (typically literature, translation studies, media and journalism, or business). The result of this is that when a communicative function is associated with a language features in the reference materials for post- primary level and this language feature is used in educational materials, the alignment has been done *a priori*. Materials alignment and course alignment is a separate branch of corpus-linguistic research unto itself, which would be a totally separate piece of research. Regarding Irish-medium materials, no specific language features or skills are prescribed in the syllabi for the subjects themselves and the syllabi for Irish as a subject only state that the teaching of subjects through the medium of Irish be supported as much as possible (e.g. An Roinn Oideachais agus Scileanna (2010a), p. 15, 27, 41).

Table 14: Language features prescribed for Irish in Mainstream Education and CEFR-GA that have been selected for Analysis

Test #	Language Feature	Mainstream Prescription	CEFR-GA Prescription
1	Greetings and Salutations	First prescribed for both infant classes in schools in English-speaking communities, specific class not specified for Irish-immersion and Gaeltacht primary schools: (An Roinn Oideachais agus Scileanna, 1999a), (p.20, 80) [e.g. *beannú do dhaoine* (Greeting people)*;* *slán a fhágáil,* (saying goodbye)]	First prescribed for CEFR-GA level A1: Ionad na dTeangacha (p. 11, 16)
2	Introductions	First prescribed for 1st and 2nd classes in schools in English-speaking communities, specific class not specified for Irish-immersion and Gaeltacht primary schools: (An Roinn Oideachais agus Scileanna, 1999a), (p.20, 80) [e.g. *bualadh le daoine* (meeting people); *cur in aithne* (introduction)]	Prescribed for CEFR-GA level A1: Ionad na dTeangacha , (p. 11, 16)

3	Noun-adjective collocations	First prescribed for 1st and 2nd classes for all school types: (An Roinn Oideachais agus Scileanna, 1999a), (p.43, 105) [e.g. *focail cháillitheacha* (modifying words)]	Prescribed for CEFR-GA level A2: Ionad na dTeangacha , (p. 9, 13, 35, 38)
4	Conditional-mood verbs	First prescribed for 3rd and 4th classes, specific class not specified for Irish-immersion and Gaeltacht primary schools: An Roinn Oideachais agus Scileanna (1999a), p.22, 82. [e.g. *féidrearthacht* (possibility), *éiginnteacht* (uncertainty), *Cead a lorg*, [Seeking permission)]	Prescribed for CEFR-GA level A2: (Ionad na dTeangacha , p. 33, 45)

5	"Má" and "Dá", Conditional- mood particles for "if".	First used as an example for junior- cycle post-primary level, used with varying levels of assistance or prompts depending on the level of ability: (An Roinn Oideachais agus Scileanna, ND), p.28 of 46. Can also be used to perform communicative functions prescribed for 3rd and 4th classes, specific class not specified for Irish-immersion and Gaeltacht primary schools: (An Roinn Oideachais agus Scileanna, 1999a), p.22, 82. [e.g. *féidrearthacht* (possibility), *éiginnteacht* (uncertainty), *Cead a lorg*, (Seeking permission)]	Prescribed for CEFR-GA level B1: (Ionad na dTeangacha , p. 55.)

6	Autonomous Forms of Verbs	First used as an example for senior-cycle post-primary level, cf. (An Roinn Oideachais agus Scileanna, 2010a)(p.17, 39); in (An Roinn Oideachais agus Scileanna, 2012b) p.16, as well as (An Roinn Oideachais agus Scileanna, 2013) p.16. [e.g. *Ag lorg nó ag tabhairt eolais* (Seeking or giving information)]	Prescribed for CEFR-GA level B1: Ionad na dTeangacha , (in examples on p. 10, 14, 20, 37, etc.)
7	Indirect Relative Clauses	No examples were found in An Roinn Oideachais agus Scileanna (2010a). The types of example used in the document are likely too short to facilitate using the indirect relative clause structure. Direct relative clauses have been given as examples in An Roinn Oideachais agus Scileanna (2010a) (p.11, 17, 20, 23, 25). Some prescribed language functions grammatically require an indirect relative clause be used. For example: Interactions about locations using a translation of "where" (p.39). Other prescribed language functions are greatly enriched by using the indirect relative clause. For example: discussing literature (p.43).	Prescribed for CEFR-GA level B2: Ionad na dTeangacha , (in examples on p. 64, 74, 75, 78, 84, etc.)

Where the results are presented in each of the seven subsections below, the level of education for which each language feature has been first prescribed is marked with a grey background, as opposed to white.

The sub-corpora used for this analysis are from the same courses and levels within these courses. The details of the sub-corpora used in this analysis are detailed in Table 15 and Table 16.

Table 15: Specification of the Mainstream Irish sub-corpora

Sub-corpus	Number of Files	Total Word Count
EduGA-Irish-PrJC	150	126,386
EduGA-Irish-PrSC	18	93,434
EduGA-Irish-PPJC	18	256,760
EduGA-Irish-PPSC	77	326,869

Table 16: Specifications of the CEFR-GA sub-corpora

Sub-corpus	Number of Files	Total Word Count
EduGA-CEFR-A1	369	349,702
EduGA-CEFR-A2	281	302,660
EduGA-CEFR-B1	313	352,863
EduGA-CEFR-B2	247	356,863

In each of the following subsections the language features being analysed are introduced, followed by the results of the analysis and a brief discussion. The overall conclusions and discussion are presented in subsection 5.1.8.

5.1.1. Test 1: Greetings and Salutations

Greetings and Salutations have been prescribed for absolute beginners in both course types, as shown in item 1 of Table 14. This is commonplace across almost all language-learning courses. Greetings and salutations are presented as formulaic pieces of language that beginners can use to initiate an interaction in a number of settings, both formal (with a teacher or adult) and informal (with friends and other children).

Minimal amounts of conjugation or declension are required when using these greetings and salutations. The greetings and salutations in Table 17 are given as examples in the reference documents. The frequency analysis of a set of *Greetings and Salutations* was conducted using the following items:

Table 17: Greetings and Salutations used in Frequency Analysis

Hello	Goodbye	How are you?
Dia dhuit / duit Go mbeannaí Dia tú / duit Dia is Muire dhuit / duit Nár laga Dia thú / tú Rath Dé ort Bail ó Dhia ort Dia (go deo) leat / linn / libh Beannacht Dé leat / linn / libh Dé do bheathasa / bhur mbeathasa **Beannacht(aí)...**	Slán, Slán agat, Slán go fóill, Slán leat, Slán tamall, Go dté tú slán, Go n-éirí an bóthar leat, Beidh mé ag caint leat, Beimid ag caint, Ádh mór	Conas atá (tú)? Cad é mar atá agat? Conas ataoi? Conas atá agat? Cén chaoi a bhfuil tú?

This language feature was first prescribed for absolute beginners, therefore, materials in EduGA-Irish-PrJC and EduGA-CEFR-A1. The frequencies presented in Table 18 represent frequencies in materials for Mainstream Irish classes and the frequencies reported in Table 19 represent frequencies in materials for CEFR-GA classes.

Table 18: Frequency of Greetings and Salutations in Mainstream Irish Sub-corpora

Normalised Frequency in EduGA-Irish-PrJC per 50k words	**Normalised Frequency in EduGA-Irish-PrSC per 50k words**	**Normalised Frequency in EduGA-Irish-PPJC per 50k words**	**Normalised Frequency in EduGA-Irish-PPSC per 50k words**
40	45	18	17

The standard deviation for the frequency of Greetings and Salutations in Mainstream Irish materials is **12.63**, and the mean of the frequencies is **30**. Therefore, one standard deviation above the mean is **42.63**. The frequency of Greetings and Salutations are statistically significant in EduGA-Irish-PrSC and the frequencies in all other sub-corpora were not statistically significant in this context.

Table 19: Frequency of Greetings and Salutations in CEFR-GA Sub-corpora

Normalised Frequency at EduGA-CEFR-A1 per 50k words	**Normalised Frequency at EduGA-CEFR-A2 per 50k words**	**Normalised Frequency at EduGA-CEFR-B1 per 50k words**	**Normalised Frequency at EduGA-CEFR-B2 per 50k words**
374	39	15	15

The standard deviation (SD) of the frequencies found in four CEFR-GA levels is **152.30** and the mean of the four CEFR-GA frequencies is **110.75**. Therefore, one standard deviation above the mean is **263.05**. Therefore, the frequency of Greetings and Salutations are statistically significant in CEFR-GA-A1 and the frequencies in the other CEFR-GA levels were not statistically significant in this context.

The frequency of this language function at the levels for which it was first prescribed was greater than the standard deviation from the mean. The educational materials for primary-school pupils, as per the results in Table 18 and Table 19, showed a normalized frequency that was roughly 10% of that found in CEFR-GA sub- corpora. The reason for this is the different types of materials, where the primary school materials are typically readers (short story books) rather than the lessons and conversational tasks one might find in CEFR- GA materials. These results indicate that, with regards to this language feature, substantially more repetition is built into the CEFR-GA materials than the primary-school materials.

5.1.2. Test 2: Introductions

The following constructions that are used to introduce people are the examples given in the reference documents (cf. Table 14) for beginner levels in Mainstream Education and in CEFR-GA Level A1, with the specific names from the reference documents removed. Having the ability to introduce both yourself and other people are seen as key skills for conversations in formal, informal, professional, or social settings.

(23) Is mise [insert person's name] Is
I (emphatic form)[insert person'sname] "My name is [insert person's name]"

(24)[insert person's name] is ainm dom

[insert person's name] is name to.1SG

"[insert person's name] is my name"

(25) Cén t-ainm atá ort?
What-DET.SG name that-is on.2SG?
"What is your name?"

(26) Cad is ainm duit?
What is name to.2SG?
"What is your name?"

(27) Tugtar [insert person's name] orm.
Give-IMPS [insert person's name] on.1SG
"I am called [insert person's name]

(28) Glaoitear [insert person's name] orm

Call-IMPS [insert person's name] on.1SG

"I am called [insert person's name]

In addition to these examples, all forms of the word *ainm*, "name", were also sought in case of other constructions fulfilling the aim of this language function. None were found. This language feature was first prescribed for beginners, therefore, for materials in EduGA-Irish-PrJC and EduGA-CEFR-A1.

Table 20: Frequency of Introductions in subsets materials for Mainstream Irish

Normalised Frequency in EduGA-Irish-PrJC per 50k words	**Normalised Frequency in EduGA-Irish-PrSC per 50k words**	**Normalised Frequency in EduGA-Irish-PPJC per 50k words**	**Normalised Frequency in EduGA-Irish-PPSC per 50k words**
13	14	29	13

The standard deviation of the frequencies found in materials for Mainstream Irish is **6.8**, and the mean of the frequencies is **17.25**. One standard deviation above the mean is **24.05**. Therefore, the frequency of Introductions at the level for which they were first prescribed is not statistically significant in this context.

This result indicates that this language feature is not suitable for measuring complexity in the manner hypothesised in materials for Mainstream Irish. Further reason for this is the statistical significance that occurred in EduGA-Irish-PPJC materials where the frequency of Introductions was greater than one standard deviation above the mean.

Table 21: Frequency of Introductions in subset of CEFR-GA materials

Nomalised frequency and EduGA-CEFR-A1 per 50k words	**Nomalised frequency and EduGA-CEFR-A2 per 50k words**	**Nomalised frequency and EduGA-CEFR-B1 per 50k words**	**Nomalised frequency and EduGA-CEFR-B2 per 50k words**
386	25	21	10

The standard deviation for the frequencies of Introductions in CEFR-GA materials is **159.15**, and the mean of the CEFR-GA frequencies is **110.5**. Therefore, the frequency of Introductions at the level for which they were first prescribed is statistically significant in this context. This result indicates that this language feature may be suitable for measuring complexity in the manner hypothesised in CEFR-GA materials.

The difference between the results in Table 20 and Table 21 can be explained by the type of content in the materials where, similar to Test 1 (cf. Section 5.1.1), the type of content in Mainstream materials would be more story focused whereas the type of content in the CEFR-GA materials would be more activity focused. These activities can be accompanied by prompts or preceded by a brief lesson. While the same language feature was prescribed for both types of language-learning class, these results indicate the extent of the impact pedagogy has on the interpretation of the reference documents. The CEFR-GA-A1 frequency is high, and the normalized frequencies in Tests 2 found the language features occurred more than ten times more frequently than in any other sub-corpus. This appears to be the result of both repetition and the narrower range of topics in CEFR-GA-A1 when compared to subsequent levels.

5.1.3. Test 3: Noun-adjective Collocations

Modifying a noun with an adjective was first prescribed for 1st and 2nd class in Mainstream Irish (EduGA-Irish- PrJC) and for CEFR-GA level A2 (EduGA-CEFR-A2). Having the ability to modify nouns is beneficial for a great number of communicative functions, across all levels of ability. The analysis conducted here has been conducted based on part-of-speech tagging, where a noun is followed by an adjective.

Table 22: Frequency of Noun-adjective collocations in Mainstream Irish materials

Sub-corpus	**Normalised Frequency in EduGA-Irish-PrJC per 50k words**	**Normalised Frequency in EduGA-Irish-PrSC per 50k words**	**Normalised Frequency in EduGA-Irish-PPJC per 50k words**	**Normalised Frequency in EduGA-Irish-PPSC per 50k words**
Freq	1,544	1,606	1,395	1,539

The standard deviation of the frequencies found in materials for Mainstream Irish **77.39** and the mean of the frequencies is **1521**. One standard deviation above the mean is **1598.39**. Therefore, the frequency of noun- adjective collocations at the level for which they were first prescribed, however, statistical significance is almost achieved and then statistical significance is achieved at the subsequent level, EduGA-Irish-PrSC.

Table 23: Frequency of Noun-adjective collocations in CEFR-GA materials

Sub-corpus	CEFR-GA Level A1 per 50k word	CEFR-GA Level A2 per 50k word	CEFR-GA Level B1 per 50k word	CEFR-GA Level B2 per 50k word
Freq	1,157	1,017	1,152	2,126

The standard deviation for the CEFR-GA frequencies is **444.07,** and the mean is **1363**. All frequencies from sub-corpora were within a standard deviation from the mean **1807.07**. In this case, the hypothesis was not confirmed. It is noteworthy that the frequency of noun-adjective collocations increased considerably in materials for CEFR-GA level B2.

The frequency of noun-adjective collocations was relatively consistent across all levels in Mainstream Education, as well as CEFR-GA levels A1 through B1. CEFR-GA-B2 was the only sub-corpus to achieve statistical significance in this case. The frequencies found are indicative of a language feature that is introduced to a limited extent, with expansion or elaboration of the language feature at the higher level of CEFR-GA-B2. These results can be contextualized with further analysis and the appropriate methodology for this would need to be investigated; two possibilities are a semantic analysis of nouns or noun-adjective collocates, or an analysis of the vocabulary range with respect to nouns and adjectives at these levels. These results are discussed in more detail in the conclusions to this analysis, Section 5.1.8.

5.1.4. Test 4: Conditional Mood Verbs

The ability to conjugate verbs in the conditional mood is first prescribed for 3rd and 4th class in Mainstream Irish (aka: EduGA-Irish-PrSC) and for CEFR-GA level A2 (EduGA-CEFR-A2). The analysis conducted here has been conducted based on part-of-speech tagging, where a verb has been conjugated to the conditional mood. The conditional mood is typically taught with a view to performing the following communicative functions: the feasibility or plausibility of an event[82]; a hypothetical situation occurring in the past, present, or future[83]; co-operating with others[84], use of imagination[85], as well as planning and deliberating[86].

Table 24: Frequency of Conditional Mood Verbs in Mainstream Irish materials

Sub-corpus	Normalised Frequency in EduGA-Irish-PrJC per 50k words	Normalised Frequency in EduGA-Irish-PrSC per 50k words	Normalised Frequency in EduGA-Irish-PPJC per 50k words	Normalised Frequency in EduGA-Irish-PPSC per 50k words
Freq	173	293	171	331

The standard deviation of the frequencies of found in materials for Mainstream Irish **71.27**, and the mean of the frequencies is **242**. One standard deviation above the mean is **313.27**. Therefore, the frequency of conditional-mood verbs at the level for which they were first prescribed is not statistically significant in this context. Statistical significance was not achieved until EduGA-Irish-PPSC

Table 25: Frequency of Conditional Mood Verbs in CEFR-GA materials

Sub-corpus	CEFR-GA Level A1 per 50k word	CEFR-GA Level A2 per 50k word	CEFR-GA Level B1 per 50k word	CEFR-GA Level B2 per 50k word
Freq	20	46	66	199

The standard deviation for the CEFR-GA frequencies is **40.89** and the mean is **49.50**. One standard deviation above the mean is therefore **90.39**. This shows the frequency of conditional mood verbs at CEFR-GA Level B2 to be significant.

None of the materials for mainstream Irish achieved statistical significance and statistical significance was only achieved two levels later than prescribed in the CEFR-GA B2 materials (cf. Table 25). The absence of statistical significance in materials for mainstream Irish coupled with the steady increase in frequency found in CEFR-GA materials indicates that materials creators perceive an element of the analysed structure to be overly complex for 3rd and 4th class in mainstream Irish as well as level A2 in CEFR-GA classes. This appears to be an example of the materials creators' interpreting the prescription of a language feature and adapting the materials to the needs of learners accordingly. It is noteworthy that this language feature is gradually introduced in linear manner in CEFR-GA materials. On the other hand the materials for mainstream Irish do not achieve statistical significance at any level, the highest frequency was found in **EduGA-Irish-Prim**, and the frequency found in **EduGA-Irish-PPJC** is more than one standard deviation below the mean. This pattern is erratic, to say the least.

[82] e.g. If that were to happen I would be very excited.
[83] e.g. If you won the lottery, what would you spend the money on?
[84] e.g. If you do this for me, then I will do that for you.
[85] e.g. If I could fly, I would go on holidays to France every weekend!
[86] e.g. If we get enough money then we can buy some sweets in the shop before going to the cinema.

5.1.5. Test 5: Conditional Mood Particles

Irish has two conditional particles "*má*" and "*dá*" [Translation: if], which are typically used in combination with a dependant form of an indicative verb in order to express conditionality. "Má" is typically combined with the present tense, but not exclusively, whereas "Dá" must be combined with a conditional-mood form of the verb. Examples 19-22 show typical examples of their uses.

(29)Má	thagann	siad	anseo			
	If	comes	they	here		
	"If they come here"					

(30)Má	itheann	sí	a	dinnéar		
	If	eats	she	her	dinner	
	"If she eats her dinner"					

(31)Dá	ndéanfá	an		obair	aréir	
If	done-COND-2SG			the	work	lastnight
"If you had have done the work last night"						

(32)Dá	gcaithfeá bheifeá	cótá tirim		báistí	
If	wear-COND-2SG beCOND-2SG dry			coat	rain
"If you had have worn a rain coat you would have been dry"					

The first of the tests completed here relates to the measurement of "má" and the second relates to the measurement of "dá". These conditional particles are first used as prescribed examples for the junior-cycle level of post-primary education (cf. An Roinn Oideachais agus Scileanna (ND)) and for CEFR-GA level B1 in the TEG syllabi (cf. Ionad na dTeangacha (ND-d)). Communicative functions that express feasibility or possibility are prescribed for 3rd and 4th classes in mainstream Irish lessons for learners in English-medium schools, and a specific class not specified for Irish-immersion and Gaeltacht primary schools: (An Roinn Oideachais agus Scileanna, 1999a), (p.22, 82).

Table 26: Frequency of conditional particles in Mainstream Irish materials

	Normalised Frequency in EduGA-Irish-PrJC per 50k words	Normalised Frequency in EduGA-Irish-PrSC per 50k words	Normalised Frequency in EduGA-Irish-PPJC per 50k words	Normalised Frequency in EduGA-Irish-PPSC per 50k words
Má	36	72	22	26
Dá	28	59	23	42

The standard deviation of the frequencies of "má" found in materials for Mainstream Irish is **19.71**, and the mean of the frequencies is **39**. One standard deviation above the mean is **58.71**. This language feature was first prescribed for EduGA-Irish-PrSC, therefore, the frequency of conditional-mood verbs at the level for which they were first prescribed is statistically significant in this context. The majority of uses at this level were combined with the substantive verb *bí* "to be", the majority of which were in the present or past tense.

The standard deviation of the frequencies of "dá" found in materials for Mainstream Irish **13.97**, and the mean of the frequencies is **38**. One standard deviation above the mean is **51.97**. Where the language feature, conditional-mood particle "dá", is first prescribed for EduGA-Irish-PrSC (3rd and 4th class) if the school is a *Gaelscoil* or in a *Gaeltacht* school (it is with these schools that the issues in Section 1.2 relate) or for EduGA-Irish-PPJC. The frequency recorded in EduGA-Irish-PrSC is statistically significant in this context.

Table 27: Frequency of conditional particles in subset of CEFR-GA materials

	CEFR-GA Level A1 per 50k words	CEFR-GA Level A2 per 50k words	CEFR-GA Level B1 per 50k words	CEFR-GA Level B2 per 50k words
Má	28	12	21	33
Dá	0.5 (e.g. less than once per 50k words)	2	9	15

The standard deviation associated with "má" in CEFR-GA materials is **7.89** and the mean score is **23.5**. One standard deviation above the mean is **31.39**, which means all frequencies other than B2 fall within one standard deviation from the mean. The standard deviation score for “dá” in CEFR-GA materials is **5.79** and the mean score is **6.63**. One standard deviation above the mean is therefore **12.42**, which means all frequencies other than B2 fall within one standard deviation from the mean. What do these results mean?

It is noteworthy that the frequencies of both particles found in materials for mainstream education drop by more than one standard deviation between EduGA-Irish-PrSC and EduGA-Irish-PPJC, on the other hand, the frequencies of both particles in CEFR-GA materials (other than “má” at level A1) build from one level to the next. This frequency difference between mainstream Irish and CEFR-GA materials is similar to the frequencies found in Test 4 (cf. Section 5.1.4). The differences in frequency may be the result of a semantic difference, where “má” is typically used in implicative or predictive senses and “dá” is typically used in predictive or speculative senses. Tests 3, 4, and 5 have indicated a similar phenomenon at play, particularly in CEFR-GA materials. Language features for learners who are not absolute beginners are being introduced to a limited extent, followed by a consistent increase in frequency at subsequent levels. As previously stated, this indicates materials creators are consistently having to interpret the extent to which learners ought to be exposed to prescribed language features.

5.1.6. Test 6: The Autonomous Forms of Verbs

In this test the frequency of autonomous verb forms are analysed, the frequencies analysed in this section have been extracted based on part-of-speech tags. Irish has an autonomous verb form that can be used to express either the passive voice or the impersonal pronouns, “one” or “you”. It can also be used when an agent is not specified, such as “No smoking” or “Don’t litter”. The autonomous form of verbs first appears as an example of prescribed communicative functions for the senior-cycle level of post-primary schools and for CEFR-GA level B1. The following is an example of how the autonomous verb is used.

(33) Rugadh agus tógadh John i Londain
Born-AUTO[87] and took-AUTO John in London

John was born and raised in London

(34) Déantar an obair gach lá
Done-AUTO the work every day

The work is done every day[88]

This test includes more groups of frequencies and their associated standard deviations than previous tests because each of the conjugations are different and tense- or mood-specific. The results are presented in Table 28 in a manner that reflects the different means and standard deviations that have been calculated for each tense, with the conditional mood included as a tense. This language feature was first used in examples for materials in EduGA-Irish-PPSC and for materials in EduGA-CEFR-B1.

[87] The Impersonal is specified in Leipzig Glossing Rules (e.g., URL: https://en.wikipedia.org/wiki/List_of_glossing_abbreviations) as IMPRS, however, autonomous forms do not appear to be catered for. The gloss AUTO is not used for another meaning or specification, therefore, I adopt it here.

[88] Use of the passive voice is considered a relatively advanced language feature in a number of languages. The English Grammar Profile describes acquiring the ability to use it as a huge leap in learners' ability, one that typically occurs at CEFR-EN B1, and goes on to describe a scenario where "they are able to use a wider variety of verbs with a greater variety of passive forms" (Cambridge University Press (2015)).

Table 28: Frequency of autonomous forms of verbs in Mainstream Irish materials

	Normalised Frequency in EduGA-Irish-PrJC per 50k words	Normalised Frequency in EduGA-Irish-PrSC per 50k words	Normalise Frequency in EduGA Irish- PPJC per 50k words	Normalised Frequency in EduGA-Irish-PPSC per 50k words	Standard Deviation (StDev)	Mean + StDev
Freq: Past Perfect	34	58	20	22	**15.11**	33.5 + 15.11 **48.61**
Freq: Present Continuous	13	42	18	53	**15.11**	34 + 15.11 **49.11**
Freq: Future Tense	4	10	10	14	**3.56**	9.5 + 3.56 **13.06**
Freq: Conditional Mood	7	7	12	20	**5.32**	11.5 + 5.32 **16.82**

The frequency of autonomous verbs at the level for which they were first prescribed is statistically significant in all cases other than the past perfect. The relatively inflated frequency of past tense autonomous verbs found in primary-level materials was manually checked and appears to be a feature of the story-telling genre that is found across a wide variety of the texts.

Table 29: Frequency of autonomous forms of verbs in CEFR-GA materials

	CEFR-GA Level A1 per 50k words	CEFR-GA Level A2 per 50k words	CEFR-GA Level B1 per 50k words	CEFR-GA Level B2 per 50k words	Standard Deviation	Mean + StDev
Freq: Past Perfect	4	4	21	71	**27.45**	25 + 27.45 **52.45**
Freq: Present Continuous	140	12	39	123	**54.19**	78.5 + 54.19 **132.69**
Freq: Future Tense	10	2	3	40	**15.45**	13.75 + 15.45 **29.2**
Freq: Conditional Mood	0	1	1	25	**10.53**	6.75 + 10.53 **17.28**

The frequency of autonomous verbs in mainstream education was statistically significant for the autonomous verb in the present tense, the future tense, and the conditional mood. The inflated frequencies, relative to other tenses, were a feature of the story-telling content that was a feature of these materials.

Gloss (1) in this section (Section 5.1.6) is an example of the type of sentence found in these materials. None of the CEFR-GA materials showed statistical significance at the level for which they were first prescribed, however, significance was consistently achieved at CEFR-GA level B2, indicating they are gradually introduced in materials for CEFR-GA courses. The inflated frequency of present-continuous tense autonomous verbs found in materials level A1 was manually checked and, while the vast majority were not included in the body of any lesson tasks, it was also repeatedly used as part of the introduction and/or explanation that's built into lessons. This introduction was expressly written to direct learners, therefore, it is an integral part of the materials. This relatively high frequency, compared to present continuous frequencies at other levels, has raised the mean and standard deviation values to a level that causes frequencies for level B2 to be below the threshold for statistical significance. If the frequency of present continuous autonomous had have been 12, for example, one standard deviation above the mean would have been 92.01 which would have meant the frequency found at B2 achieved statistical significance. While this comment is speculative it has not been included without reason; this verb form has been purposely prescribed for learners of intermediate to upper-intermediate ability for Irish.

5.1.7. *Test 7: Indirect Relative Clauses*

Relative clauses are widely accepted to be markers of complex language (cf. Diessel (2004)). A frequency analysis of this type of structure was conducted based around two relative particles "a" and "ar", which are followed by an appropriate dependent verb form. These particles can, however, mark relative-clause constructions and "a" is a homonym for a number of other function words in Irish. Therefore, part-of-speech tags were used to extract the frequencies presented in this test's results. Examples of indirect relative clauses did not appear in reference documents for mainstream education, however, it can be inferred that it is required at the highest levels when communicative functions such as *cur síos a lorg is a chur in iúl* "seeking and articulating a description" and *míniú a lorg is a thabhairt* "seeking and giving an explanation" are specified. The following tasks have been specified for learners who would be using EduGA-Irish-PPSC, namely, the ability to understand pieces of written literature, journalism, and literary criticism (An Roinn Oideachais agus Scileanna, 2010a). The ability to lengthen sentences and join short sentences with a number of conjunction types is prescribed for 5th and 6th classes (An Roinn Oideachais agus Scileanna, 1999a). Combining this fact with the stated aim of building upon these abilities at post-primary level makes it highly likely relative-clause constructions are both needed to perform communicative functions at this level and required at this level.

As noted in Section 3.1.3 Walsh (2007) finds the majority of relative clause errors occur when the indirect relative clause was required, after which it is stated that the indirect relative clause is considered one of the most complex and difficult constructions for learners of Irish (Mac Murchaidh, 2002). Examples of the indirect relative clause have been included in CEFR-GA syllabus for B2. The following is an example of the former of an indirect-relative clause in the present tense:

(33)Sin an fear a n-oibríonn mac leis sa scoil Tha

Table 30: Frequencies of relative-clause particles in Mainstream Irish materials

Normalised Frequency in EduGA-Irish-Prim per 50k words	Normalised Frequency in EduGA-Irish-Prim per 50k words	Normalised Frequency in EduGA-Irish-PPJC per 50k words	Normalised Frequency in EduGA-Irish-PPSC per 50k words
50	61	84	93

The standard deviation of the frequencies of found in materials for Mainstream Irish **17.25**, and the mean of the frequencies is **72**. One standard deviation above the mean is **89.25**. Therefore, the frequency of relative- clause particles are statistically significant in materials for the most advanced level included.

Table 31: Frequencies of relative-clause particles in CEFR-GA materials

CEFR-GA Level A1 per 50k words	CEFR-GA Level A2 per 50k words	CEFR-GA Level B1 per 50k words	CEFR-GA Level B2 per 50k words
50	56	87	136

The standard deviation of the frequencies of found in materials for Mainstream Irish **36.45**, and the mean of the frequencies is **96.25**. One standard deviation above the mean is **132.7**. Therefore, the frequency of relative-clause particles at the level for which they were first prescribed is statistically significant in this context.

The frequency profile of this language function suggests the feature is introduced gradually. The frequency of this language feature at lower is the result of a formulaic communicative function that is prescribed for beginners, namely, when asking where something is or where a person lives. For example:

(34)Cén	áit	**a**	**bhfuil** cónaí	ort?		
	Which+the	place	[relative particle]	is	living	on+you?
	Where do you live?					

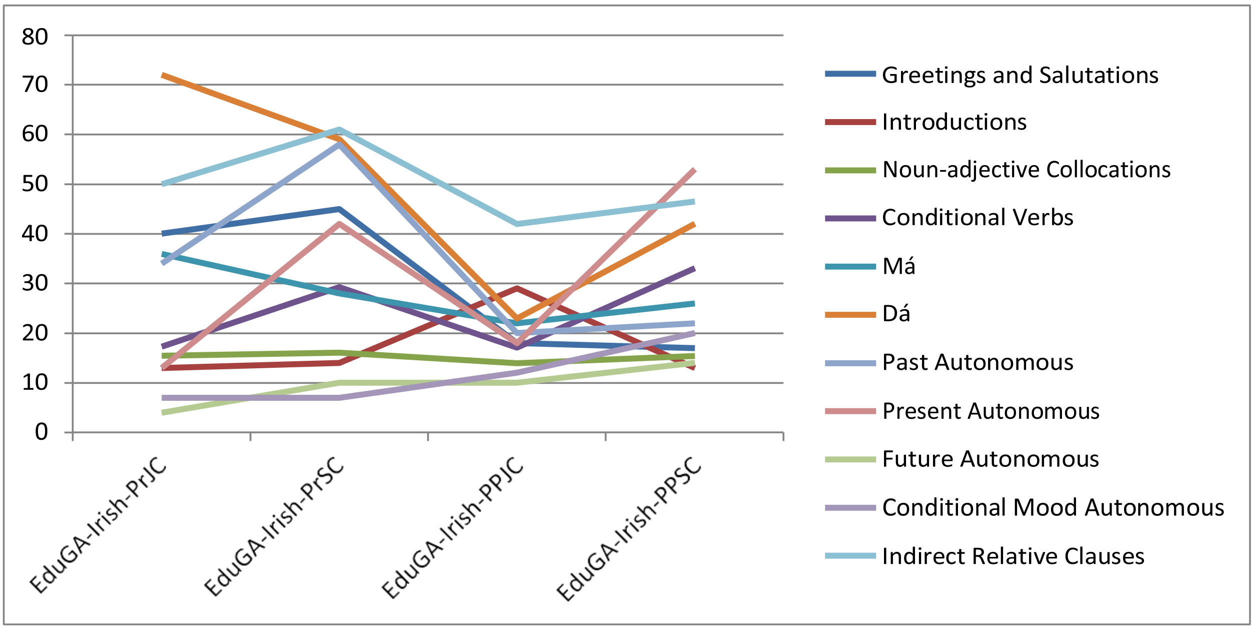
The frequency of relative clauses increases from one level to the next in the CEFR materials and in mainstream materials, with the frequency achieving statistical significance at the most advanced level of the sub-corpora for both mainstream education and CEFR-GA.

5.1.8. Overall Conclusions from Tests

Statistical significance was achieved by less than half of the language features analysed, with Mainstream Education faring particularly badly in this regard. However, where some language features did not achieve statistical significance at the level for which they were first prescribed their frequencies did continue to grow. This continued increase in frequency was indicative of a language feature that was partially introduced at first, and then elaborated at subsequent levels in terms of frequency. This elaboration may have included a broader range of vocabulary, however, it was not within the scope of the present research to test this. This approach is consistent with the building-block approach to language learning prescribed for both Mainstream Education and CEFR-GA (An Roinn Oideachais agus Scileanna (1999a), An Roinn Oideachais agus Scileanna (ND), An Roinn Oideachais agus Scileanna (2010a), Council of Europe (2001)). The language features that did achieve statistical significance were either lexically specified in the reference documents or the language feature had a limited vocabulary or set of constructions.

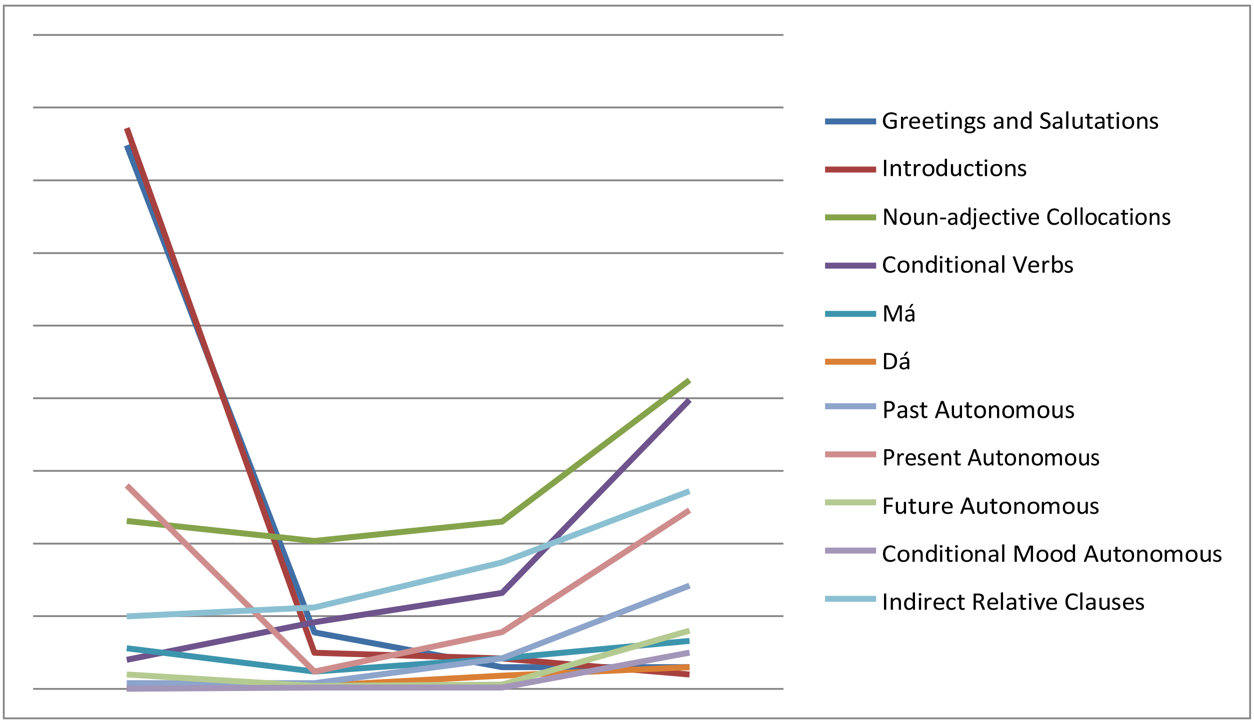
Figure 7 represents the frequencies of prescribed language features in materials for Mainstream Education from one level to the next. These frequencies have been graphed in order to highlight trends in frequency change. In order to remove the unneccesary white space that would have divided the results of Tests 3 (noun-adj collocations) and 4 (conditonal-mood verbs) from all other results two decimal points were added to those results for this representative. Figure 8 represents the corresponding features in CEFR-GA sub- corpora. Figure 7 highlights the way the frequencies of a number of language features rise and fall from one sub-corpus to the next. While this is not necessarily a problem, the fact that it consistently occurs in EduGA- Irish-PPJC likely relates to the need for improved integration between primary and post-primary levels, as outlined in Ní Mhaonaigh (2013). Given the issues impacting Irish, presented Section 1.2, where some materials were considered overly simple and others either overly complex or overly difficult, these results appear to confirm the inconsistency that was identified in Mac Donnacha et al. (2004), Mac Donnacha (2005), de Brún (2007), and Ó Muircheartaigh (2009).

Figure 7: Frequency of Prescribed Language Features in Materials for Mainstream Education



These results are particularly noteworthy because of the frequency trends found in CEFR-GA materials (cf. Figure 8), where the frequencies of the same prescribed language features rise or fall in a consistent manner and as one would intuitively expect based on their prescription. The frequency of language features for absolute beginners, such as *Greetings and Salutations* or *Introductions* in this case, drop in frequency at higher grade levels. These frequencies are indicative of the amount of repetition that is done at this level and of the limited ability of learners. More specifically, learners have not yet learned how to develop a conversation much further than greetings and introductions. Other than the anomalous present-tense autonomous verbs, the remaining prescribed language features show a steady increase in frequency from one level of ability to the next. This is therefore being taken to indicate more interconnectivity between the contents of CEFR-GA materials from one level to the next when compared to materials for Mainstream Irish.

Figure 8: Frequency of Prescribed Language Features in Materials for CEFR-GA courses



This is significant because it seems intuitive that consistently increasing a learner's exposure to examples of a language feature would lead to better learning outcomes than irregular exposure to the same language features. This steady increase in frequency agrees with the way reference documents describe the learning process; as a building-block process, where each level of ability builds upon that which came before it.

Tests 4, Test 5, and Test 6 show little frequency change from one level to the next, suggesting the frequency of the language functions tested are less criterial to a specific level than other language functions. However, these language functions appear to apply to a more general grouping of levels of ability; similar to the Basic, Independent, and Proficient User categories given on page 23 of Council of Europe (2001).

In Test 7 (Indirect Relative Clauses) the frequency of the language features gradually increases from one level to the next with some examples in materials for beginners, despite this language feature reportedly being among the most complex in the language. However, other prescribed features have an extremely low or null frequency at beginner levels, Tests 5 and 6 for example. This prompted manual inspection of the examples from beginner levels or lower grades. The result of Test 7 aligns with the findings in Diessel

(2004)(presented in Section 3.1.3) in two ways. The forms of the relative clause that are introduced early on as essentially set phrases and the when the frequency increases with each level of ability the vocabulary range also increases. At the lowest levels of ability, almost all examples are the present-tense dependent form of the substantive verb "*bhfuil*", which is used in the frequently-asked question *Cén áit a bhfuil cónaí ort*, or, "Where do you live?" (Gloss 22, Section 5.1.7).

The frequency of noun-adjective collocations was surprisingly similar at all Mainstream Irish levels (Primary: 1,585, Junior-cycle: 1,395, Senior-cycle: 1,539), and at CEFR-GA levels A1, A2, and B1 (A1: 1,157, A2: 1,017, B1: 1,152) with a marked increase at CEFR-GA level B2 where 2,126 examples were found per 50,000 words. Noun collocations have received considerable attention in studies in the domain of education but on other languages (Biber (1988), Biber, Gray, and Staples (2016), Chan and Liou (2005), Jafarpour et al. (2013), Sun and Wang (2003), Uçar and Yükselir (2015)). Therefore, it is a language feature that appears to require further analysis, in order to explain the frequencies found across levels, and a number of methodologies appear to be in place depending on research aims.

Noticeably fewer examples of language features were directly related with topics and communicative functions across all levels of ability in the reference documents for Mainstream Irish ((An Roinn Oideachais agus Scileanna, 1999a), (An Roinn Oideachais agus Scileanna, ND), (An Roinn Oideachais agus Scileanna, 2010a)) when compared to the reference documents for CEFR-GA courses (cf. Ionad na dTeangacha (ND-a), Ionad na dTeangacha (ND-c), Ionad na dTeangacha (ND-d), Ionad na dTeangacha (ND-e)). This may have contributed to some of the results and to the issue with materials, because materials creators necessarily have to infer examples to a greater extent when creating materials for Mainstream Irish.

Finally, the highest frequency of prescribed language features in Mainstream Irish was primarily aligned with frequencies in CEFR-GA level B1, with the frequencies found at CEFR-GA level B2 surpassing those found in post-primary level senior-cycle materials, namely: in Section 5.1.3, Test 3 (noun-adjective collocations); in Section 5.1.4, Test 4 (Conditional-mood verbs), in Section 5.1.6, Test 6 (Autonomous forms of verbs); and in Section 5.1.7, Test 7 (indirect relative clauses).

5.2. An analysis of word and sentence length

In this section an analysis of average word length and average sentence length were conducted. The results clearly show that average word length is not a viable methodology for Irish-language complexity studies.

Average sentence length, on the other hand, provides results that are more closely aligned with levels of education.

5.2.1. Average Word Length.

Section 3.2 has explored a number of studies where the complexity of a document and found average word length is typically used in association with other metrics. In some cases (namely, Saggion et al. (2012) and Rayner and Duffy (1986) average word length for a sentence is calculated as follows:

1. "The quick brown fox jumps over the lazy dog"[89], 38 letters ÷ 9 words = 3.888
2. "Scaoileann sé Binncheol amach as an gcás ansin."[90] , 39 letters ÷ 8 words = 4.875

However, word length was also counted according to the number of syllables in a word in a number of other studies. In the *Coh-Metrix* Tests developed in Graesser et al. (2004) and in the *F-K Test* developed in Flesch (1948), Kincaid et al. (1975) (for details see Flesch (1981)) the number of syllables in words are used with a large number of other variables to calculate a document's complexity score. A similar approach was adopted in two other tests, namely the Gunning Fog Index developed in Fry (1968) and the SMOG[91] Test developed in McLaughlin (1969), however, the latter two tests only use words of three syllables or more in their calculations and the frequently omit proper nouns, "familiar jargon" (Fry, 1968), and compound words.

3. "The quick brown fox jumps over the lazy dog", 11 syllables ÷ 9 words = 1.222[92]
4. "Scaoileann sé Binncheol amach as an gcás ansin.", 12 syllables ÷ 8 words = 1.5.

A syllabification feature was developed by Eric Zoerner for the ABAIR speech synthesis projec[9]t[3], however, it was not within the scope of the present research to test and use the syllabification feature.

The results presented in Table 32 are based on the number of letters per word and on all tokens in each of the listed sub-corpora. If this metric is reliable and the complexity of educational materials was appropriately assigned, then a gradual increase in average word length from one level of education to the next is expected.

[89] An English sentence that was invented because it includes every letter in the English, it is frequently used as an example of English.
[90] A sample sentence from a key scene in a play that has featured on the Leaving Certificate syllabus, for several decades now. The play is called *An Lasair Choille,* and it was written by Caitlín Maude and Mícheál Ó hAirtnéide. [91] "Simple Measure of Gobbledygook"
[92] The number of syllables may change depending on dialect, both syllable counts are based on neutral dialects from Ireland.
[93] URL: https://www.abair.tcd.ie/?page=people&lang=gle (Last accessed: 02/06/2019)

Table 32: Average Word-Length Scores for Subsets of the EduGA Corpus

Subset of the EduGA Corpus	Sub-corpus Codename	Average Word Length
Primary-level junior Irish	EduGA-Irish-Prim	4.29
Post-primary level junior-cycle Irish	EduGA-Irish-PPJC	**4.27**
Post-primary level senior-cycle Irish	EduGA-Irish-PPSC	4.50
Primary-level Science	EduGA-Sci-Prim	3.99
Post-primary level junior-cycle Science	EduGA-Sci-PPJC	5.75
Post-primary level senior-cycle Science	EduGA-Sci-PPLC	**5.30**
Post-primary level junior-cycle Business Studies	EduGA-BusS-JC	5.08
Post-primary level senior-cycle Business Studies	EduGA-BusS-LC	**4.94**
Primary-level History	EduGA-Hist-Prim	4.69
Post-primary level junior-cycle History	EduGA-Hist-PPJC	4.72
Post-primary level senior-cycle History	EduGA-Hist-PPLC	**3.96**
CEFR-GA, A1	EduGA-CEFR-A1	4.01
CEFR-GA, A2	EduGA-CEFR-A2	4.68
CEFR-GA, B1	EduGA-CEFR-B1	**4.63**
CEFR-GA, B2	EduGA-CEFR-B2	**4.61**
CEFR-GA, C1	EduGA-CEFR-C1	4.81
CEFR-GA, C2	EduGA-CEFR-C2	4.84

Across all subjects the average word length in at least one level is lower than the average word length in at least one preceding (or lower) level. This phenomenon has been marked in Table 32 with bold font. These results can be interpreted in two ways.

The results may validate the findings of Mac Donnacha (2005) and others regarding materials that are either overly simple or overly complex, aligning with the results presented in Section 5.1 to a certain degree, where the frequency of prescribed language features were found to be more inconsistent in mainstream education than the frequencies found in CEFR-GA.

The results may also show this methodology does not transfer from other languages without changes. This may indicate the measure is not suitable for studies of Irish-language complexity.

I now present some examples that illustrate why I believe these results show the methodology is unreliable for Irish. Irish verbs are very regular in comparison to English verbs, therefore, the conjugational changes described here apply to almost every Irish verb – including some irregular verbs. The present tense suffix can add as many as four letters to the root of the verb. The same is true of the future tense and conditional mood, where up to five letters can be added to the root of the conjugated verb. Compare plant with its translation cuir in the present tense:

(33)*Cuireann* *sé* *síolta*
Plants he seeds
He plants seeds

The majority of consonant-initial Irish verbs can also gain a one-letter lenition or eclipse at the beginning of the word, when used in the interrogative, negative, and conditional moods. Vowel-initial verb forms cannot be lenited, but are eclipsed in a more limited set of examples from the three named moods than the consonant-initial verbs. Plural forms can be identical to the nominative singular form or up to six letters longer, for example:

(34)*Páirc* *agus* *a* *bpáirceanna*
Park and their parks
Park and their parks

Irish morphology dictates that a noun has more unique forms associated with it than English and most modern Western-European languages (see examples in Table 6: Inflected noun forms in Irish and English). These morphological changes that are based on casing are typically grouped into five declensions in Irish. Each declension acts as a guide to how the genitive and plural forms of its member nouns are formed.
Therefore, second-declension nouns that share certain characteristics with páirc will also receive a five-letter suffix in order to make them plural. When combined with the present-tense verb cuireann, we can end up with a relatively simple sentence that contains two long but simple words.

(35) *Cuireann sé síolta sna páirceanna*

Plant he seeds in:the parks

He planted seeds in the parks

Compare this to the past-tense version of the same sentence, where seeds were only planted in one park:

(35) Chuir sé síol sa pháirc

Planted he seed in:the park

He planted a seed in the park

The present-tense Irish sentence has an average word length of 5.8 letters, and its English translation has an average word length of 4 letters. The past-tense Irish sentence has an average word length of 3.8 letters, and its English translation 3.28 letters. The shift in tense and number caused a difference of 2 between the two Irish sentences, and a difference of 0.72 between the two English sentences. While these examples have been engineered, they plausibly reflect the difference between a sentence from a piece of literature, when written mostly in the past tense, and a descriptive current-affairs sentence that is written in the present tense.

The focus of this research is on developing a set of metrics in order to measure ways in which educational materials become more complex. On the one hand, this metric has identified the inconsistencies in complexity that were presented as issues impacting Irish-language learning in Section 1.2. However, the fact that no sub-corpus (cf. Table 32) showed gradual increases in average word lengths suggests the metric needs to be adapted before it can work for Irish-language data. An adaptation would need to take the language's morphology into account so that present-tense verbs and nominative-plural nouns would not significantly skew the results of an average word-length analysis and of a syllable analysis.

This metric appears to be unreliable in the context of Irish-language complexity studies, highlighting the need for additional fundamental research of this sort on Irish-language texts.

5.2.2. Average Sentence Length

Section 3.2 presented a number of studies where the average number of words in sentences was used to calculate the complexity of a document. Sentence length is generally considered to be a proxy for syntactic complexity in readability tests and complexity analyses (Berendes & Vajjala, 2018). The software used to implement the analysis has also been discussed in Section 3.2, therefore, this section focuses on the results and discussion.

Having the ability to construct and understand longer sentences is referred to in syllabi and curricula for mainstream education as a marker of developing language ability (An Roinn Oideachais agus Scileanna (2010a), An Roinn Oideachais agus Scileanna (ND), An Roinn Oideachais agus Scileanna (1999a)). Length of sentence is also frequently used in a number of the algorithms that are used as readability tests, for example: the *Flesch-Kincaid* (F-K) *readability tests* that were introduced in Section 5.2.1 (Flesch (1948), Kincaid et al. (1975), and Flesch (1981)), as well as the *Coh-Metrix* readability test (Graesser et al., 2004), and the *Fry Readability Test* developed in Fry (1968) also factors the average number of words per sentence into its calculations. In the SMOG (Simple Measure of Gobbledygook) tests developed in McLaughlin (1969) a similar approach is adopted, the number of polysyllabic words per sentence is used in the calculations.

A number of other studies of complexity also consider longer sentences to be inherently more complex. For example, Tomasello and Diessel (2000) and study the emergence of complex features in the L1 among children. While their study identifies a number of other factors that contribute to the complexity of a construction, the complexity they observed goes from joining multiple set phrases in order to achieve a communicative goal to the use of relative clauses instead of two single-clause sentences. Numerous other studies continue to find that domains such as academic writing contain longer sentences than media articles (cf. Herrero-Zorita and Moreno Sandoval (2016)). It is clear that this metric has been applied to a wide range of complexity analyses, with consistently reliable results yielded.

As noted in Section 4.3.2 sentences have been tagged in EduGA. The output takes the following XML format:

```
<BigFile>

<s>Rith siad an rás.<s>

<s>The quick brown fox jumps over the lazy dog.</s>

<s>Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor
incididunt ut labore et dolore magna aliqua.</s>

</BigFile>
```

The sub-corpora used for this analysis are detailed in Table 33. Where sub-corpora from Table 13 have been omitted from this analysis (cf. Table 33) it is because too many manual corrections were required. It was discovered that this was not the fault of the sentence tokenization software, but was simply the result of sentence boundaries relying on graphical features in textbooks rather than punctuation. These graphical features included coloured boxes or speech bubbles, as well as sentences that were included in tables.

Table 33: Results and Sub-corpora used for Sentence Analysis

Sub-corpus	Number of Words in Sub-corpus	Number of Sentences	Average Number of Words per Sentence
EduGA-Irish-Prim	219,820	24,411	9
EduGA-Irish-PPJC	256,760	23,616	11
EduGA-Irish-PPSC	326,869	18,238	18
EduGA-Irish-UniU	1,231,550	20,168	18
EduGA-CEFR-A1	349,702	49,546	7
EduGA-CEFR-A2	302,660	40,671	8
EduGA-CEFR-B1	352,863	39,631	9
EduGA-CEFR-B2	356,863	33,436	11
EduGA-CEFR-C1	656,127	50,981	13
EduGA-CEFR-C2	***217,079***	***18,444***	***12***
EduGA-Hist-Prim	91,889	8,187	14
EduGA-Hist-PPJC	66,270	9,334	14
EduGA-Hist-PPSC	***26,565***	***2,009***	***13***
EduGA-Sci-Prim	134,629	9,921	13
EduGA-Sci-PPJC	181,956	13,773	13
EduGA-Sci-PPSC	21,143	1,645	13
EduGA-BusS-PPJC	57,741	8,853	7
EduGA-BusS-PPSC	133,331	9,349	14

The results show that this metric best distinguishes beginner-level materials for language learning courses from materials at advanced levels. Similar to the results presented in Section 5.1, this metric shows the CEFR-GA materials to be a more cohesive unit than the materials for mainstream Irish. The average number of words per sentence in mainstream Irish jumps from 11 at junior cycle to 18 at senior cycle and at university level. As evidenced by the other results in Table 33 this difference is substantial, especially when compared to CEFR-GA materials where the average length of sentences only increased by either one or two words for each level. What is not clear from the results is whether sentences in junior-cycle materials should contain more words or sentences in senior-cycle materials should contain fewer words. Sentences in CEFR- GA materials for levels C1-C2 were 30-33% shorter on average than sentences in senior-cycle materials, suggesting the sentences in senior-cycle materials may be too long. On the face of it, moving to an average sentence length of 13-14 words would also align the syntactic complexity of senior-cycle Irish with the syntactic complexity of History, Science, and Business Studies of the same level. However the History, Science, and Business Studies materials are probably intended to be more informative than the Irish- language materials. The Irish-language materials are probably intended to be more discursive than the aforementioned Irish-medium materials, therefore containing longer sentences. As a result, another comparison was sought. In order to begin to answer this question and contextualize the results 200 sentences were collected from two non-education sources: the first one hundred from a news and current affairs publication that is analogous to a newspaper called *Tuairisc*[94]; the second one hundred from a magazine called *Nós* that publishes cultural, sporting, technology, lifestyle, and travel articles[95]. These sentences were sampled from ten different articles in each publication type.

Table 34: Average Sentence Length in Two Irish-language Media Publications

Average Sentence Length in Tuairisc (news and current affairs publication)	**Average Sentence Length in Nós (magazine)**
24 words	22 words

The average length of sentences in media articles and language-learning materials is significantly different, suggesting learners may find it difficult to handle the syntactic complexity of either type of media publication. Given the goals of mainstream language learning are both to communicate with the language community and to facilitate academic ability, the only conclusion to be drawn from these results is that the average sentence in junior-cycle materials ought to be longer than 11 words rather than the average sentence in senior-cycle materials being shorter than 18 words. These results suggest the syntactic complexity of CEFR-GA materials may be overly similar from one level to the next, and that if changes were to be made then they ought to focus on making the difference between levels more pronounced than they are currently. Table 35 illustrates this suggested change.

[94] URL: www.tuairisc.ie (last accessed 15/06/2019)
[95] URL: https://nos.ie/ (last accessed 15/06/2019)

Table 35: Suggested Changes to Average Sentence Length

CEFR-GA Level and Average Sentence Length	Suggested Change to Average Sentence Length
CEFR-GA level A1: 7	CEFR-GA level A1: 8
CEFR-GA level A2: 8	CEFR-GA level A2: 10
CEFR-GA level B1: 9	CEFR-GA level B1: 12
CEFR-GA level B2: 11	CEFR-GA level B2: 15
CEFR-GA level C1: 13	CEFR-GA level C1: 18
CEFR-GA level C2: 12	CEFR-GA level C2: 21

Another feature of the results was the ceiling effect (or a cessation of increases in frequency) that was found in the averages at higher levels of ability, namely senior-cycle Irish and undergraduate Irish, and CEFR-GA level C1 and C2. This may have occurred because different elements of the language complexity change. For example, the focus of the materials may have been on increasing the semantic complexity between these levels. This may have been the cause behind the plateau in frequencies at higher levels in language-learning materials and across numerous levels in Irish-medium materials. Testing this would require a complexity analysis of the content of educational materials.

5.3. The Development and Implementation of a Term-based Complexity Analyses

Many of the issues with the language in educational materials that were identified in the literature relate to words that are unknown or difficult to understand. Examples of terms causing issues or difficulties with educational materials have featured throughout the literature (cf. Mac Donnacha (2005), Pétervári et al. (2014), de Brún (2007), and Walsh (2007)) (See Sections 1.2 and 3.1.3). Terminology has also been prescribed across a number of reference documents (cf. Sections 2.1 and 2.2) either directly as terminology or as specialized or topic-specific vocabulary, which typically equates to terminology. Therefore the aim here is to analyse term usage with a view to developing a term-based complexity measure that addresses these issues. The core elements of this analysis have been specified in Section 3.4.3, and are summarised here before the results are presented.

This analysis combines the use of a database of terms with statistical measures in order to analyse the semantic complexity of lexical items in sub-corpora in EduGA. More specifically, the statistical measures combine lemma frequencies from a corpus of general Irish with a score that has been calculated using the statistical method known as TF-IDF. This analysis has been developed during the course of this research, therefore, key steps in the development are outlined in this section.

Before moving on, some necessary clarifications and definitions are required regarding the content. Firstly, the distinction made between ***complexity*** and ***difficulty*** in the **List of Abbreviations and Key Concepts** is particularly important to the present analysis, where complexity was defined as an absolute element of a language feature and difficulty as a feature that is unique to each individual. Complexity is typically defined as Rescher (1998) defines it; "a matter of the number and variety of an item's constituent elements and of the elaborateness of their inter-relational structure". Terms are therefore complex because of the amount of information to which they refer, as well as the inter-relationships between those pieces of information.

Differentiating the complexity of terms, if possible, would be a study unto itself, however, we can say that terms are more complex than non-terminological words. The complexity of terms is therefore treated as a constant in the present research, one term is considered to be as complex as the next. Terms are difficult because of their complexity and because it can be difficult to acquire them resulting from low frequencies relative to non-terminological words.

Semantic complexity can relate here to the number of elements, and inter-relationships between those elements, associated with the concepts to which we refer by using terms. For example: when a biologist uses the term "photosynthesis" (s)he is referring to a process used by a variety of flora whereby light energy is used to combine (or synthesise) carbon dioxide and water; thus creating the carbohydrates that are needed to grow, reproduce, or complete other activities. This is an example of a specific meaning being associated with a specific term. An example of a non-specific meaning is the use of "process" at the beginning of the definition of photosynthesis. In this non-specific sense, process refers to a series of actions taken in order to achieve a particular result. As a result, the word "process" can be used in this non-specific manner in almost any domain of knowledge. In this sense each term carries a certain semantic complexity, but the difficulty associated with each term changes from one interlocutor to the next. Terms are typically acquired by both first- and second-language learners later than non-terminological words because they are both complex and difficult.

5.3.1. Sub-corpora Analysed

Table 36: Selected Sub-Corpora and Token Frequencies for Term-based Analysis

Subject	Sub-corpus	Code name	Number of Docs	Number of Tokens	Total Tokens
Irish	Irish for primary level	**EduGA-Irish-Prim**	173	219,820	803,449
	Irish for Junior Cycle Post-primary level	**EduGA-Irish-PPJC**	10	256,760	
	Irish for Senior Cycle Post-primary level	**EduGA-Irish-PPSC**	76	326,869	
Science	Science for primary level	**EduGA-Sci-Prim**	18	134,629	337,728
	Science for Junior Cycle Post-primary level	**EduGA-Sci-PPJC**	8	181,956	
	Science for Senior Cycle post-primary level	**EduGA-Sci-PPSC**	2	21,143	

The sub-corpora used during the development of this methodology are listed in Table 36. Materials for learning Irish are being prioritised, because they are at the core of the Irish-medium and Irish-language education system. Science materials are being included because scientific materials will typically be predominantly factual, and likely contain a larger frequency of terms. Another factor considered when sub- corpora were being selected was the time it took to score the corpora and then conduct the analysis. Each of the scoring stages were essentially additional annotation stages.

5.3.2. Identifying Terminology: Development of a Methodology

At first the number of terms in each sub-corpus was sought, and the percentage of lemmas[96] that were matched with terms. Section 3.4.3 presents the reasoning behind using lemma for the lexical matching process and why the matching process was done with the lemmas rather than the token. In summary, lexical matching is an efficient and useful one-for-one matching process. By matching words one for one it will only retrieve exact matches, rather than similar words. This will however cause it to miss any inflected forms of words that may be wanted. The results in Table 37 show that virtually the same percentage of lemmas matched as terms in each of the sub-corpora. From my experience working on terminology projects and my teaching experience, there are two issues with the figure of 39% terminology in educational materials. The first is that if four words in ten are considered to be terminological, this sounds more like a technical manual or an encyclopaedia. The second is that the same percentage was found across all grade levels, where one would expect more common words and fewer terms in materials for primary level when compared to senior cycle at post-primary level.

Table 37: Summary of results for lexical matching using the Téarma database

EduGA Subcorpus	Number of Lemmas in Subcorpus	Number of Lexical Matches in Téarma?	Percentage Téarma Matches
EduGA-Irish-Prim	166,021	64,689	38.96%
EduGA-Irish-PPJC	233,790	91,870	39.29%
EduGA-Irish-PPSC	27,550	10,664	38.70%

The reason for this result was “overmatching”, an outcome frequently arising from lexical matching. Overmatching in this case refers to unnecessary or inappropriate matches occurring as well as the appropriate matches. The software’s inability to discriminate between the different senses, terminological or non-terminological, associated with lemmas is the core issue. Manually checking all matched terms would be unrealistic, and unnecessary. Approximately 20% of the matches were manually checked in order to ascertain whether the matches that had been made were terminological or otherwise. The following observations were made:

Category 1 matches include the result we are hoping for, where the term and domain were unambiguously matched. This category also includes terms that were matched, but where the lexically matched item could refer to one of a number of parts of speech. For example:

Table 38: Examples of Matches with Ambiguous Parts of Speech

Irish Term	Concept #	Domain of Knowledge	POS	English Translation
grian	1	Ealaín > Griangrafadóireacht [*Art > Photography*]	verb	solarize
	2	Muirí > Seoltóireacht > Loingseoireacht > Cúinsí Farraige, Réada Fisiciúla [*Nautical » Sailing » Navigation » Sea Conditions, Physical Objects*] Eolaíochtaí Nádúrtha & Matamaitic > Réalteolaíocht [*Natural Sciences & Mathematics > Astronomy*]	noun	sun
grian-	3	Eolaíochtaí Nádúrtha & Matamaitic > Réalteolaíocht [*Natural Sciences & Mathematics > Astronomy*]	prefix	solar- (prefix) solar (adjecti ve)

[96] As noted in Section 3.4.3, terms were matched with lemmas so that variant forms didn't cause the matching process to miss terms because a variant to the lemma, or root form, or a variant form was used.

These examples can be disambiguated using parts of speech or cluster analysis. The final observation in this category relates to a term that was accurately matched, but where the domain of knowledge to which the term belongs is ambiguous. For example:

Table 39: Examples of Matches with Ambiguous Domains of Knowledge

Irish Term	Concept #	Domain of Knowledge	POS	English Translation
méadaigh	1	Matamaitic > Céimseata > Céimseata Chomhordanáideach [*Mathematics > Geometry > Coordinate Geometry*]	verb	increase
	2	Tíreolaíocht [*Geography*]	verb	increase
	3	Rialta [*Government*]	verb	gain (increase)
	4	Gnó [*Business*]	verb	extend (of influence)
	5	Eolaíochtaí Nádúrtha & Matamaitic [*Natural Sciences & Mathematics*] Airgeadas [*Finance*]	verb	augment
	6	Eolaíochtaí Nádúrtha & Matamaitic [*Natural Sciences & Mathematics*]	verb	enlarge
	7	Ríomhairí, Ríomheolaíocht [*Computers, Computer Science*]	verb	increase (of productivity, etc.)
	8	Ríomhairí, Ríomheolaíocht [*Computers, Computer Science*]	verb	blow up (increase)

Category 2 matches include the result we are not hoping for, where a hononym of a term has been used to express its non-terminological sense. These matches relate to the likes of "throw" as a general-purpose word, for propelling an item through the air, is what was generally meant in EduGA materials. Specific technical meanings were not found associated

with Judo to describe a method of legally lifting and/or unbalancing an opponent in order to gain an advantage. Another example of this is "light", which sometimes refers to a general brightness or type of illumination and in other more technical contexts refers to a form of electromagnetic radiation. Finally, an Irish example is *iompar* ("to carry", in its general sense), which may be used in a technical or non-technical sense in educational materials, therefore falling into Category 2 (cf. Table 40).

Table 40: Examples of Matches that are Ambiguously Terminological

Irish Term	Concept #	Domain of Knowledge	POS	English Translation
iompar	**1**	*Bitheolaíocht* [Biology]	noun	Gestation
	2	*Na Meáin ›Craoltóireacht* [The Media › Broadcasting]	noun	transportation
	3	*Oideachas* [Education]; *Polaitíocht* [Politics]; *Eolaíocht Shóisialta › Socheolaíocht* [Social Science › Sociology]	noun	conduct
	4	*Geolaíocht* [Geology]; *Innealtóireacht Shibhialta › Bóithre* [Civil Engineering › Roads]	noun	transport
	5	*Sábháilteacht ›Sábháilteacht Uisce* [Safety ›Water Safety]	noun	Carry

The homonyms of terms matched in Category 2 refer to at least 2 concepts, one terminological and one general. Therefore, one way of controlling for Category 2 matches may be to factor in the terminological polysemy of terms, in other words, the number of terminological concepts a term refers to. This value is used here to establish whether or not highly polysemous terms have a higher probability of having a homonym that refers to a non-terminological concept.

5.3.3. Terminological Polysemy

Rayner and Duffy (1986) reinforce the cause for considering polysemy in this context, where they found that polysemous words were more cognitively complex via eye-fixation tests. The Téarma Database was used as a source for the terminological productivity of terms, by this I mean, where the Téarma database tells us a term can refer to ten distinct concepts then this term is attributed a value of ten, on the other hand, where a term refers to a single concept then this term is attributed a value of one. The terminological productivity of each term according to the Téarma database was added to the results of the lexical matching process as an additional column of metadata. In a language-learning situation the frequency of a word in a corpus of general language may be considered a proxy for the likely difficulty of a word. By this I mean, a learner is less likely to be familiar with words that are used infrequently when compared to frequently used words. The next step needed is to establish a way of categorizing the words used in materials in a way that relates to the context of general Irish. I believe this to be the case because one of the primary aims of language learning is to acquire the ability to communicate with the wider language community – outside the classroom. The estimated percentage of vocabulary that's necessary for second-language learners to understand written texts is thought to be between 95% (Laufer (1989)) and 98% (Hu and Nation (2000)). Therefore if the average learner does not know between 5% and 2% of words in a text, they will still be able to understand the text as a whole. The less frequently a word is used, the less likely it is that a learner will have acquired it. Lower- frequency words that have not been encountered yet, or those that have been encountered but not frequently enough to be fully acquired, are among the more difficult words for learners to understand. Anecdotally speaking, at lower levels of education or ability the words included in this 2%-5% may occur relatively frequently but still be unknown because of the learner's inexperience. In a corpus of NCI's size, 30 million words, containing 266,973 unique lemmas, these percentages equate to between 13,348 lemmas (5%) and 5,339 lemmas (2%) (these NCI frequencies are absolute frequencies). The 13,348 lowest-frequency unique lemmas in NCI all have a frequency of 1 in NCI. 13,348 is not a rank, it is a count of unique lemmas. For context, the first unique lemma to have a frequency of 2 in NCI occurs in the 132,209th position of the rank list, 50% of the way through the 266,973 lemmas of the corpus. The frequency of matched terms are henceforth sought in NCI in order to contextualize their use in EduGA. Results that consider both polysemy and frequency in a corpus of general language can be visualised in the following format:

Table 10: Example of data layout

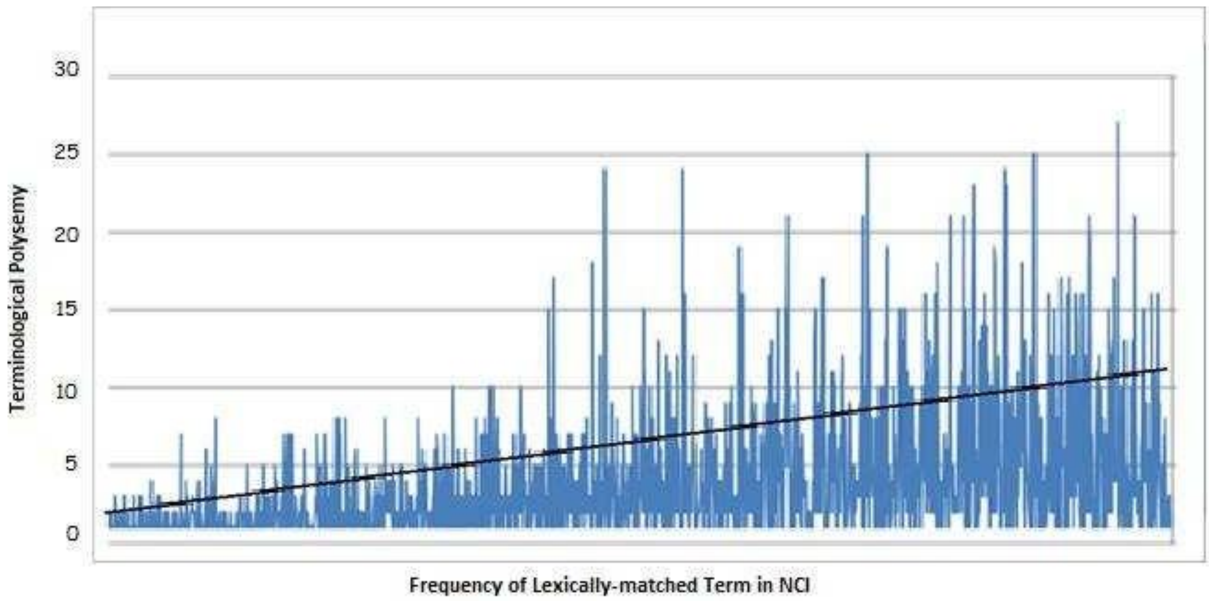
Sub Corpus	Doc Name	Lemma	Freq	Téarma Match	# of Téarma Concepts	NCI Match	NCI Freq
EduGA Hist PPSC	**StairLC01**	*meánaois* “mid-life”	35	√	**1**	√	634
		cliath “harrow, break harrow, staff, stave, hurdle, wattle”	30	√	**6**	√	578
		plandáil “plant, plantation”	25	√	**3**	√	274
		stailc “strike, walk-out”	25	√	**2**	√	438
		meánaoiseach “medieval, mediaeval”	22	√	**1**	√	147

When term candidates were sought in Science materials from post-primary level up to senior-cycle for which lexical matches were also sought in NCI[97] so that their frequency in general language can be added to the analysis. Figure 9 illustrated the terms found in EduGA-Sci-PPSC in terms of their terminological polysemy as well as their NCI frequency. Null matches in either database have been omitted from this example. The black line in Figure 9 is a trend line[98] and is included to assist in the reading of a large number of values. The direction of the trend line is based on changes in values that are calculated using five values as pivots.

[97] NCI is a 30-million word corpus of general Irish, URL: https://focloir.sketchengine.co.uk/run.cgi/index (last accessed: 17/06/2019)

[98] Trend Line (technical analysis). URL: https://en.wikipedia.org/wiki/Trend_line_%28technical_analysis%29

Figure 9: Terminological Productivity of Terms in EduGA-Sci-PPSC (Post-primary level senior-cycle materials)



A general correlation was found to occur between a drop in terminological productivity and a drop in the frequency of the matched term in NCI during the course of this research. In the majority of situations the terms that were less polysemous were also unambiguously terminologica[9]l[9], however, the combination of terminological polysemy and general frequency could not be used on a term-by-term basis to disambiguate terms. Figure 9 shows terms ordered across the x-axis according to their frequency in NCI, with the highest frequency in NCI to the left. The terminological productivity associated with these terms is graphed on the y- axis. The general trend is marked by the black line, showing that less productive terms only occur less and less frequently in a corpus of general language. However, Figure 9 shows that combining these frequencies in a measure will not predict complex terms (in other words, the polysemous and infrequently-used terms) accurately enough on a term-by-term basis to be used as a complexity metric. The following are some examples; the lemmas *caith*[100] (21 concepts) and *clár*[101] (19 concepts) are both highly polysemous and occur at relatively high frequencies in NCI of 667~ occurrences per million words. In general, this high frequency is the result of these words' general meanings rather than being the result of their terminological uses. On the other hand, the word *eolas* (2 concepts) is not highly polysemous and also occurs ~20k times per 30-million words. At the other end of the spectrum *greamaigh*[102] (18 concepts) occurs just 442 times per 30-million words, and *gearmáiniam*[103] (1 concepts) occurs just once per 30-million words. *Greamaigh* also has a general non-terminological meaning associated with it, while *gearmáiniam* refers to the scientific concept "germanium". As a result of these fluctuations this measure is only informative in a general sense, for a sub- corpus or an entire corpus. The approach will not help disambiguate terminological sense from non- terminological sense, and if term-by-term feedback was to be generated by using this approach then it would likely contain too much noise for it to be practicably useful. Finally, the wider context of an entire sub- corpus appears to be necessary for this type of measure, which is not aligned with the aims of the present research.

This analysis would benefit from a method of identifying terminological uses over general uses of words. This can be done by identifying which words are most important to the meaning of a document, also known as the most topical words in a document. In some cases, such as in a piece of literature, the most important words in a document will be the main protagonists. Jurafsky and Martin (2018) show how "Romeo" in Shakespeare's *Romeo and Juliet* was identified as an important word using the TF-IDF scoring algorithm. In this example "Romeo" was found to have the same frequency (TF) as the word "action" in a corpus of Shakespeare's plays, however, because "Romeo" only appears in one play where "action" appears in thirteen "Romeo" achieves a much higher IDF score therefore giving "Romeo" a much higher TF-IDF score in the one document it appears in. By combining the TF-IDF weighting with term matching I will be able to identify which of the words that have been matched with terms are important in each document. Where a match from Category 2 has occurred the prominence of word will be dampened by virtue of its ubiquity as opposed to being promoted by its overall frequency.

[99] For example: The term "cut" from "cut and paste" from the domain of computing in comparison to the term "support vector machine" from the same domain, as well as examples previously mentioned here like "photosynthesis".

[100] Entry for "caith" in Téarma. URL: https://www.tearma.ie/q/caith/ (last accessed: 22/06/2019)

[101] Entry for "clár" in Téarma. URL: https://www.tearma.ie/q/cl%C3%A1r/ (last accessed: 22/06/2019)

[102] Entry for "gramaigh" in Téarma. URL: https://www.tearma.ie/q/greamaigh/ (last accessed: 22/06/2019)

5.3.4. Terminological Analysis

The following Complexity Scores are calculated by following these steps, each of which is detailed in this section:

Step 1: Lemma-Only Versions of the Educational materials (aka: LOVEly materials) are used for this analysis in order to facilitate the lexical matching and to group all forms terms under its lemma. The following is an example of how the data changes:

Table 41: Converting Documents into LOVEly Documents

Original Token	*Chonaic an bhean na mná a bhfaca an bhean eile* Saw the woman the women REL-PTCL saw the woman other "The woman saw the women who saw the other woman" (1) *chonaic*: 1, (2) *an*: 2, (3) *bhean*: 2, (4) *na*: 1, (5) *mná*: 1, (6) *a*: 1, (7) *bhfaca*: 1, (8) *eile*: 1	
	10 **tokens**, 8 **types**	
LOVEly Materials	*Feic an bean an bean a feic an bean eile* See the woman the woman REL-PTCL see the woman other "The woman see the woman who see the other woman" (1) *feic*: 2, (2) *an*: 3, (3) *bean*: 3, (4) *a*: 1, (5) *eile*: 1	
	10 lemma **tokens**, 5 lemmas **types**	

Step 2: The sub-corpora of LOVEly materials are processed one sub-corpus of documents at a time, with TF- IDF scores attributed to each lemma type in each document. The program reads the input as tokens, and outputs types with TF-IDF scores. However, because the program receives LOVEly documents as input in this research the program is receiving lemma tokens as input and it gives lemma types with TF-IDF scores as output.

[103] Entry for "gearmáiniam" in Téarma. URL: https://www.tearma.ie/q/gearm%C3%A1iniam/ (last accessed: 22/06/2019)

While the “lemma types” in the LOVEly documents can be described as “lemmas” or “types”, depending on the context. They are called “lemma types” in this special context so that the terms “lemma” and “type” can be used in reference to the original documents.

The lemma types in Document D all receive an individual score, this score is calculated in the context of both the document and the sub-corpus in which the lemma type has occurred. This is done for every document in every sub-corpus (cf. Table 36). Therefore, once scored, the lemma type “bean” (as above) has a hypothetical score of 0.234 in Document A in EduGA-Irish-PPJC and a hypothetical score of 0.432 in Document B in EduGA-Irish-PPJC.

Step 3: The TF-IDF scored lemma types from the LOVEly materials are lexically matched with terms from the Téarma database.

Step 4: The TF-IDF scored lemma types from the LOVEly materials are lexically matched with the lemma- frequencies from NCI. This step is taken in order to add the context of general-language frequencies to the results, where infrequent words in a corpus of general language are assumed to have a higher probability of causing difficulties for learners.

Step 5: Analyse the results. When analysing results, it must be remembered that terms refer to concepts that are considered more complex than the general concepts non-terminological words refer to.
Infrequently-used words are considered more difficult for learners than frequently-used words. Therefore, when infrequently-used terms are more topical in one sub-corpus over another, the sub-corpus with the higher-scoring infrequently-used terms is considered more semantically complex.

Table 42 details the sub-corpora used for this analysis. The mainstream Irish sub-corpora were chosen because of the reported issues with these materials. The number of Tokens in each sub-corpus does not change whether we use the original materials or the LOVEly materials. EduGA-Irish-PrJC and EduGA-Irish- PrSC were combined for the purposes of this analysis. This was done because An Roinn Oideachais agus Scileanna (1999a) unifies the topic of materials for primary school by prescribing the same ten topics (cf. Section 2.1) for all eight primary-level classes. The Science sub-corpora were chosen because the informative nature of the genre is, in a general sense, more likely to contain terminology than any other sub-corpus.

Table 42: Number of Lemma types and Matched Terms per Sub-corpus

Subject	Sub-corpus	Code name	No. of Docs	No. of Tokens	No. of Lemma Types	No. of Terms Matched in Téarma
Irish	Irish for primary level	**EduGA-Irish-Prim**	168	219,820	50,980	32,760
	Irish for Junior Cycle Post-primary level	**EduGA-Irish-PPJC**	10	256,760	21,295	9,350
	Irish for Senior Cycle Post-primary level	**EduGA-Irish-PPSC**	76	326,869	38,265	23,444
Science	Science for primary level	**EduGA-Sci-Prim**	18	134,629	11,755	3,593
	Science for Junior Cycle Post-primary level	**EduGA-Sci-PPJC**	8	181,956	8,089	3,762
	Science for Senior Cycle post-primary level	**EduGA-Sci-PPSC**	2	21,143	3,040	1,688

In this section examples of lemma tokens are presented followed by a detailed walkthrough of each step of the TF-IDF calculation. The examples are first presented in Table 43 alongside their frequencies in two documents and in EduGA-Irish-Prim.

Table 43: Selection of Token-lemma counts per Document in the EduGA-Irish-Prim Sub-corpus

Sub-Corpus	Lemma & Translation	Total No. of Docs containing Lemma	Freq of Lemma in the Sub-corpus	Document CodeName	Freq of Lemma Token in the Doc	Total No. of Lemmas in the Doc	Total No. of Tokens in the Doc
EduGA Irish Prim	náisiúnta (national)	6	20	364SeidS	2	1,516	12,355
				366SeidS	11	2,432	28,956
	Eorpach (European)	3	37	367SeidS	1	1,932	19,949
				1324BNGA	35	1,399	18,126
	feamainn (seaweed)	3	5	229SeidS	2	1,716	16,421
				370SeidS	2	1,367	16,340
	ceintiméadar (centimetre)	3	5	300SeidS	3	191	1,051
				372SeidS	1	1,256	14,610
	surfáil (surfing)	2	2	229SeidS	1	1,716	16,421
				370SeidS	1	1,367	16,430
	zaipire (zapper)	2	18	231SeidS	9	1,345	14,758
				372SeidS	9	1,256	14,610

Uses and discussion of the strengths and weakness of TF-IDF are included in Section 3.3, with its application to the present study discussed in Section 3.4.3. The acronym stands for Term Frequency – Inverse Document Frequency. The two halves of the acronym are separated by a hyphen rather than a minus, while TF × IDF more accurately represents the operation the algorithm is best known as TF-IDF. TF-IDF is a frequently-used algorithm in natural language processing for term extraction and information retrieval. The algorithm applies a score that is used as a weighting or ranking score in order to identify the most topical words in each of the documents in a corpus. This weighting score uses the frequency of each token in a document. This document freqency is offset by the frequency of the same token in the corpus. Therefore, near-ubiquitous words like “the”, “and”, or “be” will receive a very low score and, in the case of this document, words like “complexity” and “corpus” would receive a relatively high score. I now present the manner in which TF-IDF is calculated by first showing the values used for TF, followed by the values used by IDF.

Lemma-token is henceforth referred to using LT

TF(LT) = (Number of times LT appears in a doc) / (Total number of lemma-tokens in the doc).

The scores shown in the following example have been rounded to the eighth decimal. The eighth decimal place was chosen for this study because it was found that scores could be adequately differentiated from this point. This decimal point may not be applicable to another study, as there is no standard decimal place for TF-IDF appraisal. The program used to calculate TF-IDF scores allows for any number of decimal places, however, the 20th decimal place for all scores was zero and a substantial number of scores begin to score zero sooner than the 20th decimal place. It is worth noting at this point that lemmas (such as particles) that are *extremely* common at both a document and sub-corpus level frequently receive a tf-idf score of zero reoccurring. The following values are from EduGA-Irish-Prim.

1. **364SeidS**, TF(náisiúnta) = (2 / 12,355) **= 0.00016188**.
   366SeidS, TF(náisiúnta) = (11 / 28,956) **= 0.00037989**.

2. **367SeidS**, TF(Eorpach) = (1 / 19,949) **= 0.00005013**.
   1324BNGA, TF(Eorpach) = (35 / 18,126) **= 0.00193093**.

3. **229SeidS**,TF(feamainn) = (2 / 16,421) **=0.000121795**.
   370SeidS, TF(feamainn) = (2 / 16,340) **= 0.000122399**.

4. **300SeidS**, TF(ceintiméadar) = (3 / 1,015) **= 0.00295567**.
   372SeidS, TF(ceintiméadar) = (1 / 14,610) **= 0.00006845**.

5. **229SeidS**, TF(surfáil) = (1 / 16,4210) **= 0.00006090**.
   370SeidS, TF(surfáil) = (1 / 16,340) **= 0.00006120**.

6. **231SeidS**, TF(zaipire) = (9 / 14,758) **= 0.00060984**.
   372SeidS, TF(zaipire) = (9 / 14,610) **= 0.00061602**.

While TF scores are different from one document to the next, the IDF score for each LT is calculated at the sub-corpus level – therefore it is a constant for each sub-corpus. The IDF values are calculated as follows:

IDF(LT)

log(Total number of documents in the sub-corpus / Number of documents in the sub-corpus with LT in it)

1. **EduGA-Irish-Prim,** IDF(náisiúnta) = log(173 / 6), or **IDF(náisiúnta) =** 3.36153212.
2. **EduGA-Irish-Prim,** IDF(Eorpach) = log(173 / 3), or **IDF(Eorpach) =** 4.05467931.
3. **EduGA-Irish-Prim,** IDF(feamainn) = log(173 / 3), or **IDF(feamainn) =** 4.05467931.
4. **EduGA-Irish-Prim,** IDF(ceintiméadar) = (173 / 3), or **TF(ceintiméadar) = 4.05467931**
5. **EduGA-Irish-Prim,** IDF(surfáil) = log(173 / 2), or **IDF(surfáil) =** 4.4601440.
6. **EduGA-Irish-Prim,** IDF(zaipire) = log(173 / 2), or **IDF(zaipire) =** 4.46014440.

The final TF-IDF weighting score is the product of the TF and IDF quantities.

1. **364SeidS, 0.00016188 *** 3.36153212 gives **TF-IDF(náisiúnta) = 0.00054416.**

366SeidS, 0.00037989 * 3.36153212 gives **TF-IDF(náisiúnta) = 0.00127700.**

2. 367SeidS, 0.00005013 * 4.05467931 gives **TF-IDF(Eorpach) = 0.00020325.**

1324BNGA, 0.00193093 * 4.05467931 gives **TF-IDF(Eorpach) = 0.00782929.**

3. 229SeidS, 0.000121795 * 4.05467931 gives **TF-IDF(feamainn) = 0.00049384.**

370SeidS, 0.000122399 * 4.05467931 gives **TF-IDF(feamainn) = 0.00049629.**

4. 300SeidS, 0.00295567 * 4.05467931 gives **TF-IDF(ceintiméadar) = 0.00049384.**

372SeidS, 0.00006845 * 4.05467931 gives **TF-IDF(ceintiméadar) = 0.00049629.**

5. **229SeidS, 0.00006090 *** 4.4601440 gives **TF-IDF(surfáil) = 0.00027161.**

370SeidS, 0.00006120 * 4.4601440 gives **TF-IDF(surfáil) = 0.00027296.**

6. **231SeidS, 0.00060984 *** 4.46014440 gives **TF-IDF(zaipire) = 0.00271997.**

231SeidS, 4.46014440 ***** 4.46014440 gives **TF-IDF(zaipire) = 0.00274752.**

TF-IDF scores are calculated from each lemma token, in each document, in each sub-corpus. Each document is a different size and each sub-corpus is a different size, particularly from one subject to the next. Therefore, it would not be appropriate to compare the TF-IDF scores of two different sub-corpora. For example, saying that TF-IDF(náisiúnta) in 366SeidS is 0.00127700 is greater than TF-IDF(surfáil) in 370SeidS is 0.00027296 does not indicate “náisiúnta” is more topical than “surfail”, because each TF-IDF score is based on a different set of values that bear little or no relationship to one another. Each document has a total combined TF-IDF score that is document specific, by extension each sub-corpus also does. TF-IDF have been normalised in the present study in order to make them comparable. Every TF-IDF score is converted to a percentage of the total TF-IDF score of each document, therefore, TF-IDF(náisiúnta) hypothetically contributes 0.2% of the total TF-IDF score of the document 366SeidS and TF-IDF(surfáil) hypothetically contributes 1.3% of the total TF-IDF score of the document 370SeidS.

As noted in **Step 3** lexical match with each scored lemma-type (LT) is sought in the Téarma database, if the lemma is not found in Téarma an X is added to the output or a √ is added if a match is found. This step is important as it acts as a preliminary filter for extremely simple words (such as: *bí* "be", *an* "the", *agus* "and"). In the context of EduGA-Irish-Prim, this reduces the number of document-level lemma-types to be analysed from 50,980 to 32,760. The remaining words are generally more complex than the non- terminological words or function words, and the higher they score the more topical to the document they are. An additional dimension is required in order to contextualise the contents of the documents.

In **Step 4** a lexical match for the document-specific lemma-types is also sought in NCI, and the NCI lemma- frequency is added to the output for the analysis. If the lemma is not found in NCI, naturally, a frequency of zero is returned. The NCI frequencies were then split into ranges that are used to contextualize the use of topical terms in educational materials.

As previously stated in this section, the 13,348 lowest-frequency lemmas in NCI all have a frequency of 1 in NCI. 13,348 does not refer to rank, it is a count of unique lemmas. For context, the first unique lemma to have a frequency of 2 in NCI occurs in the 132,209th position of the rank list, 50% of the way through the 266,973 lemmas of the corpus. The raw NCI lemma frequencies have been used to establish five frequency ranges (henceforth: FreqRange).

- FreqRange 1: Terms in educational materials that have a frequency between 0-1 in NCI. Considering these are lemmas, rather than tokens, I have grouped 0-frequency and single-occurrence lemmas together
- FreqRange 2: Terms in educational materials that have a frequency between 2-10 in NCI
- FreqRange 3: Terms in educational materials that have a frequency between 11-100 in NCI
- FreqRange 4: Terms in educational materials that have a frequency between 101-1,000 in NCI
- FreqRange 5: Terms in educational materials that have a frequency between 1,001-10,000 in NCI

Table 44 shows the NCI frequencies of the sample terms, with their frequency ranges between brackets. When comparing sub-corpora such as EduGA-Irish-Prim and EduGA-Irish-PPJC, the percentage TF-IDF scores of all LTs in each range can be used to inspect how much the meaning and topic of a document depends on terms of varying frequencies. These frequency ranges are therefore being used as a proxy for the difficulty a learner will probably have with a word; resulting from the learner's exposure, or lack thereof, to that word.

Table 44: Sample Tf-idf Scored Lemmas, Term Matching, and NCI lemma frequency

Lemma Token	NCI Frequency	Lexical Match found in Téarma	Doc Code Name	TF-IDF Score
náisiúnta	11,131 (FreqRange ALL)	√	364SeidS	**0.00054416**
			366SeidS	**0.00127700**
Eorpach	9,543 (FreqRange 1001-10000)	√	367SeidS	**0.00020325**
			1324BNGA	**0.00782929**
feamainn	993 (FreqRange 101-1000)	√	300SeidS	**0.00049384**
			372SeidS	**0.00049629**
ceintiméadar	95 (FreqRange 11-100)	√	229SeidS	**0.00049384**
			370SeidS	**0.00049629**
surfáil	20 (FreqRange 2-10)	√	229SeidS	**0.00027161**
			370SeidS	**0.00027296**
zaipire	0 (FreqRange 0-1)	√	231SeidS	**0.00271997**
			372SeidS	**0.00274752**.

Once **Step 4** has been completed the analysis can begin. Before providing the results in Section 5.3.5, a quick recap of the steps will clarify what sorts of words *can* appear in the results. This analysis focusses on terms because of the inherent complexity associated with them. Automatically extracting terms from documents is not a straighforward and easy process, therefore, a number of processing steps are required. Firstly, the LOVEly materials were created, meaning no forms other than the lemma were processed or will be found in the results. Secondly, every lemma was processed for TF-IDF scoring, the output of which provides a list of lemma-types that have been scored according to their topicality on a per document basis. The same lemma could appear in multiple documents and receive multiple different scores in a sub-corpus. Thirdly, a match for each lemma type is now lexically sought in the Téarma database. Lemma-tokens that were not matched with a term in Téarma are filtered out of the list of TF-IDF scored terms that will be analysed. All of the examples given in this section (*náisiúnta, Eorpach, feamainn, ceintiméadar,* and *zaipire*) feature in the results, but the likes of *agus* or *an* (“and” or “the” respectively) would not. Finally, each lemma-type is associated with a second value: the frequency of the corresponding lemma in a corpus of general Irish. The result of these steps is a list of words that have an inherent complexity, each of which can now be ranked according to the likelihood a learner has acquired it.

5.3.5. Results

Subsections 5.3.5.1 and 5.3.5.2 present tables containing examples of LTs extracted from subsets of EduGA, this is followed by an analysis of the distribution of TF-IDF in each of the EduGA subset. Results are presented as a total of all percentage TF-IDF scores in each frequency range. Figure 10 and Figure 11 summarise the results for the sub-corpora of each subject alongside one another in order to facilitate comparison.

5.3.5.1. Educational Materials for Teaching Irish in Mainstream Education

Table 45, Table 46, and Table 47 present the distribution of TF-IDF scores in each frequency range for the EduGA-Irish-Prim, EduGA-Irish-PPJC, and EduGA-Irish-PPSC sub-corpora. These tables also show how much terms vs non-terminological words have contributed to the combined TF-IDF scores of each FreqRange.

Table 45: EduGA-Irish-Prim (Primary-level Irish Materials)

NCI Frequency Range	Tf-idf Scores Term-Lemmas	No. of Terms per 50k	Percentage of TF-IDF (from Terms)	Cumulative Percentage of TF-IDF (from Terms)	Tf-idf Scores All Lemmas
0-1	0.17505731	11	**0.18%**	**0.18%**	12.74006054
2-10	0.33842596	23	**0.16%**	**0.34%**	15.39211667
11-100	4.65838728	253	**4.47%**	**4.81%**	28.43542260
101-1,000	28.96326691	1,567	**25.12%**	**29.93%**	66.04384283
1,001-10,000	72.70360152	3,338	**45.36%**	**75.29%**	120.75853070
ALL	96.56674807	7,446	**24.71%**	**100%**	155.81636170

Table 46: EduGA-Irish-PPJC (Post-primary level junior cycle Irish Materials)

Frequency Range	Tf-idf Scores Term-Lemmas	No. of Terms per 50k	Percentage of TF-IDF (from Terms)	Cumulative Percentage of TF-IDF (from Terms)	Tf-idf Scores All Lemmas
0-1	0.00354835	8	**0.15%**	**0.15%**	0.36468502
2-10	0.01859628	11	**0.64%**	**0.79%**	0.55589718
11-100	0.14190373	116	**5.06%**	**5.85%**	1.06841568
101-1,000	0.70765742	498	**23.37%**	**29.22%**	2.06496099
1,001-10,000	2.01919314	916	**54.19%**	**83.41%**	3.62529608
ALL	2.42087351	1,832	**16.59%**	**100%**	4.19075529

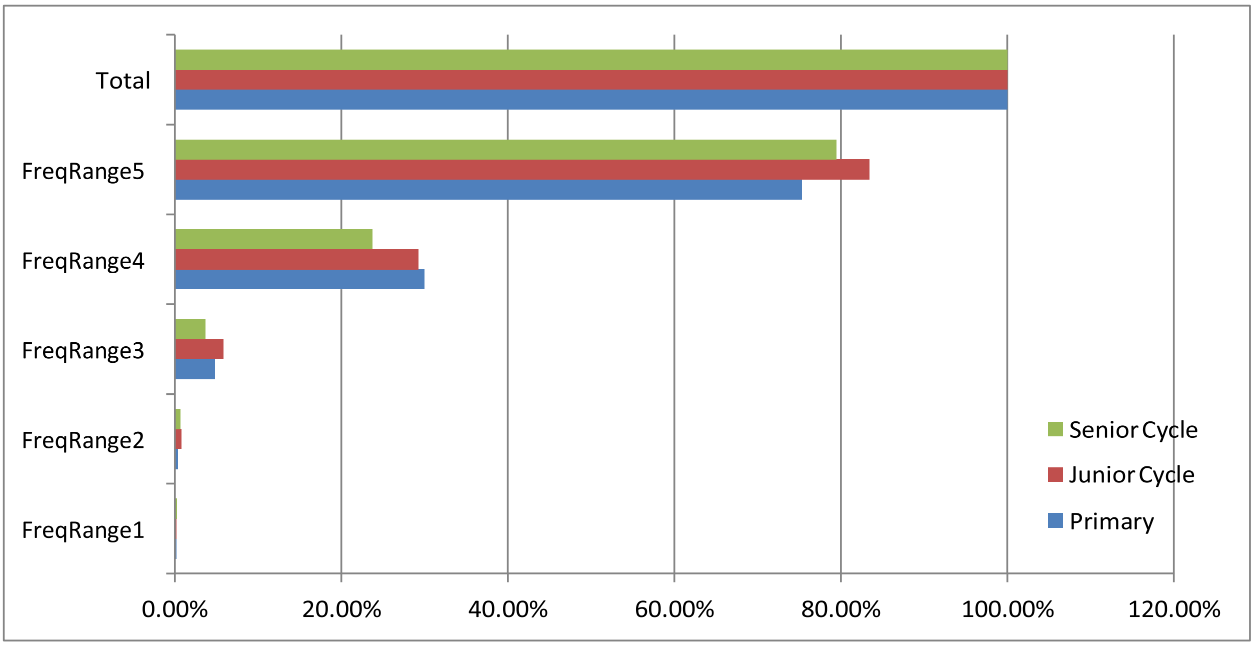
Table 47: EduGA-Irish-PPSC (Post-primary level senior cycle Irish Materials)

Frequency Range	Tf-idf Scores Term-Lemmas	No. of Terms per 50k	Percentage of TF-IDF (from Terms)	Cumulative Percentage of TF-IDF (from Term)	Tf-idf Scores All Lemmas
0-1	0.10447535	10	**0.27%**	**0.27%**	3.13240473
2-10	0.26321192	29	**0.39%**	**0.66%**	4.86524731
11-100	1.44160251	121	**3.01%**	**3.67%**	9.47680910
101-1,000	9.27607245	704	**20.02%**	**23.69%**	21.65464377
1,001-10,000	31.09433292	1,741	**59.43%**	**79.45%**	47.05978983
ALL	39.13896675	3,607	**20.55%**	**100%**	57.46551362

The results show the largest number of term matches in any of the sub-corpora in EduGA-Irish-Prim, followed by a severe fall in the volume of matched terms in EduGA-Irish-PPJC, and at least twice this frequency in EduGA-Irish-PPSC. However, we can see that the cumulative percentage of TF-IDF scores for FreqRanges 1-3 and again at FreqRange 5 is greater in EduGA-Irish-PPSC than it is in EduGA-Irish-Prim. This exemplifies how frequency alone can be misleading, and that the values require contextualization.

Ultimately these differences in score and frequency are minor and, other than noting the considerable drop in the frequency of terms in EduGA-Irish-PPJC, it appears as though the semantic complexity of all three sub- corpora is relatively similar. One would expect terms to carry a higher cumulative TF-IDF score at FreqRanges 1 and 2 in EduGA-Irish-PPSC than EduGA-Irish-PPJC and EduGA-Irish-Prim, because the expectation is for materials to become more complex as learners acquire more and more of the language. Figure 10 shows that this is not the case, therefore indicating that this metric has found the semantic complexity of Irish- language materials remains the same from primary level through to senior-cycle post-primary level.

Figure 10: Percentage Difficulty Scores for Irish Materials



If the success of combining lexically-matching with TF-IDF scoring was to be appraised, where the aim was to remove noise and focus on *bone fide* term, then one indication of its success can be seen at FreqRange 1 in Table 45 where the TF-IDF score is almost 100 times smaller after terms were matched, in

Table 46 where the TF-IDF score is almost 100 times smaller after terms were matched, and in Table 47 we see the TF-IDF score is almost 30 times smaller after terms were matched.

Table 48 contains the top 10 TF-IDF-scoring examples for FreqRanges 1-5 in EduGA-Irish corpora. It shows topics are likely tailored to teenagers, which is to be expected, because of terms like “text messaging”, “cyberbullying”, and “googling”. Food also features as a topic at various levels, with “salami”, “lasagne”,”ice pop” and “(potato) mash”. As does health, with the terms “plaster”, “meidicine/cure”, “(IV) drip”, and “inhaler”. The frequency ranges in which some of these concepts have occurred show NCI is out of date. Of course this is not a criticism of NCI, significant technological changes have occurred since publication (Kilgarrif et al., 2006). There is a focus in the primary-level materials on topics that have not been repeated at post-primary level, namely the Greek legends and myths as well nature studies or pre-science. the former topic can be seen in topical terms such as the character “Halcyone” and the weapon “boar spear” and from the latter topic can be seen in terms like “queen bee”, “centipede”, “bear”, and “aridity”. These are just a few of the examples that have been objectively extracted from materials and can be given as tangible feedback to materials creators and teachers.

Table 48: Top-scroing terms from each frequency range for Irish Materials

NCI Frequency Range	EduGA-Irish-Prim	EduGA-Irish-PPJC	EduGA-Irish-PPSC
0-1 **Very rare in NCI**	fiacla (*teeth*) prioslóir (*bib*) figín (*plectrum*) téaduirlis (*stringed instrument*) torcshleá (*boar spear*) tionsclaigh (*industrialize*) Ailcín (*Halcyone*) salamaí (*salami*) zaipire (*zapper* [TV remote control]) claisín (*groove / throating*)	blag (*blog*) téacsteachtaireacht (*text message*) súnámaí (*tsunami*) tonnmharcaíocht (*surfing*) sionc (*sync*) tuarthach (*predictive*) cibearbhulaíocht (*cyber bullying*) carrchaladh ([car] *ferry*) Garda (*guard*) cliathreathaí (*hurdler*)	scrúduimhir (*exam number*) aibhsithe (*highlighted*) podchraoladh (*podcast*) gúgláil (*googling*) maistreán (*mash* [food]) trastíre (*cross-country*) diltálaire (*drip* [medical]) eadrannach (*intermittent*) griandó (*sunburn*) cibearbhulaíocht (*cyber bullying*)
2-10 **Rare in NCI**	tuire (*aridity*) bíp (*beep*) greimlín (*plaster*) criospaí (*crisps*) cráinbheach (*queen bee*) móibíl (*mobile*) tóireadóir (*probe*) céadchosach (*centipede*) cróitín (*pen for small animals / creep*) cliathrás (*hurdle race / hurdles*)	criospaí (*crisps*) steiteascóp (*stethoscope*) táimhe (*inertia*) reoiteog (*ice pop*) ruifíneach (*hoodlum*) poncúlacht (*punctuality*) pléascóg (*banger* [fireworks]) trasnach (*intersect*) láine (*volume*) pinniúr (*court / alley* [sports])	trasrian (*crossing* [traffic]) téacsáil (*texting*) análóir (*inhaler*) déchiallach (*equivocal*) análaitheoir (*respirator*) scuabáil (*sweeping*) seobhaineachas (*chauvinism*) boghdóireacht (*archery*) comhthuiscint (*mutual understanding*) reoiteog (*ice pop*)

11-100 **Infrequent in NCI**	lámhainn (*glove*) scaif (*scarf*) hamstar (*hamster*) crián (*crayon*) slipéar (*slipper*) pizza (*pizza*) circín (*orchid*) cleasaí (*trickster / joker*) scátáil (*skater*) luascán (*swing*)	athfhás (*regrowth*) seisc (*sedge*) neamhord (*disorder*) lasagne (*lasagne*) cruaite (*hardened*) saorvéarsaíocht (*free verse*) bácáilte (*baked*) scaimh (*swarf / filings*) freanga (*warp*)	sínteán (*stretcher*) deisigh (*fix / mend*) teanglann (*language lab*) súiste (*flail*) ibh (*imbibe*) faocha (*winkle / periwinkle*) díluacháil (*devalue*) siombalachas (*symbolism*) niamhrach (*brilliant* [vivid]) neamhphearsanta (*impersonal*)
101-1,000 **Moderately frequent in NCI**	béar (*bear*) balún (*balloon*) stoca (*sock*) luch (*mouse*) leite (*porridge*) sliotar (*sliotar* [ball]) luaidhe (*lead*) sciob (*steal*) scáthán (*mirror*) cuileog (*fly* [insect])	ológ (*olive*) nóinín (*daisy*) meadaracht (*metre* [poetry]) matán (*muscle*) cearrbhach (*queen bee*) sláintiúil (*healthy*) docht (*rigid*) íobartach (*victim*) folús (*vacuum*) oscailteacht (*openness*)	céadfa (*sense*) liric (*lyric*) goilliúnach (*hurtful / sensitive*) dícheallach (*dilligent*) cispheil (*basketball*) scíth (*shield*) físeán (*video*) gortaigh (*wound / hurt*) lúcháir (*joy / rejoice*) bradán (*salmon*)

1,001 - 10,000 **Common in NCI**	lón (*lunch*) cóta (*coat*) sneachta (*snow*) caisleán (*castle*) taisteal (*travel / travelling*) halla (*hall*) ridire (*knight*)	véarsa (*verse*) scaoil (*release*) dán (*poem*) halla (*hall*) tábla (*table*) líonadh (*fill / completion*) ola (*oil*)	iascaireacht (*fishing*) clúdach (*clúdach*) tuismitheoir (*parent*) cuntas (*account*) clog (*clock*) luach (*value*) scoláire (*scholar*)
	cailleach (*witch*) aimsir (*weather*) réalta (*star*)	spás (*space*) brú (*pressure*) leigheas (*cure / medicine*)	comhartha (*sign / symbol*) cúirt (*court*) greim (*bite*)

Where the calculation of semantic complexity is expressed as a series of values, if meaningful change is to be implemented where necessary then I believe it is important that examples be provided as a means to guiding these changes.

1. From my experience the majority of the terms included in Table 48 could be acquired or understood by learners without a great deal of difficulty. One of the issues with Irish-language educational materials cited in Section 1.2 was that the materials were overly simple, indeed, this is what these results indicate.
2. A relatively small amount of low-frequency terms are used in materials for teaching Irish, as seen in FreqRanges 1, 2 and 3, and only a little over 5% of the total topicality score for the subject. Furthermore, some of the low-frequency terms from the domain of IT and Computers would now be frequently-used terms.
3. Without term matching, 9 of the highest scoring lemmas in EduGA-Irish-PPSC are people’s names[104]. This is indicative of a type of genre that is discursive. These materials have been manually checked and do include a number of dialogues.
4. The top scoring terms clearly come from a range of different topics, a number of which teenagers might relate to. For example: *téacsteachtaireacht* “text message”, *blag* “blog”, *cibearbhulaíocht* “cyber-bullying”, and a number of sporting terms like *tonnmharcaíocht* “surfing” and *cliathrásaíocht* “hurdles / hurdle racing”. It has been noted that some of the technology-based terms may no longer fall into FreqRange 1, it is notable however that the highest scoring terms in FreqRanges 1 & 2 contributed an extremely small percentage of the topicality scores an all EduGA-Irish sub-corpora.

5.3.5.2. *Educational Materials for Teaching Science*

Table 49, Table 50, and Table 51 present the distribution of TF-IDF scores in each frequency range for the EduGA-Sci-Prim, EduGA- Sci-PPJC, and EduGA-Sci-PPSC sub-corpora. These tables also show how much terms vs non-terminological words have contributed to the combined TF-IDF scores of each FreqRange.

Table 49: EduGA-Sci-Prim (Primary Level Science Materials)

NCI Frequency Range	Tf-idf Scores Term-Lemmas	No. of Terms per 50k	Percentage of TF-IDF (from Terms)	Cumulative Percentage of TF-IDF (from Terms)	Tf-idf Scores All Lemmas
0-1	0.00399818	5	**4.16%**	**4.16%**	0.04318383
2-10	0.00656089	6	**2.69%**	**6.85%**	0.05882064
11-100	0.02243207	31	**16.57%**	**23.42%**	0.10773640
101-1,000	0.05351853	82	**32.44%**	**55.86%**	0.16039984
1,001-10,000	0.09114362	158	**39.29%**	**95.15%**	0.21131920
ALL	0.09578990	343	**4.85%**	**100%**	0.21877353

Table 50: EduGA-Sci-PPJC (Post-primary level Junior Cycle Science Materials)

NCI Frequency Range	Tf-idf Scores Term-Lemmas	No. of Terms per 50k	Percentage of TF-IDF (from Terms)	Cumulative Percentage of TF-IDF (from Terms)	Tf-idf Scores All Lemmas
0-1	0.00347200	45	**2.82%**	**2.82%**	0.03769257
2-10	0.01527994	79	**9.6%**	**12.42%**	0.05861564
11-100	0.04856591	234	**27.09%**	**39.51%**	0.10991043
101-1,000	0.09278761	340	**35.88%**	**75.50%**	0.15937670
1,001-10,000	0.12133464	290	**23.23%**	**98.73%**	0.19005666
ALL	0.12288004	1,044	**1.27%**	**100%**	0.19194339

[104] Eoin, Daithí, Peadar, Síle, Oisín, Caoimhín, Seán, Mícheál, and then Seán again in a different document; with the name "Seán" repeatedly featured across numerous other documents as a top-50 scorer for EduGA-Irish-PPSC. Although some names are included in the Term Database, these are not presented in results because the concepts they refer to are different overall to terms and because materials creators probably don't require this type of feedback.

Table 51: EduGA-Sci-PPSC (Post-primary level Senior Cycle Science Materials)

NCI Frequency Range	Tf-idf Scores Term-Lemmas	No. of Terms per 50k	Percentage of TF-IDF (from Terms)	Cumulative Percentage of TF-IDF (from Terms)	Tf-idf Scores All Lemmas
0-1	0.01187734	254	**8.77%**	**8.77%**	0.04145541
2-10	0.02247161	235	**16.8%**	**25.57%**	0.06315320
11-100	0.05812180	718	**40.59%**	**66.16%**	0.11060540
101-1,000	0.08489607	1,188	**30.48%**	**96.64%**	0.14319276
1,001-10,000	0.08783857	1,202	**3.36%**	**100%**	0.14760986
ALL	0.08783857	3,968	**0%**	**100%**	0.14762387

Figure 11 summarises the contents of Table 49, Table 50, and Table 51. The cumulative TF-IDF scores in each FreqRange indicate the semantic complexity of each of sub-corpus is appropriate relative to each of the other levels of education. By this I mean, within regards to science, Senior Cycle materials are more semantically complex than Junior Cycle materials, and Junior Cycle materials are more semantically complex than Primary materials. This statement is made based on terms being inherently complex, and infrequently- used words being more difficult for learners than frequently-used words.

Therefore, when infrequently-used terms are more topical in one sub-corpus over another, the sub-corpus with the higher scoring infrequently-used terms is considered more semantically complex.

Figure 11: Percentage Difficulty Scores for Science Materials

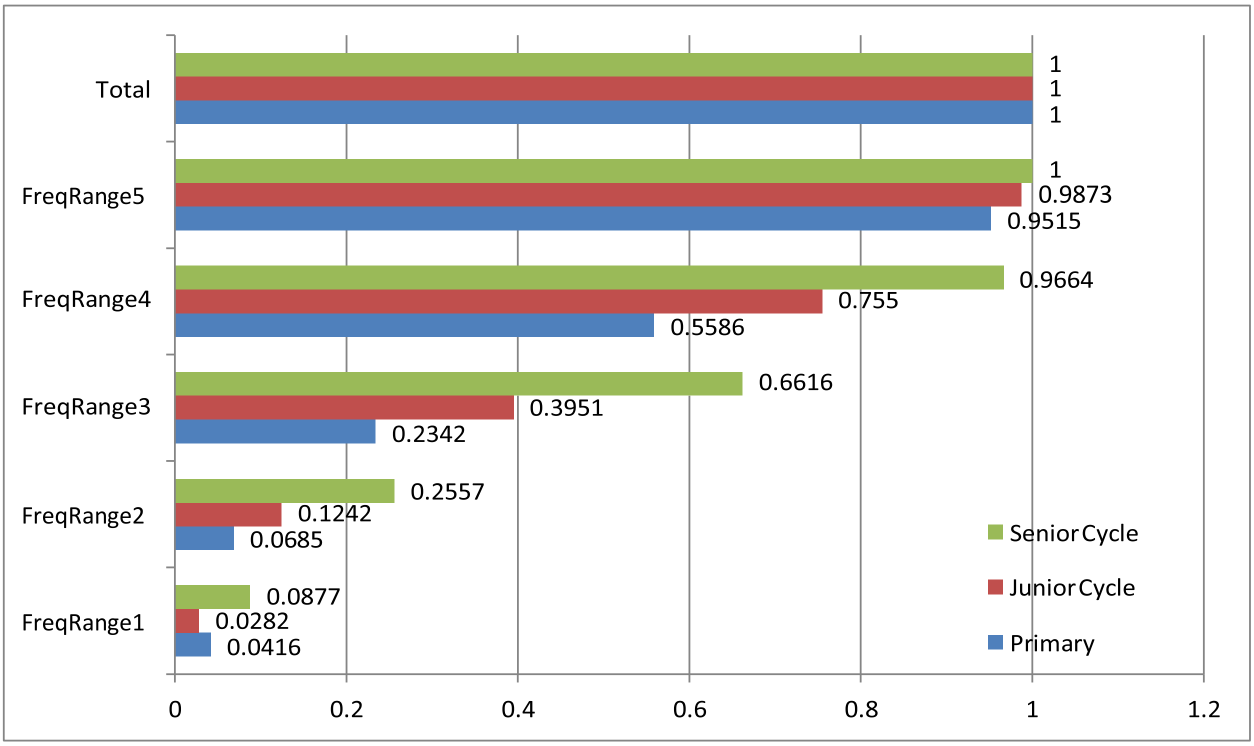


Table 52 contains the top 10 TF-IDF-scoring examples for FreqRanges 1-5 in EduGA-Sci sub-corpora. The terms are more clearly aligned with a single domain of knowledge than the terms extracted from EduGA- Irish sub-corpora.

Table 52: Top scoring terms from each frequency range for Science Materials

NCI Frequency Range	EduGA-Sci-Prim	EduGA-Sci-PPJC	EduGA-Sci-PPSC
0-1	ceadúnóir (*licenser*) fiadhúlra (*wildlife*) beathra (*life*) anglait (*monkfish*) airleacán (*harlequin*) scairpiasc (*sea scorpion / scorpionfish*) EOECNA (*UNESCO*) pasaireach (*passerine*) féarmhá (*downland*) angóra (*angora*)	imoibriúchán (*reaction*) logálaí (*log / logger*) crómatagrafaíocht (*chromatography*) craosadán (*thistle-funnel*) húiméireas (*humerus*) sionc (*synchronous / sync*) seolfheadán (*delivery-tube*) braonaire (*dropper*) aibitheach (*abiotic*) aiméibe (*amoeba*)	teanntacht (*tensility*) anóideach (*anodic*) leictriceimiceach (*electrochemical*) leictrealaíoch (*electrolytic*) treáiteach (*penetrative*) trasnasc (*cross-link*[age]) eoiteicteach (*eutectic*) catóideach (*cathodic / cathode*) ardmhinicíocht (*high-frequency*) sileacón (*silicone*)
2-10	sabhána (*savanna*) ailbíneach(*albino*) lapaire (*wader*) leabharmharc (*bookmark*) blastanas (*seasoning*) eoráiseach (*Eurasian*) gealáin (*highlights*) glónra (*glaze*) snámhacht (*buoyancy / flotation*) gealóg (*salmon fry / trout fry / fry / bunting*)	triaileadán (*test tube*) cuadraí (*quadrat*) amaláis (*amylase*) xiléim (*xylem*) fithiseán(*orbital*) litiam (*lithium*) sárocsaíd (*peroxide*) ainhidriúil (*anhydrous*) cóimheastóir (*control*) fithisigh (*orbit*)	séama (*seam*) criostalta (*crystalline*) teirmeaplaisteach (*thermoplastic*) olltáirgeadh (mass production) leictrilít (electrolyte) smúdáil (*ironing / smoothing*) insínteacht (*ductility / extensibility*) poileitiléin (*polyethylene*) anóid (anode) vinil (vinyl)

11-100	tír (*country*) scéal (*story*) bean (*woman*) duine (*person*) mór (*big*) eile (*other*) aon (*one / other*) cuir (*put / plant*) cuid (*part / portion of*) maith (*good*)	litmeas (*litmus*) iaidín (*iodine*) alcaile (*alkali*) steallaire (*syringe*) aoluisce (*limewater*) fleascán (*flask*) riospráid (*repiration*) sulfáit (*sulphate*) hiodrocsaíd (*hydroxide*) prótón (*proton*)	cruachan (*induration, hardening, setting*) caighdeánú (*standardization*) lúbtha (*looped / bowed*) naprún (*apron*) athraitheach (*changeable*) soladach (*solid*) ailtéarnach (*alternate*) eitre (*chink / groove*) ribeach (*capillary*) calóg (*flake*)
101-1,000	postas (*postage*) tarrtháil (*rescue*) piobar (*pepper*) speiceas (*species*) dialann (*diary*) grinneall (*bed* [of sea/river]) lacha (*duck*) spúinse (*sponge*) láib (*mud*) turcaí (*turkey*)	eascra (*beaker*) promhadán (*test tube*) imscrúdú (*investigation*) sóisearach (*junior*) leictreon (*electron*) seastán (*platform / pedestal*) táscaire (*indicator*) adamh (*atom*) tuaslagán (*solution*) cailciam (*calcium*)	strus (*stress*) meaisín (*machine*) teilgean (*projecting*) scriú (*screw*) slabhra (*chain*) voltas (*voltage*) rubar (*rubber*) íobartach (*sacrificil / victim*) druilire (*drill*) stua (*arch / arc*)

1,001 - 10,000	spraoi (*play*) stáisiún (*station*) farraige (*sea*) iasc (*fish*) earrach (*spring* [season]) fiáin (*wild*) fómhar (*autumn*) samhradh (*summer*) geimhreadh (*winter*) cara (*friend*)	léann (*scholarship / education*) tábla (*table*) eolaíocht (*science*) gealach (*moon*) íomhá (*image*) grian (*sun*) teastas (*certificate*) gorm (*blue*) cnámh (*bone*) réalta (*star*)	tábla (*table*) ráta (*rate*) Gearmánach (*German*) cogadh (*war*) cosanta (*protected / guarded*) ualach (*load / burden*) gunna (*gun*) Sasana (*England*) teip (*fail / failure*) inmheánach (*internal*)

A number of the same principles apply to the examples included in Table 48 and Table 52, with regards to the topics covered being evident from the highest scoring terms in FreqRanges 1 and 2 in particular. There appears to be less domain variation in the EduGA-Sci materials, as you would expect, when comparing the LTs in Table 48 with the LTs in Table 52. The science sub-domains can be inferred from the extracted LTs, with biology and/or nature studies appearing to be more topical in EduGA-Irish-Prim materials at all FreqRanges. A mixture of Chemistry and Physics appears to be more topical at FreqRanges 1 through 4 in the materials for EduGA-Sci-PPJC and EduGA-Sci-PPSC.

1. It is significant that, as shown in Figure 11, the difficulty scores at senior-cycle materials reaches 100% of the topicality scores within the 5 frequency ranges. This is significant because it shows that all terms with a non-zero topicality score have a frequency of 10,000 or higher in NCI. It also shows that EduGA-Sci-PPSC materials contain a significant amount of infrequently-used terms.
2. In general terms the FreqRanges taken from NCI have given more good examples of what would be called "difficult words" in the EduGA-Sci analysis, however, it appears as though terms like "seasoning" (cookery) ought to be in a different range to "xyleme" (chemistry).
3. With regards to primary-level science materials (EduGA-Sci-Prim), 95.15% of the TF-IDF scores is achieved within the first five frequency ranges because of the number of content words relative to function words. The sentences in these materials are short and declarative, and their function is similar to that of a vocabulary list without actually being a vocabulary list. the following structures are particularly prominent examples, *Tá an coinín beag* "The rabbit is small", *Tá an béar mór* "The bear is big" or *Seo an fheirm* "This is the farm", *Seo an feirmeoir* "This is the farmer".

5.3.5.3. *Conclusions and Comments*

1. This approach has successfully filtered out a large amount of noise, especially in the case of EduGA- Irish in order to facilitate the analysis of *bone fide* terms, however, it is not without its limitations. Some of the term matches found in EduGA-Irish and EduGA-Sci could still constitute noise, particularly *maith* “good” and *mór* “big” that were found to be among the most topical terms in FreqRange 3 in EduGA-Sci-Prim because of how rarely adjectives of this nature are used in Science materials.
2. The frequency ranges successfully contextualized matched terms where a domain had not changed significantly since NCI was published, however, this exercise has emphasised how the corpus is out of date in its coverage of domains *Technology and IT* and *Culinary and Cooking*.
3. Interpretation of the examples included in the results indicate there is a significant difference between the semantic complexity of terms used in EduGA-Irish and EduGA-Sci, the latter being far more complex. This aligns with issues identified in the literature in Section 1.2, where some Irish-language materials were said to be too simple and where some Irish-medium materials were said to be too complex and/or difficult.
4. One of the strengths of this approach is the ability to drill down from a corpus level, to a document level, to the word level. This means language-based feedback can easily be generated alongside the objective scoring of materials. Both elements are equally important, as the needs of each classroom of learners is quite unique a teacher may feel it is appropriate to temporarily increase the semantic complexity of the materials used in class.
5. From my experience as a teacher, it appears as though the examples in Table 52 are more likely to be considered difficult by learners than those extracted from EduGA-Irish materials for Table 48, particularly those in EduGA-Sci-PPJC and EduGA-Sci-PPSC. Another similar observation of the results in Table 52 that is based on my experience is that there may appear to be a substantial leap in difficulty between EduGA-Sci-Prim and EduGA-Sci-PPJC. I base this on a topical comparison between the species of fish or habitat descriptions found in EduGA-Sci-Prim and the processes, elements, and apparatus for Chemistry that were extracted from EduGA-Sci-PPJC. However, crucially, the objective change in semantic complexity illustrated in Figure 11 shows the extent of this shift in an objective and reproducible manner. The same can be said of the differences between the total TF-IDF scores attributed to terms in EduGA-Sci when compared to EduGA-Irish. Where the cumulative TF-IDF score of terms

accounted for only ~1% of the TF-IDF score at FreqRanges 1 and 2 in EduGA-Irish-Prim, EduGA-Irish-PPJC ,and EduGA-Irish-PPSC the cumulative TF-IDF scores at FreqRanges 1 and 2 are very different in EduGA-Sci. The cumulative TF-IDF score of FreqRanges 1 and 2 in EduGA-Sci-Prim and EduGA-Sci-PPJC accounts for ~10% of the total TF-IDF score for those levels, and for 25% in EduGA- Sci-PPSC. By virtue of a larger percentage of the topicality score depending on low-frequency terms, this objectively shows EduGA-Sci materials are more semantically complex and difficult for learners than EduGA-Irish materials at these FreqRanges.

In conclusion, the terms with the largest percentage of the TF-IDF were the most important and topical to the meaning of the document, as well as being inherently complex language features by virtue of being terms. The examples given in Table 48 and Table 52 are the type of feedback that can be given to materials creators by using this methodology. It is clear that the lemmas included in each frequency range need to be updated using a more modern corpus before giving teaches and materials creators access to this tool.

The aim here was to provide objective feedback to materials creators within a given context, rather than simply listing what is wrong or right for each grade level. This feedback can then be interpreted by materials creators, and changes made to their materials if or where they see fit in accordance with the needs of their learners.

6. Conclusions, Discussion, and Future Work

This thesis set out to develop a set of complexity metrics for educational materials written in Irish by analysing the way language in educational materials becomes more complex from one level of education to the next. The aim was for the analyses upon which the complexity metrics are based to be objective, therefore, a corpus of educational materials had to be compiled. The following research questions were posed in Section 1.4.

i) How are items prescribed in the curriculum or syllabus reflected in educational materials? By this I mean, do prescribed language features occur at a frequency that is statistically significant at the level of education for which they are first prescribed?
ii) Can complexity measures used on other languages be used to objectively evaluate the complexity of Irish-language and Irish-medium educational materials?
iii) Can terminology be used to measure the semantic complexity of educational materials?

A key consideration of this research was the creation of resources so as to facilitate future research in this field. As such, wordlists have been included in Appendix 1: Extended Results and Referential Datasets. This thesis also includes as three pieces of empirical analysis on Irish-language data. These analyses are the first of their kind for Irish, and put the issues outlined in Section 1.2 in empirical terms for the first time. This research paves the way for improvements in curriculum and syllabus design as well as the material creation by providing objective and statistical feedback for the writers of such documents. The second piece of analysis in this research paves the way for empirically-based complexity studies of Irish texts, as well providing a means of empirically describing difficulty. The third piece of analysis established and implemented a novel methodology for extracting and analysing Irish terminology, a process that had not been done automatically for Irish. This methodology will certainly be of benefit to the terminological and lexicographical work currently being undertaken for Irish.

6.1. Statistical Signficance of Prescribed Language Features

The seven prescribed lexico-grammatical language features that were analysed were prescribed for both mainstream education, from primary level to post-primary senior cycle, and for CEFR-GA courses, from level A1 to B2. Prescribed language features were chosen for this analysis because of their relevance to the present research, and because I could be certain they would feature in the educational materials. Less than half of the prescribed language features occurred at a frequency that was statistically significant at the level for which the language feature was first prescribed. Statistical significance is defined here being greater than one standard deviation above the mean.

In a number of cases, in materials for both mainstream education and CEFR-GA courses, the prescribed language feature achieved statistical significance at the level subsequent to the level for which it was first prescribed. This was indicative of a language feature being introduced in a limited manner, before being elaborated at subsequent levels. In the other cases it was not clear if a pedagogically-motivated tactic was being employed. When the results of all seven analyses for mainstream education were considered as a group it became clear that materials for junior cycle at post-primary level were somewhat of a problem child. I describe them in this manner because the frequency of language features was lower for junior cycle at post-primary than they did at primary level. By contrast, the same language features showed consistent increases in frequency from one level of education to the next in the CEFR-GA corpora. Figure 12 shows these erratic frequencies in the materials for mainstream education, while Figure 13 shows the steady changes in frequency in CEFR-GA materials. These results do not mean this method cannot be used to assess the grade level of educational materials, they indicate that materials for mainstream education may not provide reliable referential frequencies for conditional-mood verbs and for the autonomous forms of verbs, whereas materials for CEFR-GA courses appear to provide more reliable referential frequencies against which other educational materials could be compared.

Figure 12: Frequency of Prescribed Language Features in Mainstream Education

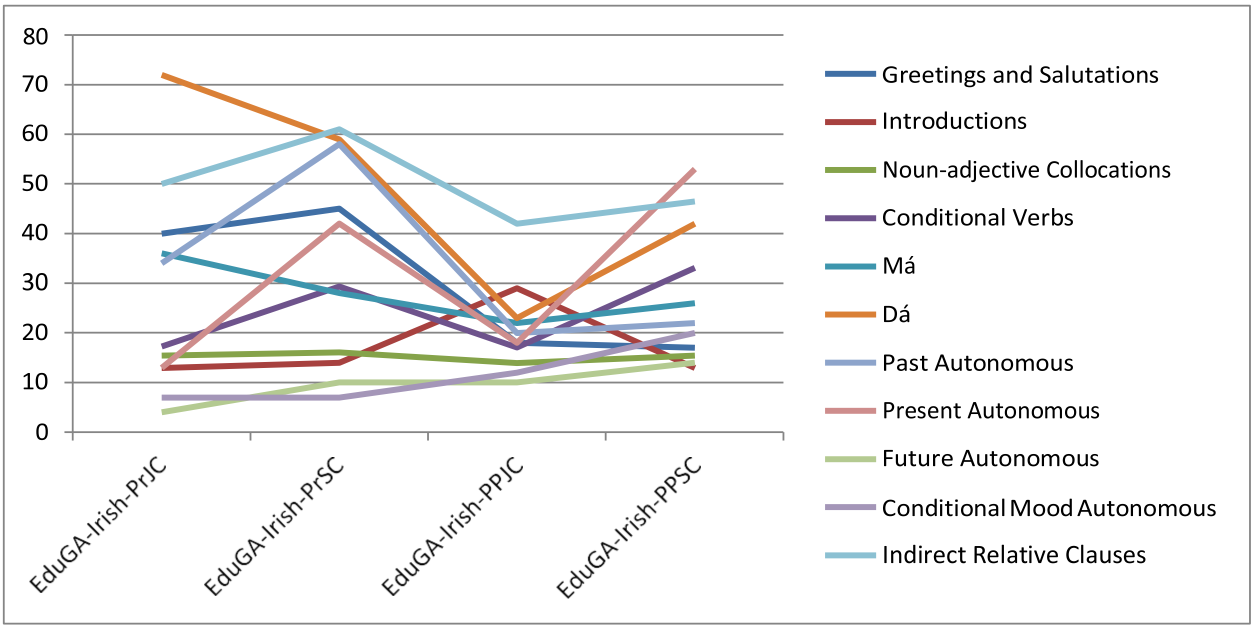
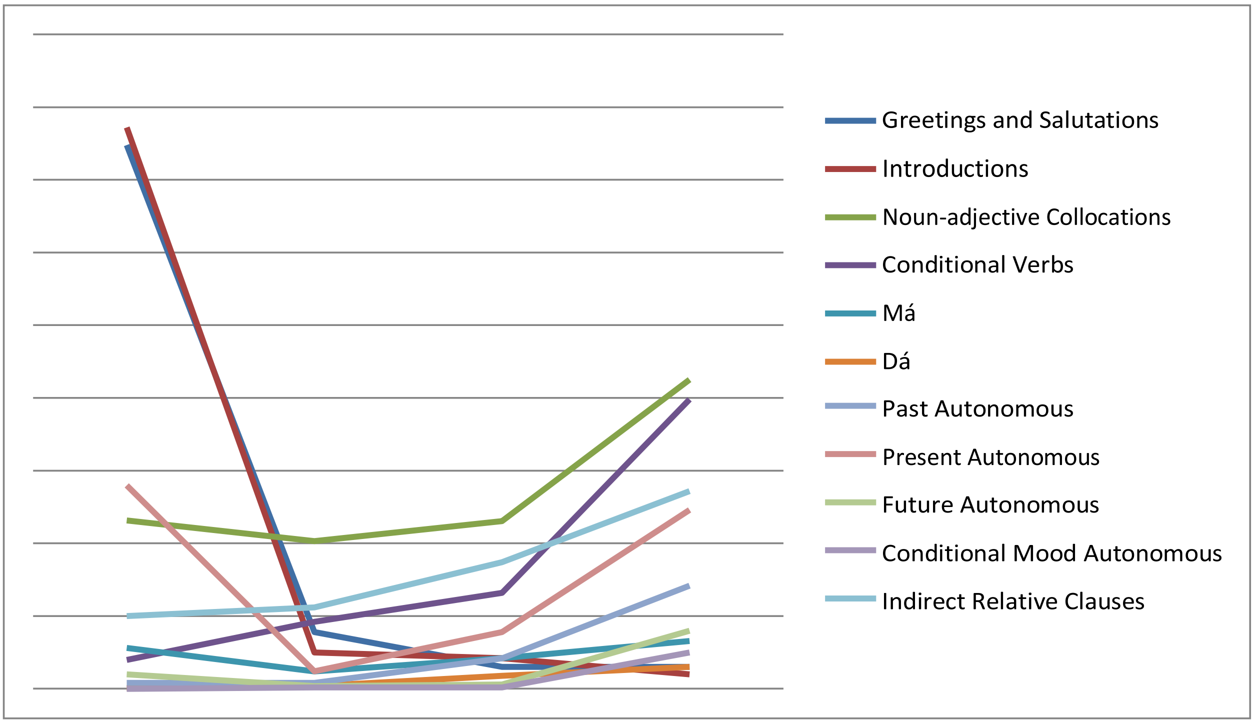


Figure 13: Frequency of Prescribed Language Features in CEFR-GA Materials



If prescribed features are to be used to assess the complexity of a document then mainstream materials are not suitable as reference points. On the other hand, CEFR-GA materials do appear to provide reliable reference points.

6.2. Length of Word and Length of Sentence

Length of word and length of sentence were chosen as analytic steps for the present research because they are widely used in complexity studies of other languages, and because they contribute values to the algorithms behind a number of readability tests.

Length of word was found to be an unrealiable indicator of complexity for Irish-language materials where the number of letters in a word was used as the basis for the analysis. This was found to be the case because Irish-language morphology can double the length of a number of verbs in the present or future tense, for example, and because a number of plurals are formed in a manner that also doubles the length of the word.

Length of sentence was found to be a more reliable indicator of complexity for Irish-language materials, with the average length of sentences generally increasing with each increase in grade level. The average length of sentences in materials for junior cycle at post-primary level (11 words per sentence) was found to be far closer to primary level (9 words per sentence) than senior cycle at post-primary level (18 words per sentence). Upon establishing the average length of sentences in news (24 words per sentence) and magazine articles was (22 words per sentence) it was concluded that the results for junior-cycle post-primary materials were once again problematic, and that they should ideally by higher rather than sentences in senior-cycle post-primary materials being shorter.

The average length of sentences in CEFR-GA materials increased by either 1 or 2 words with each level. once again showing a cohesiveness between levels of education that was not being matched in the mainstream materials. It must, however, be noted that the largest average sentence length was only 13 words, which falls considerably short of the average length of sentences in the two types of media articles. Whether there were pedagogical reasons for these relatively short sentences remains to be seen.

I conclude that this metric does appear to be a reliable indicator of syntactic complexity for educational materials, however, the average length of sentences may need to be raised slightly across all CEFR-GA levels before these average sentence lengths can be used referentially when assessing educational materials. Once again, the results gathered from analysing materials for junior cycle at post-primary level have found the materials to be somewhat of a problem child – showing no cohesion with primary-level materials or senior-cycle materials.

6.3. Term-based Complexity Measure

Terminology was chosen as the basis for this analysis because of the inherent level of complexity associated with terms relative to common non-technical words. It has been prescribed as a complex feature in some reference documents, and can be inferred from others where it is stated that the learner will need to acquire specialized vocabulary. This research benefits from a well-developed database of terms being available as a resource, a factor that cannot be overstated because of Irish's status as an under-resourced language.

The development done during the course of this research has brought the term-based analysis to a point where it successfully filters out noise from non-terminological words. The methodology was developed in a manner that considered the local context, within the sub-corpus of educational materials, as well as the general wider context, by adding a dimension that was based on a corpus of general Irish.

The methodology objectively showed the extent to which science materials are more semantically complex than Irish-language materials. This difference was found to be significant, given that primary-level science materials were found to be considerably more complex than materials for senior cycle at post-primary level until FreqRange 4. One of the benefits of this complexity metric is its ability to provide feedback to materials creators, feedback they can decide to act on or not. I conclude that terminology can be used as a basis for measuring semantic complexity in a relativistic manner.

6.4. Discussion and Future Work

When collecting data for EduGA it became clear that there was no shortage of data; time constraints were the primary reason more data was not included. Therefore, the option to use EduGA as a monitor corpus that continues to grow as new educational materials are published has been considered for the future.

However, with structural changes occurring in post-primary education the primary focus of future research may be on diachronic studies.

Some of the literature appeared to use the terms difficult(ty) and complex(ity) interchangeably. The present research defines “difficulty” as a learner-centric phenomenon that is unique to each person. “Complexity” is defined as a universally accepted means of objectively ranking language features. This highlighted the fact no fundamental research had been conducted on complexity for Irish up to now. By this I mean, for example, that it is widely accepted that relative clauses are complex structures in Irish (cf. Mac Murchaidh (2002)) as well as other lanuages but the various features of the clause structure that contribute to its complexity have not been explored. This research has shown how complexity can be objectively measured, therefore teased apart from difficulty.

While some state-of-the-art NLP tools are available for Irish, it became apparent during the course of this research that a great number of the tools available to researchers of other European languages have not yet been developed, or adapted, for Irish. While I was able to use some language-independent tools, some of the processing and pre-procesing tasks completed would not have been possible if it were not for the programming skills I have recently required out of necessity. I believe a standard NLP toolkit for Irish-language linguists could be developed so that this barrier to learning, research, and analysis did not result in worthwhile research not being completed.

During the course of this research several methodologies have been tested on Irish-language data for the first time, as well as the term-based analysis that was developed during the course of this research. It is important that momentum is not lost and that research in this area continues. An advisable next step appears at this point to be testing the trasferability of complexity metrics from other languages.

With three distinct strands of analysis brought to conclusion it is my aim to communicate these results to the Department of Education in a manner that helps inform their review processes, with a view to helping improve curriculum and syllabus design by adding a new type of knowledge to this review process. The results of this study could also be used to assist in the generation of a larger set of examples for each language function at each level of education, from which teachers and materials creators can then draw.

I also aim to communicate these results to the creators of materials in a manner that helps them make positive developments to their materials as well as giving them the confidence to deviate from the status quo where appropriate. Finally, both the results of this research and its methodologies need to be understoof by teachers and trainee teachers because they are the group who needs access to tools, information, and support most of all. I say this with consideration for the variety of learners types, issues, and needs in every classroom. Once communicated, I believe this support would be best delivered via a web application into which text could be entered and feedback generated.

Appendix 1: Extended Results and Referential Datasets

- Further results from analysis in Section 5.1

Figure 14: Introductions

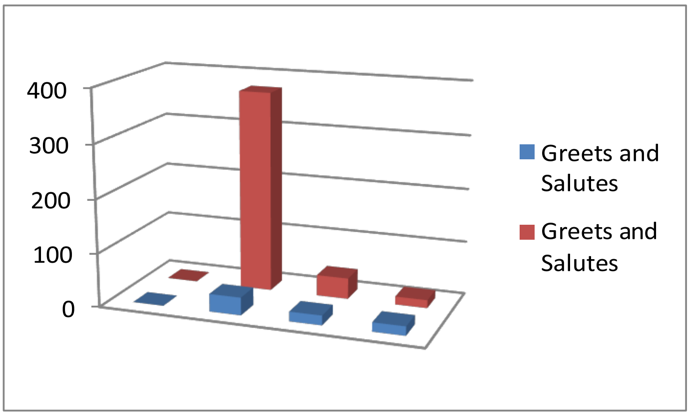


Figure 15: Greetings and Salutations

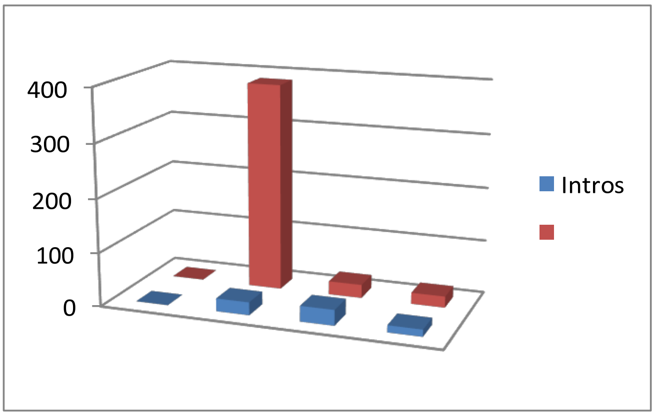


Figure 16: Noun-Adjective Collocations

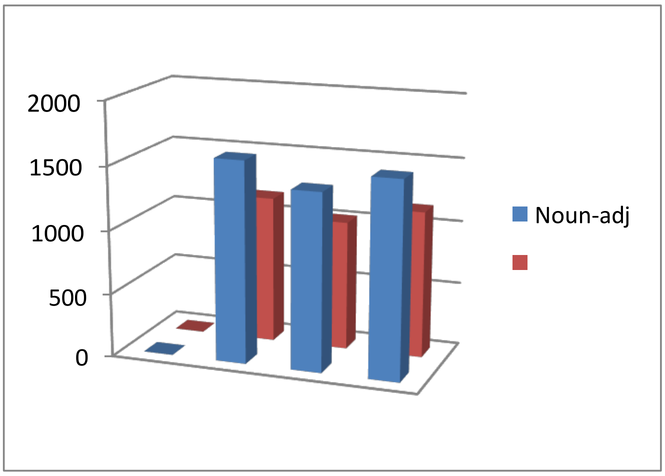


Figure 17: Conditional-Mood Verbs

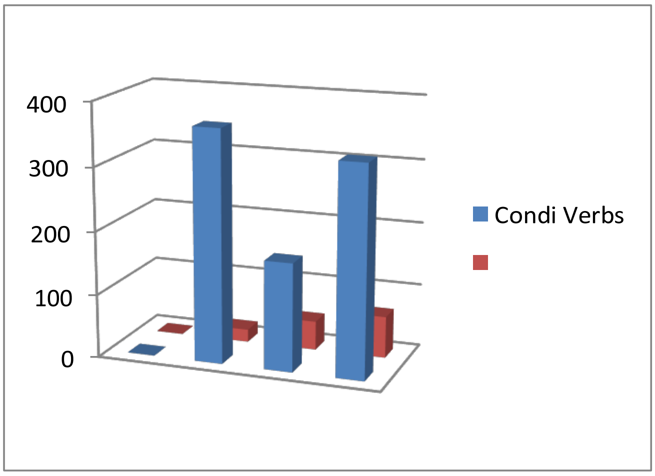


Figure 18: Conditional-Mood Particles in Mainstream Materials

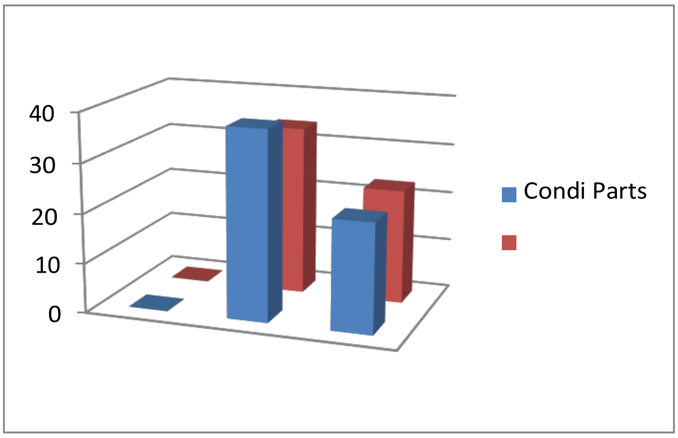


Figure 19: Conditional-Mood Particles in CEFR-GA Materials

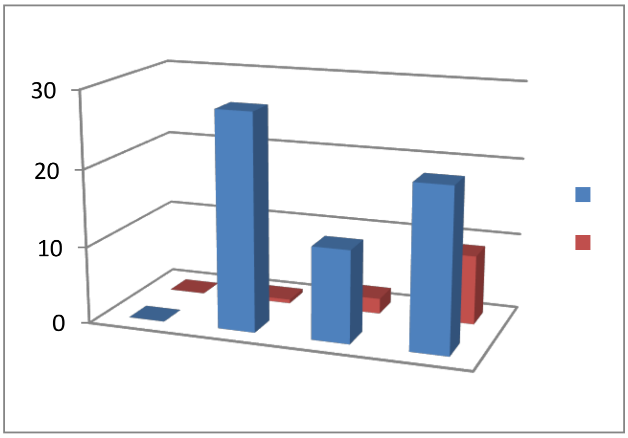


Figure 20: Autonomous Verbs in Mainstream Materials

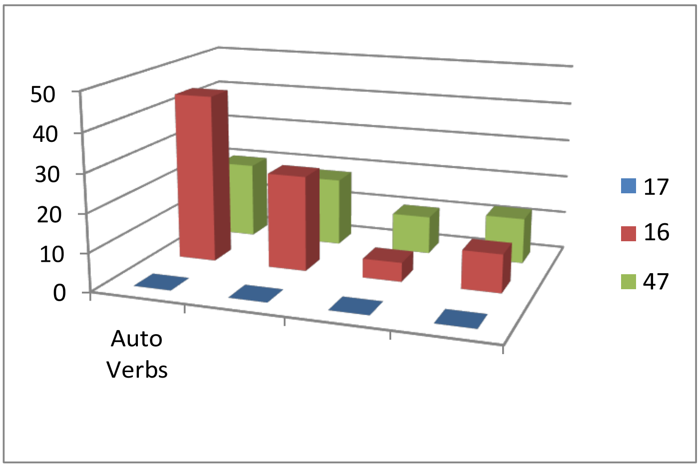


Figure 21: Autonomous Verbs in CEFR-GA Materials

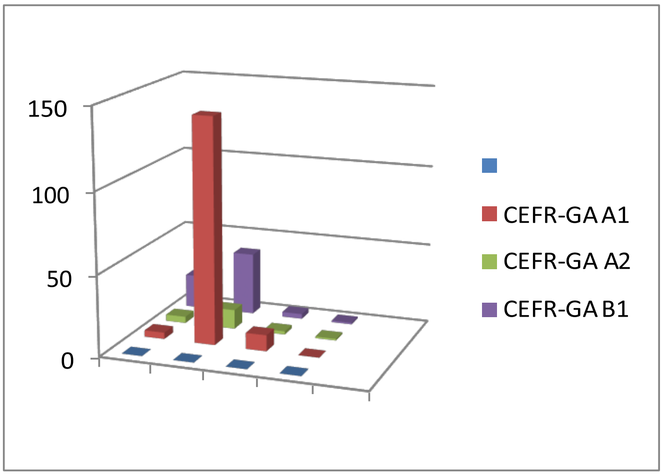


Figure 22: Indirect Relative Clauses

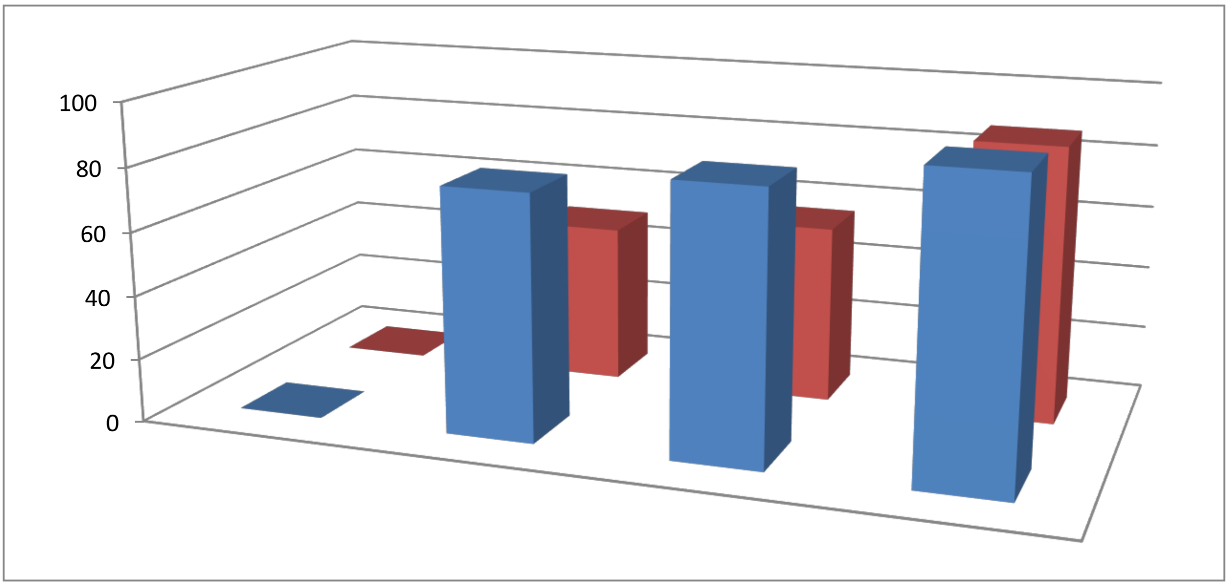


Table 53: Word lists for EduGA-Irish-Prim

EduGA-Irish-PrJC **Rank – Frequency – Token**			**EduGA-Irish-PrSC** **Rank – Frequency – Token**		
1	2994	an	1	7560	an
2	2767	a	2	6427	a
3	1560	ar	3	4006	ar
4	1473	agus	4	3119	agus
5	1254	na	5	2604	na

6	1113	arsa	6	2553	ag
7	750	ag	7	2325	bhí
8	706	sé	8	2315	sé
9	610	go	9	2132	go
10	598	ní	10	1528	i
11	591	tá	11	1243	le
12	587	bhí	12	1056	raibh
13	523	seo	13	984	is
14	503	is	14	982	sa
15	480	do	15	969	mé
16	477	le	16	918	ní
17	446	mé	17	911	ach
18	430	é	18	901	sí
19	404	tú	19	857	é
20	402	ná	20	788	leis
21	399	sin	21	783	do
22	398	i	22	759	seo
23	396	leis	23	688	tá
24	372	gaeilge	24	664	sin
25	368	áiseanna	25	617	siad
26	346	amach	26	598	ina
27	341	aon	27	574	mo
28	310	atá	28	531	in
29	299	ann	29	528	de
30	298	sa	30	527	tú
31	297	ach	31	484	as
32	296	sí	32	476	eile
33	268	maith	33	467	arsa
34	267	in	34	449	ann
35	258	bhfuil	35	443	féin
36	249	mo	36	438	ó
37	246	teagaisc	37	430	mar
38	245	féin	38	423	nuair
39	244	bunscoileanna	39	415	amach

40	244	oideachais	40	415	nach
41	229	raibh	41	414	bhfuil
42	226	liam	42	413	aon
43	225	ó	43	406	air
44	222	isteach	44	394	nó
45	221	baile	45	391	lá
46	221	átha	46	370	ná
47	219	cliath	47	366	atá
48	214	liom	48	365	gach
49	212	nó	49	352	faoi
50	209	ina	50	343	ba
51	203	eile	51	317	scríobh
52	198	daidí	52	314	isteach
53	197	anois	53	310	iad
54	182	nuair	54	300	ansin
55	170	gearóid	55	298	acu
56	168	ansin	56	296	í
57	163	chun	57	293	gur
58	158	ceart	58	281	cén
59	157	as	59	279	dtí
60	156	den	60	264	cé
61	155	fháil	61	263	anois
62	153	chuid	62	262	chuir
63	152	cá	63	252	den
64	151	gan	64	252	thug
65	150	obair	65	250	leat
66	149	mhaisigh	66	249	aige
67	145	chonaic	67	240	duine
68	144	sráid	68	238	arís
69	143	chur	69	238	rud
70	143	gach	70	236	dá
71	141	mise	71	236	maith
72	139	seisean	72	234	chun
73	137	leagan	73	227	scéal

74	137	mór	74	223	chomh
75	136	éirinn	75	222	agat
76	135	scríobh	76	220	san
77	135	ón	77	218	agam
78	130	caibidil	78	217	dul
79	127	bíodh	79	216	liom
80	126	cead	80	212	mór
81	126	mamaí	81	212	níor
82	126	roimh	82	209	don
83	125	baineann	83	209	dúirt
84	125	chéad	84	199	thosaigh
85	123	ré	85	199	tháinig
86	123	slí	86	197	á
87	122	bunaithe	87	195	gan
88	122	cosaint	88	189	chonaic
89	122	gaeltachta	89	189	scoil
90	122	gcomhad	90	188	bheadh
91	122	leictreonach	91	184	bheith
92	122	lán-ghaeilge	92	184	faoin
93	122	meicniúil	93	181	chuid
94	122	mhodh	94	181	páistí
95	121	í	95	181	síos
96	119	cad	96	180	bhaile
97	119	chlóbhuail	97	180	focail
98	119	ealaíne	98	177	anseo
99	119	níl	99	174	níos
100	118	agam	100	172	dó

Table 54: Wordlists for EduGA-Irish-PPxx

EduGA-Irish-PPJC Rank – Frequency – Token			EduGA-Irish-PPSC Rank – Frequency – Token		
1	10484	an	1	16493	an
2	9529	a	2	13080	a
3	5741	ar	3	8315	ar
4	4777	na	4	7826	agus
5	4411	agus	5	5917	ag
6	3793	i	6	5842	na
7	3597	ag	7	5108	go
8	2565	go	8	4783	bhí
9	2440	sé	9	4122	sé
10	2389	is	10	3072	i
11	2253	le	11	2826	tá
12	2073	bhí	12	2663	is
13	2058	sa	13	2646	le
14	1888	tá	14	2590	sa
15	1769	the	15	2075	raibh
16	1654	tú	16	2073	leis
17	1647	mé	17	2055	é
18	1487	sí	18	2050	sí
19	1453	ní	19	1891	siad
20	1442	do	20	1813	ní
21	1347	seo	21	1765	in
22	1346	mo	22	1691	sin
23	1325	in	23	1624	ach
24	1098	siad	24	1398	seo
25	1082	scríobh	25	1374	mé
26	931	leis	26	1247	the
27	868	cén	27	1211	atá
28	865	gach	28	1187	ina
29	852	nuair	29	1181	mar
30	797	é	30	1169	pictiúr
31	785	atá	31	1114	de

32	768	sin	32	1107	bhfuil
33	763	raibh	33	1026	ann
34	757	bhfuil	34	959	féin
35	725	thíos	35	941	tú
36	723	ach	36	884	air
37	678	ann	37	878	do
38	636	aonad	38	847	nach
39	610	ó	39	835	san
40	601	scoil	40	829	faoi
41	583	liom	41	825	as
42	550	aon	42	823	duine
43	547	faoi	43	821	síos
44	544	amach	44	813	eile
45	537	mar	45	809	gur
46	525	as	46	807	nuair
47	522	de	47	803	acu
48	512	cuir	48	789	mo
49	503	ceisteanna	49	762	dá
50	496	to	50	707	ó
51	493	ina	51	652	don
52	490	san	52	634	ba
53	487	nó	53	617	déan
54	477	trí	54	614	aige
55	472	habairtí	55	609	gach
56	470	cad	56	609	í
57	464	lá	57	598	liom
58	456	labhair	58	594	amach
59	446	cleachtadh	59	586	ná
60	446	sibh	60	565	leat
61	425	chun	61	560	cén
62	425	of	62	559	nó
63	416	don	63	556	cur
64	414	aimsir	64	534	lá
65	411	cuid	65	532	iad

66	411	my	66	531	aon
67	411	teach	67	526	fear
68	410	maith	68	523	dul
69	404	cá	69	502	ceisteanna
70	401	he	70	502	chun
71	399	seomra	71	502	gan
72	398	leat	72	500	mór
73	397	obair	73	487	tháinig
74	396	duine	74	464	den
75	394	féin	75	461	chuir
76	392	níos	76	460	dúirt
77	386	í	77	458	bheith
78	370	dul	78	454	isteach
79	368	and	79	450	chéile
80	367	isteach	80	441	scríobh
81	364	gaeilge	81	433	daoine
82	360	foghlaim	82	429	á
83	355	eile	83	427	maith
84	353	rang	84	425	dhéanamh
85	352	bhaile	85	418	chéad
86	346	air	86	406	uirthi
87	346	was	87	404	níl
88	343	níor	88	403	agam
89	342	chuig	89	403	rud
90	342	ná	90	402	dó
91	341	nach	91	401	ansin
92	340	dá	92	399	níor
93	331	bíonn	93	397	arís
94	330	dó	94	397	thug
95	330	faoin	95	393	níos
96	327	mór	96	392	chur
97	327	scoile	97	386	cad
98	303	cloisfidh	98	385	leo
99	301	daoine	99	383	muid

100	299	cé	100	382	orthu

Table 55: Wordlists for EduGA-CEFR-A1/A2

EduGA-CEFR-A1 **Rank – Frequency – Token**			**EduGA-CEFR-A2** **Rank – Frequency – Token**		
1	7536	an	1	10613	an
2	7343	a	2	8473	a
3	6032	the	3	6034	the
4	4169	i	4	5581	i
5	4092	tá	5	4221	you
6	3829	you	6	3920	ar
7	3561	is	7	3853	tá
8	2933	in	8	3598	is
9	2678	go	9	3283	in
10	2375	ar	10	3198	go
11	2251	do	11	3096	ag
12	2038	to	12	2947	mé
13	2035	ag	13	2673	tú
14	1989	agus	14	2551	to
15	1947	tú	15	2478	agus
16	1874	mé	16	2244	na
17	1836	sé	17	2084	do
18	1654	na	18	2075	sé
19	1465	and	19	2073	and
20	1287	maith	20	1738	le
21	1258	le	21	1539	bhfuil
22	1221	bhfuil	22	1492	ní
23	1169	of	23	1324	it
24	1162	are	24	1212	of
25	1074	mo	25	1202	mo
26	1014	what	26	1181	cén
27	1013	have	27	1173	atá
28	977	it	28	1150	maith

29	938	ní	29	1148	leat
30	874	your	30	1147	sa
31	841	é	31	1048	bhí
32	840	atá	32	1005	are
33	823	seo	33	989	have
34	819	agat	34	977	é
35	792	sa	35	929	on
36	764	leat	36	925	liom
37	748	on	37	851	seo
38	740	liom	38	838	agam
39	706	cén	39	836	with
40	698	with	40	812	as
41	696	at	41	810	cad
42	654	as	42	810	duine
43	630	cad	43	804	sin
44	621	how	44	795	at
45	611	irish	45	779	leis
46	594	or	46	774	that
47	587	dé	47	771	your
48	581	where	48	756	what
49	558	téim	49	755	agat
50	547	that	50	731	obair
51	533	this	51	643	sí
52	531	agam	52	621	how
53	531	words	53	619	for
54	530	obair	54	589	yes
55	530	teach	55	585	words
56	510	comhrá	56	584	very
57	508	person	57	576	mar
58	507	yes	58	573	níl
59	506	listen	59	561	ach
60	500	for	60	557	this
61	500	no	61	552	raibh
62	482	can	62	533	no

63	479	bhí	63	530	dul
64	462	níl	64	528	dé
65	460	from	65	507	don
66	451	sí	66	487	cuid
67	450	people	67	485	lá
68	444	these	68	483	comhrá
69	442	leis	69	481	graphic
70	433	amach	70	477	my
71	429	dtí	71	476	dtí
72	408	seomra	72	476	féin
73	400	sin	73	473	cá
74	399	below	74	470	de
75	397	don	75	465	person
76	390	name	76	464	ansin
77	371	chónaí	77	463	like
78	369	ceart	78	451	ó
79	356	orm	79	448	these
80	351	my	80	442	can
81	349	here	81	437	am
82	347	she	82	428	work
83	346	word	83	412	amach
84	345	am	84	412	deas
85	343	like	85	410	about
86	342	two	86	402	oíche
87	340	he	87	401	chuig
88	339	cé	88	399	out
89	333	front	89	393	faoi
90	331	beidh	90	391	she
91	322	rang	91	389	aice
92	321	deas	92	388	ann
93	316	back	93	384	phrases
94	315	live	94	383	anois
95	313	class	95	383	be
96	310	pháirtnéir	96	377	beidh

97	309	leor	97	372	cé
98	298	ainm	98	367	siad
99	295	gach	99	358	seachtaine
100	295	mar	100	357	níor

Table 56: Wordlists for EduGA-CEFR-B1/B2

EduGA-CEFR-B1 **Rank – Frequency – Token**			**EduGA-CEFR-B2** **Rank – Frequency – Token**		
1	16403	an	1	16440	an
2	11535	a	2	12548	a
3	7087	ar	3	7849	ar
4	5641	na	4	6862	na
5	5433	agus	5	5666	agus
6	5317	i	6	5581	i
7	4497	go	7	4654	ag
8	4362	ag	8	4473	go
9	3967	tá	9	3472	is
10	3886	tú	10	3451	the
11	3331	is	11	2957	le
12	2884	the	12	2883	tú
13	2829	mé	13	2770	tá
14	2813	le	14	2339	in
15	2605	sé	15	2321	sa
16	2495	in	16	2131	bhfuil
17	2475	sa	17	2080	sé
18	2346	ní	18	2054	seo
19	2288	do	19	2049	do
20	2266	leat	20	1929	mé
21	1929	bhfuil	21	1859	ní
22	1909	bhí	22	1761	leat
23	1837	seo	23	1757	leis
24	1759	sin	24	1703	bhí
25	1729	duine	25	1661	atá

26	1675	atá	26	1634	sin
27	1506	leis	27	1579	mar
28	1400	cén	28	1388	cén
29	1368	agat	29	1183	duine
30	1329	sí	30	1179	raibh
31	1278	maith	31	1143	you
32	1153	é	32	1123	de
33	1143	ach	33	1110	é
34	1107	mar	34	1104	cad
35	1097	raibh	35	1048	iad
36	1073	eile	36	1035	ach
37	1066	cad	37	1015	eile
38	1063	liom	38	993	agat
39	1052	féin	39	974	maith
40	1039	agam	40	969	to
41	1032	as	41	967	ó
42	1014	nó	42	957	lá
43	992	sibh	43	956	faoi
44	982	siad	44	952	liom
45	958	faoi	45	932	as
46	932	de	46	928	nó
47	931	mo	47	920	féin
48	880	iad	48	822	bheith
49	871	you	49	769	ann
50	865	nuair	50	752	obair
51	854	thíos	51	746	daoine
52	854	ó	52	727	gach
53	824	obair	53	713	gaeilge
54	795	aice	54	706	siad
55	785	to	55	692	thíos
56	776	ann	56	659	san
57	760	chéile	57	654	of
58	758	anois	58	650	amach
59	726	san	59	649	agam

60	682	daoine	60	637	nach
61	672	níl	61	622	ansin
62	663	lá	62	596	aice
63	654	beidh	63	594	níos
64	635	nach	64	588	déag
65	598	gach	65	586	nuair
66	596	ansin	66	586	sibh
67	595	ceisteanna	67	576	aon
68	585	cé	68	569	chuid
69	583	chun	69	554	mo
70	578	bheith	70	550	anois
71	577	aon	71	549	mó
72	561	amach	72	548	chun
73	553	ba	73	540	sí
74	546	of	74	526	chéile
75	541	ceist	75	525	and
76	499	áit	76	524	scoil
77	497	dtí	77	522	ba
78	491	post	78	513	ina
79	486	gaeilge	79	507	ceisteanna
80	484	ina	80	499	den
81	480	rud	81	498	cur
82	474	chuid	82	484	ná
83	465	ná	83	479	beidh
84	464	dul	84	478	chur
85	462	teach	85	478	fáth
86	453	and	86	471	tar
87	448	fearr	87	471	ón
88	445	don	88	469	teanga
89	444	cur	89	462	don
90	442	comhrá	90	455	dá
91	439	mór	91	449	it
92	429	it	92	443	cé
93	426	níos	93	442	fáil

94	425	dhéanamh	94	441	thaitin
95	417	bíonn	95	438	dhéanamh
96	412	freagraí	96	436	amháin
97	401	minic	97	436	éis
98	399	aimsir	98	428	fearr
99	398	am	99	427	níor
100	397	mhaith	100	419	acu

Table 57: Wordlists for EduGA-CEFR-C1/C2

EduGA-CEFR-C1			**EduGA-CEFR-C2**		
Rank – Frequency – Token			**Rank – Frequency – Token**		
1	27437	an	1	10313	an
2	24265	a	2	7923	a
3	14146	ar	3	4023	na
4	11479	na	4	3907	ar
5	11347	agus	5	3486	agus
6	10022	the	6	3109	i
7	8690	i	7	2583	go
8	7249	go	8	2560	ag
9	7143	ag	9	2351	the
10	6129	is	10	1970	is
11	6064	le	11	1721	in
12	5601	in	12	1685	le
13	4583	tá	13	1632	tá
14	4475	of	14	1291	sa
15	3921	seo	15	1181	seo
16	3652	sa	16	1181	sé
17	3495	to	17	1163	sin
18	3263	de	18	1142	de
19	3174	sin	19	931	é
20	3045	sé	20	902	ní
21	2995	and	21	901	leis
22	2869	do	22	892	of

23	2865	mar	23	842	mar
24	2744	atá	24	814	bhí
25	2669	leis	25	798	bhfuil
26	2504	bhfuil	26	727	atá
27	2462	bhí	27	704	do
28	2356	tú	28	693	ach
29	2222	é	29	693	to
30	2211	ní	30	666	duine
31	1977	as	31	661	mé
32	1834	ó	32	657	ó
33	1829	nó	33	654	tú
34	1807	ach	34	580	faoi
35	1704	gaeilge	35	575	and
36	1692	faoi	36	554	féin
37	1650	mé	37	536	as
38	1535	san	38	535	siad
39	1493	eile	39	532	raibh
40	1473	leat	40	481	daoine
41	1459	obair	41	476	nó
42	1409	duine	42	473	níos
43	1380	den	43	463	ann
44	1332	ann	44	462	gach
45	1308	raibh	45	459	san
46	1306	fáil	46	450	nuair
47	1280	iad	47	447	acu
48	1265	chur	48	445	ina
49	1236	chun	49	435	leat
50	1229	féin	50	433	nach
51	1220	cén	51	429	eile
52	1218	gach	52	424	grúpa
53	1201	for	53	419	was
54	1142	aon	54	410	gur
55	1135	that	55	406	chur
56	1132	bheith	56	398	sí

57	1123	leagan	57	365	aimsir
58	1117	on	58	356	gaeilge
59	1113	thíos	59	345	iad
60	1072	daoine	60	337	don
61	1071	don	61	337	he
62	1067	nuair	62	334	dá
63	1067	siad	63	333	ná
64	1059	ina	64	332	thíos
65	1014	gur	65	323	bheith
66	1005	ba	66	322	ba
67	1000	nach	67	321	amach
68	998	ná	68	321	aon
69	991	amach	69	318	leo
70	985	bhaile	70	309	rud
71	979	agat	71	308	amháin
72	960	níos	72	308	mór
73	952	be	73	301	cén
74	929	aistriúchán	74	299	deireadh
75	919	sí	75	298	den
76	907	chuid	76	297	maith
77	902	it	77	296	chuid
78	889	cad	78	290	sna
79	868	dá	79	288	lá
80	842	mór	80	287	mó
81	839	maith	81	285	má
82	837	dhéanamh	82	284	alt
83	835	mo	83	281	ansin
84	829	acu	84	281	tar
85	829	lá	85	279	dhéanamh
86	824	you	86	277	chun
87	811	níl	87	269	scríobh
88	810	with	88	264	air
89	807	or	89	263	beidh
90	789	féidir	90	254	gan

91	783	beidh	91	253	ceart
92	777	air	92	252	féidir
93	770	gan	93	250	idir
94	764	was	94	250	she
95	758	aistriúcháin	95	248	bain
96	758	idir	96	247	it
97	756	teanga	97	246	chéile
98	755	mó	98	244	ainmfhocal
99	747	rud	99	244	you
100	721	amháin	100	243	cuir

EduGA-Sci-Prim			**EduGA-Sci-PPJC**			**EduGA-Sci-PPSC**		
Rank – Frequency – Token			**Rank – Frequency – Token**			**Rank – Frequency – Token**		
1	6152	an	1	8875	an	1	6330	an
2	5518	a	2	8348	a	2	5260	a
3	3857	ar	3	3775	ar	3	2762	ar
4	3313	agus	4	3649	agus	4	2667	agus
5	2720	na	5	3266	na	5	2020	na
6	1592	i	6	2387	i	6	1420	le
7	1516	is	7	1713	le	7	1320	i
8	1241	le	8	1531	is	8	1194	go
9	1196	go	9	1496	go	9	1169	ag
10	1109	ag	10	1474	cid	10	1068	is
11	955	é	11	1355	ag	11	873	sa
12	836	sin	12	1265	sa	12	818	de
13	726	sé	13	1223	uisce	13	727	uisce
14	707	www	14	1150	de	14	710	atá
15	700	de	15	901	atá	15	708	seo
16	678	mara	16	862	sin	16	646	é
17	674	in	17	776	seo	17	624	sin
18	632	sa	18	766	chun	18	599	mar
19	630	tá	19	732	é	19	598	do
20	595	bíonn	20	712	mar	20	596	chun
21	563	leis	21	675	leis	21	581	nó

22	534	ó	22	674	tá	22	569	tá
23	516	gach	23	665	bhfuil	23	567	léinn
24	516	nó	24	650	cuir	24	533	sé
25	509	inis	25	650	in	25	487	cuir
26	504	chuid	26	647	nó	26	486	don
27	497	iad	27	622	sé	27	485	leis
28	495	mar	28	614	bhíonn	28	476	bhfuil
29	477	san	29	611	do	29	474	in
30	463	arcáin	30	593	as	30	409	den
31	461	stáisiún	31	567	léinn	31	403	féidir
32	454	ann	32	561	den	32	392	as
33	451	dúlra	33	558	ó	33	392	tú
34	441	the	34	512	ann	34	376	cad
35	440	as	35	492	don	35	375	nuair
36	432	ach	36	468	cad	36	362	ábhar
37	421	atá	37	464	iad	37	358	fíor
38	416	chun	38	448	cén	38	347	úsáid
39	415	bhíonn	39	444	nuair	39	345	déan
40	407	do	40	437	ina	40	339	ann
41	393	líon	41	432	féidir	41	334	cén
42	385	ní	42	429	san	42	319	mac
43	356	seo	43	428	isteach	43	305	iad
44	338	ceart	44	426	tú	44	304	eile
45	334	tú	45	417	bíonn	45	302	bhíonn
46	332	siad	46	398	déan	46	301	isteach
47	323	cosaint	47	395	úsáid	47	299	dhéanamh
48	318	amach	48	390	fíor	48	297	níos
49	316	faoi	49	377	eile	49	294	amach
50	312	ceadúnóirí	50	373	amach	50	293	trí
51	308	ina	51	360	dhéanamh	51	288	gach
52	293	bhí	52	360	trí	52	282	faoi
53	288	eile	53	347	ná	53	277	ó
54	286	uisce	54	347	ábhar	54	270	ina
55	282	bhfuil	55	343	níos	55	264	bíonn

56	275	nuair	56	342	faoi	56	251	san
57	273	á	57	333	gach	57	236	síos
58	272	í	58	324	mac	58	235	aon
59	265	den	59	316	aon	59	232	mhac
60	264	mór	60	312	ón	60	231	ná
61	251	féin	61	305	síos	61	229	ón
62	247	dhéanamh	62	296	amháin	62	224	domhan
63	244	ná	63	296	chéile	63	223	dá
64	237	níos	64	295	dath	64	223	imscrúdú
65	218	air	65	293	ní	65	219	idir
66	216	cén	66	275	aigéad	66	218	ainm
67	214	bheith	67	268	idir	67	212	dtí
68	210	féidir	68	264	charbóin	68	210	chéile
69	210	of	69	264	dá	69	209	amháin
70	203	dá	70	263	cm	70	202	ní
71	196	héireann	71	254	léaráid	71	202	stets
72	186	agat	72	246	dhá	72	192	eolaíochta
73	186	mó	73	239	domhan	73	183	léaráid
74	184	obair	74	237	í	74	182	cm
75	182	chéile	75	233	mbíonn	75	177	cogg
76	180	mbíonn	76	233	mhac	76	176	jsss
77	179	nach	77	230	air	77	175	air
78	178	isteach	78	229	siad	78	175	dath
79	172	don	79	229	tuaslagán	79	172	dé
80	167	sí	80	226	dtí	80	166	bheith
81	163	aon	81	223	ach	81	166	seimineár
82	163	chomh	82	223	adamh	82	161	má
83	162	cuir	83	223	imscrúdú	83	161	siad
84	157	faoin	84	220	ainm	84	159	aigéad
85	155	rud	85	220	bia	85	159	tábla
86	154	bia	86	218	líon	86	156	ocsaíd
87	152	fad	87	217	nach	87	152	chur
88	151	daoine	88	211	eolaíocht	88	151	tástáil
89	149	leat	89	210	tábla	89	150	fáil

90	148	úsáid	90	209	athrú	90	147	modh
91	145	acu	91	209	bheith	91	147	mó
92	144	orthu	92	209	ocsaigin	92	144	nach
93	142	ón	93	206	br	93	143	ceann
94	139	ba	94	202	stets	94	143	miotal
95	137	dath	95	201	leictreon	95	142	tabhair
96	137	samhradh	96	199	ocsaíd	96	141	ach
97	133	déan	97	198	á	97	141	leanas
98	132	mé	98	197	ainmnigh	98	139	líon
99	132	thuaidh	99	194	eolaíochta	99	138	charbóin
100	131	ainmhithe	100	194	sóidiam	100	137	dhá

EduGA-Hist-Prim Rank – Frequency – Token			**EduGA-Hist-PPJC Rank – Frequency – Token**			**EduGA-Hist-PPSC Rank – Frequency – Token**		
1	4024	an	1	2948	an	1	1510	a
2	3276	a	2	2553	a	2	1027	an
3	2566	na	3	2319	na	3	779	na
4	1534	ar	4	1872	agus	4	553	i
5	1307	i	5	1731	ar	5	517	bhí
6	1240	agus	6	1242	bhí	6	509	ar
7	769	the	7	1002	ag	7	498	cén
8	672	sa	8	917	go	8	459	cad
9	633	ag	9	815	i	9	362	ag
10	602	bhí	10	656	sé	10	349	le
11	466	seo	11	642	le	11	338	agus
12	457	siad	12	533	sa	12	274	go
13	452	go	13	440	siad	13	210	leis
14	447	le	14	411	de	14	197	gceist
15	440	oibre	15	396	raibh	15	176	sa
16	425	l	16	368	in	16	162	fáth
17	401	stair	17	367	mar	17	151	tharla
18	399	é	18	344	chun	18	146	in
19	397	sé	19	319	ba	19	137	raibh
20	386	ó	20	313	leis	20	136	gur

21	380	in	21	300	sin	21	131	conas
22	376	mórthimpeall	22	299	daoine	22	130	ó
23	366	um	23	290	ach	23	126	de
24	365	orm	24	285	is	24	97	saghas
25	363	chomhairle	25	283	ina	25	97	éirinn
26	363	gaeltachta	26	282	seo	26	88	é
27	363	oideachas	27	275	é	27	80	rith
28	362	gaelscolaíochta	28	267	ó	28	76	amháin
29	358	bileog	29	232	ní	29	76	chun
30	351	chun	30	223	do	30	75	atá
31	333	de	31	219	ann	31	75	luaigh
32	313	í	32	213	nuair	32	71	rinne
33	308	m	33	208	nó	33	70	iad
34	304	do	34	199	amach	34	70	ú
35	283	is	35	195	bhliain	35	69	bhain
36	275	of	36	191	níos	36	69	do
37	247	san	37	177	féin	37	68	is
38	236	bhliain	38	176	iad	38	67	ina
39	232	tá	39	173	tá	39	62	as
40	228	they	40	168	san	40	62	mar
41	227	leis	41	162	sna	41	62	mínigh
42	220	ina	42	161	acu	42	59	cur
43	215	á	43	154	faoi	43	58	dhá
44	208	daoine	44	153	tháinig	44	56	síos
45	199	mar	45	149	eile	45	54	sé
46	198	chéad	46	147	gach	46	53	san
47	196	ba	47	146	chuir	47	53	sin
48	196	faoi	48	146	cuid	48	52	bhfuil
49	187	to	49	145	chuid	49	50	faoi
50	185	as	50	143	mhór	50	50	uí
51	185	iad	51	140	as	51	49	tar
52	183	atá	52	140	nua	52	49	éis
53	175	raibh	53	138	dá	53	47	héireann
54	168	focal	54	136	rinne	54	47	nuair

55	164	leabhar	55	134	duine	55	46	tháinig
56	159	úsáid	56	130	dtí	56	44	aoise
57	158	sin	57	129	thart	57	42	maidir
58	156	saol	58	128	ná	58	39	aon
59	146	don	59	124	chéad	59	39	den
60	143	amach	60	124	oideachas	60	39	eile
61	142	ann	61	121	den	61	39	ní
62	142	mór	62	121	mór	62	39	seo
63	141	stór	63	121	um	63	38	bhliain
64	138	snáithe	64	120	áit	64	37	ainmnigh
65	135	pháiste	65	120	éirinn	65	37	déan
66	134	lth	66	115	chomhairle	66	36	bheith
67	134	snáithaonad	67	114	air	67	36	daoine
68	131	cén	68	113	aon	68	36	í
69	125	síos	69	112	cuireadh	69	35	idir
70	125	tháinig	70	112	dhéanamh	70	35	téarma
71	123	nó	71	111	chuaigh	71	34	amach
72	121	scríobh	72	111	chéile	72	34	átha
73	119	trasna	73	111	gaeltachta	73	33	bhaile
74	117	dhéanamh	74	109	gaelscolaíochta	74	33	cliath
75	112	ón	75	109	isteach	75	32	ann
76	106	sna	76	107	rialtas	76	32	thug
77	104	eile	77	106	rí	77	32	tábhacht
78	104	tír	78	103	nach	78	32	úsáid
79	103	den	79	102	bheith	79	31	ba
80	103	éirinn	80	102	chur	80	31	féin
81	102	gach	81	102	thug	81	31	lorcáin
82	101	ainm	82	102	ón	82	30	nua
83	101	sí	83	97	atá	83	30	nó
84	100	nuair	84	96	mó	84	30	réabhlóid
85	99	féin	85	95	am	85	30	sna
86	96	bhfuil	86	94	fáil	86	28	chuir
87	96	cé	87	94	leor	87	28	cé
88	95	air	88	93	gcuid	88	28	gairid

89	94	nua	89	93	héireann	89	28	meánaoise
90	93	lá	90	88	cén	90	28	muintir
91	93	tú	91	88	faoin	91	27	chomh
92	90	duine	92	88	fuair	92	27	cháipéis
93	90	páistí	93	88	gur	93	27	cogadh
94	87	orthu	94	88	obair	94	27	cuireadh
95	83	he	95	88	rud	95	27	linn
96	82	am	96	88	thosaigh	96	27	mac
97	81	faoin	97	88	éirigh	97	27	tionchar
98	81	gur	98	86	orthu	98	26	ach
99	80	acu	99	84	fosta	99	26	hitler
100	80	cad	100	83	mbeadh	100	26	oibre

EduGA-BusS-PPJC Rank – Frequency – Token			**EduGA-BusS-PPSC Rank – Frequency – Token**		
1	2843	a	1	6603	an
2	2082	an	2	6450	a
3	1381	ar	3	3063	ar
4	1235	agus	4	2322	na
5	1042	na	5	2096	agus
6	644	i	6	1748	le
7	629	le	7	1622	is
8	601	ag	8	1480	seo
9	523	is	9	1440	ag
10	522	go	10	1362	i
11	464	seo	11	1114	leis
12	463	do	12	1030	sé
13	404	de	13	996	go
14	342	tá	14	901	gnó
15	291	é	15	890	mar
16	285	airgead	16	852	atá
17	281	in	17	840	chun
18	277	sé	18	757	don
19	266	atá	19	749	do

20	241	nó	20	720	tá
21	233	as	21	720	é
22	233	sa	22	706	marc
23	226	leis	23	682	ceist
24	219	bhfuil	24	674	roinn
25	211	chun	25	662	dhéanamh
26	207	mar	26	653	sa
27	197	ó	27	646	féidir
28	192	ioncam	28	636	siad
29	183	féidir	29	598	bíonn
30	181	árachas	30	570	níos
31	176	leibhéal	31	544	gá
32	174	sin	32	543	ní
33	172	ann	33	524	page
34	160	don	34	522	leathanach
35	155	gach	35	513	caibidil
36	152	caiteachas	36	504	de
37	149	tú	37	476	sin
38	148	leanas	38	468	in
39	147	dhéanamh	39	464	gafa
40	142	iad	40	436	bhfuil
41	137	níos	41	433	unit
42	136	earraí	42	428	má
43	134	ná	43	423	aon
44	133	airgid	44	387	chur
45	126	ní	45	384	ghnó
46	126	siad	46	381	layout
47	126	úsáid	47	371	nó
48	125	daoine	48	370	nuair
49	125	sí	49	358	as
50	122	ach	50	354	ann
51	121	airgeadais	51	333	gnólacht
52	121	duine	52	332	déan
53	121	gnó	53	332	fostaithe

54	121	mí	54	316	sí
55	120	bheith	55	306	ghnólacht
56	120	eile	56	298	gnólachtaí
57	119	chuid	57	296	ná
58	117	má	58	292	bhíonn
59	117	nuair	59	286	ó
60	114	ard	60	274	airgead
61	112	faoi	61	270	faoi
62	111	teaghlaigh	62	262	trí
63	108	chur	63	259	gach
64	108	íoc	64	253	den
65	107	amach	65	248	leanas
66	106	árachais	66	246	amach
67	102	amháin	67	246	iad
68	102	trí	68	244	íoc
69	101	ba	69	234	dá
70	100	uimh	70	231	eile
71	99	nach	71	230	idir
72	96	fháil	72	228	shampla
73	95	tta	73	224	nach
74	91	obair	74	224	nua
75	90	dhá	75	221	chuid
76	90	ina	76	220	ón
77	89	buiséad	77	216	san
78	89	den	78	212	bheith
79	87	acu	79	204	acu
80	85	leabhar	80	204	úsáid
81	83	aghaidh	81	203	rialtas
82	83	bealtaine	82	199	daoine
83	82	iomlán	83	198	aige
84	81	am	84	198	earraí
85	81	cén	85	198	tomhaltóir
86	81	pá	86	196	fháil
87	81	san	87	196	leo

88	78	bíonn	88	196	táirge
89	77	dá	89	194	bhaineann
90	77	reatha	90	192	eolas
91	76	bhíonn	91	190	ghnólachtaí
92	75	bhí	92	186	amháin
93	75	ceisteanna	93	182	airgeadais
94	75	cheannach	94	182	ngnólacht
95	75	cuntas	95	178	síos
96	75	marc	96	178	thabhairt
97	75	seirbhísí	97	174	mó
98	74	isteach	98	172	buntáistí
99	73	gnáth	99	172	déanann
100	72	rud	100	172	maidir

EduGA-Math-Prim			**EduGA-Math-PPJC**			**EduGA-Math-PPSC**		
Rank – Frequency – Token			**Rank – Frequency – Token**			**Rank – Frequency – Token**		
1	404	an	1	16147	a	1	19621	an
2	302	a	2	13635	an	2	19560	a
3	300	cent	3	10961	x	3	14784	x
4	278	euro	4	8573	i	4	11697	i
5	186	ar	5	6803	na	5	8385	agus
6	165	cé	6	6653	ar	6	8198	ar
7	152	mhéad	7	5694	ii	7	7683	na
8	147	snáithaonad	8	5573	agus	8	6110	y
9	147	snáithe	9	4500	b	9	5497	ii
10	147	uimhreas	10	4008	iii	10	5111	is
11	135	agus	11	3669	is	11	5011	b
12	114	atá	12	3581	cm	12	4240	go
13	107	na	13	3472	seo	13	4019	sin
14	88	tomhais	14	2990	y	14	3528	é
15	86	seo	15	2669	é	15	3280	iii
16	76	cuir	16	2620	iv	16	3223	faigh
17	75	déag	17	2523	go	17	3123	tá
18	66	níos	18	2445	de	18	3098	seo

19	62	i	19	1984	tá	19	3079	cm
20	58	is	20	1909	atá	20	2668	atá
21	58	ná	21	1821	sa	21	2639	le
22	57	aonad	22	1797	ag	22	2508	ag
23	55	faoi	23	1774	le	23	2484	de
24	54	mí	24	1773	faigh	24	2476	bhfuil
25	54	sa	25	1633	gach	25	2187	sa
26	50	nó	26	1416	chun	26	1935	cos
27	48	labhair	27	1364	bhfuil	27	1779	luach
28	45	cén	28	1256	ceann	28	1702	má
29	42	leis	29	1227	sin	29	1684	iv
30	38	measúnú	30	1190	cé	30	1568	gach
31	37	le	31	1152	mar	31	1554	chun
32	37	ú	32	1091	uimhir	32	1437	líne
33	35	dá	33	1039	nó	33	1375	do
34	35	don	34	1025	sé	34	1363	mar
35	34	suimiú	35	1023	líne	35	1346	nó
36	33	dheich	36	1019	má	36	1288	leis
37	33	mé	37	1014	scríobh	37	1172	sampla
38	33	tá	38	924	dá	38	1164	sé
39	32	méid	39	917	luach	39	1151	fad
40	31	fágtha	40	907	do	40	1144	in
41	31	in	41	892	cén	41	1138	dá
42	31	úsáid	42	891	iad	42	1113	achar
43	30	airgid	43	888	vi	43	1100	scríobh
44	29	agam	44	884	in	44	1080	ná
45	29	aon	45	868	úsáid	45	1077	ina
46	28	comhaireamh	46	850	ná	46	1048	iad
47	28	deich	47	819	leanas	47	1024	úsáid
48	28	mó	48	818	fad	48	981	ó
49	27	múinteoir	49	783	mhéad	49	953	ceann
50	27	trí	50	770	líon	50	942	dhá
51	26	cúig	51	751	ina	51	895	as
52	26	lá	52	749	faoi	52	895	síos

53	26	nóta	53	743	leis	53	875	v
54	25	ag	54	725	dheis	54	863	den
55	25	anailís	55	699	dhá	55	855	dtí
56	25	de	56	697	léaráid	56	855	san
57	25	sé	57	695	síos	57	831	cé
58	24	dhá	58	668	amháin	58	827	fháil
59	24	dé	59	664	nuair	59	810	léaráid
60	24	gach	60	655	km	60	808	tan
61	24	seacht	61	649	fháil	61	805	trí
62	23	cruthanna	62	647	amach	62	778	amháin
63	23	daltaí	63	637	san	63	778	lárphointe
64	22	aois	64	633	achar	64	768	ann
65	22	asal	65	628	níos	65	763	líon
66	22	ceann	66	600	trí	66	725	féidir
67	22	do	67	599	cleachtadh	67	711	dheis
68	22	eile	68	585	thíos	68	708	triantáin
69	22	naoi	69	577	den	69	699	leanas
70	22	seiceáil	70	575	ab	70	698	cothromóid
71	22	tam	71	555	dtí	71	693	dóchúlacht
72	21	ann	72	554	sampla	72	692	níos
73	21	ceithre	73	547	eile	73	681	ab
74	21	chloig	74	543	airde	74	661	airde
75	21	cruth	75	521	féidir	75	659	cén
76	21	lú	76	521	sonraí	76	656	gcás
77	21	tú	77	510	díobh	77	652	meán
78	20	chileagram	78	507	acu	78	636	faoi
79	20	meaitseáil	79	507	ann	79	622	ceart
80	20	spás	80	483	as	80	622	í
81	19	fad	81	483	ó	81	618	ach
82	19	go	82	469	chéile	82	614	nuair
83	19	oibríochtaí	83	459	mó	83	610	más
84	19	úll	84	454	céard	84	585	tú
85	18	amháin	85	452	cuir	85	584	amach
86	18	chat	86	444	uair	86	576	chiorcail

87	18	chur	87	443	caibidil	87	572	gur
88	18	isteach	88	440	chéad	88	572	ríomh
89	18	leath	89	434	ach	89	566	chéile
90	18	leathanach	90	433	bhí	90	564	pointe
91	18	ndeich	91	430	ngach	91	549	ais
92	18	ocht	92	427	ceart	92	549	uimhir
93	18	sin	93	415	graf	93	546	eile
94	18	uair	94	412	téacs	94	534	sonraí
95	18	é	95	410	trialacha	95	528	mhéad
96	17	bhfuil	96	394	méid	96	520	idir
97	17	bhosca	97	386	tarraing	97	520	thíos
98	17	kg	98	384	ais	98	496	caighdeánach
99	17	tacair	99	379	triantáin	99	487	thugtar
100	16	chlog	100	377	téarma	100	486	chéad

EduGA-Corp-PPJC Rank – Frequency – Token			**EduGA-Art-PPxx Rank – Frequency – Token**			**EduGA-SPHE-PPJC Rank – Frequency – Token**		
1	4054	a	1	1349	an	1	6883	a
2	3305	an	2	1101	a	2	4152	an
3	1957	agus	3	654	ar	3	3432	agus
4	1707	ar	4	632	the	4	3304	ar
5	1531	na	5	569	agus	5	2861	na
6	1088	ag	6	542	na	6	1725	ag
7	882	liathróid	7	406	i	7	1715	go
8	842	le	8	380	ag	8	1715	le
9	738	i	9	362	go	9	1369	i
10	642	go	10	296	in	10	1323	do
11	596	is	11	291	le	11	1062	is
12	496	bheith	12	227	of	12	1048	leanaí
13	425	úsáid	13	220	do	13	773	seo
14	411	leis	14	202	sé	14	741	nó
15	392	sa	15	202	tá	15	739	de
16	388	seo	16	198	and	16	720	tá
17	352	níos	17	196	is	17	688	leis

18	339	de	18	188	to	18	679	sé
19	336	mbliana	19	167	sin	19	672	mar
20	309	é	20	164	sa	20	626	tú
21	291	iad	21	152	seo	21	616	bheith
22	290	mar	22	145	leis	22	591	atá
23	289	dhéanamh	23	125	bhí	23	588	sa
24	279	isteach	24	123	mar	24	567	bhfuil
25	254	eile	25	120	as	25	562	chun
26	251	ina	26	120	de	26	558	siad
27	245	in	27	120	obair	27	542	é
28	241	sé	28	114	century	28	539	in
29	234	m	29	114	mé	29	506	dhéanamh
30	234	tascanna	30	114	ní	30	458	duine
31	233	lámh	31	107	grúpa	31	423	eile
32	226	taobh	32	93	atá	32	422	faoi
33	221	siad	33	93	múinteoir	33	418	chur
34	219	scileanna	34	86	é	34	413	gach
35	219	tá	35	84	dhéanamh	35	409	bhí
36	217	do	36	83	faoi	36	395	ní
37	215	atá	37	81	on	37	391	daoine
38	212	nó	38	78	tú	38	383	ina
39	200	amach	39	77	bhaile	39	380	sin
40	199	aois	40	77	leat	40	370	féidir
41	199	bhfuil	41	76	bhfuil	41	359	ach
42	199	í	42	74	beidh	42	358	ann
43	194	páistí	43	72	chuid	43	350	iad
44	192	chur	44	72	haois	44	345	gníomhaíocht
45	190	ach	45	71	ba	45	337	níos
46	188	amháin	46	71	he	46	331	nuair
47	184	féidir	47	70	that	47	325	féin
48	181	gach	48	69	was	48	319	sí
49	181	rith	49	67	agat	49	314	acu
50	173	gan	50	67	you	50	279	leo
51	171	den	51	66	ach	51	277	ó

52	171	duine	52	66	ceist	52	277	úsáid
53	164	baineannaigh	53	66	gach	53	268	raibh
54	164	fireannaigh	54	66	scoile	54	266	nach
55	161	chun	55	65	ealaíne	55	266	ná
56	158	bualadh	56	64	chun	56	256	lena
57	157	faoi	57	64	don	57	254	bíonn
58	154	ansin	58	60	ná	58	251	as
59	154	cluichí	59	57	sí	59	237	rang
60	154	thabhairt	60	56	cén	60	236	cad
61	151	léim	61	56	déag	61	235	agat
62	148	spórt	62	56	trí	62	234	leat
63	145	don	63	55	eile	63	233	aon
64	144	ba	64	54	bheith	64	233	bhíonn
65	144	chos	65	53	iad	65	228	obair
66	139	dul	66	52	thíos	66	223	orthu
67	138	acu	67	51	aon	67	220	má
68	138	ó	68	51	bhur	68	211	páistí
69	136	chóir	69	51	daltaí	69	210	amach
70	135	san	70	51	ealaín	70	206	ba
71	133	caitheamh	71	51	nó	71	202	iarr
72	132	imreoir	72	51	raibh	72	200	scoil
73	128	ann	73	50	chur	73	199	den
74	128	cad	74	50	duine	74	199	gur
75	128	spás	75	50	féin	75	192	thabhairt
76	126	mó	76	49	san	76	186	maith
77	125	bíonn	77	49	school	77	185	dá
78	123	lámha	78	49	tuismitheoir	78	178	am
79	123	talamh	79	48	fáil	79	176	don
80	122	daoine	80	47	gan	80	176	rud
81	121	liathróidí	81	46	dtí	81	175	san
82	119	imreoirí	82	46	his	82	174	cén
83	118	lena	83	46	ó	83	171	bhaineann
84	118	rolladh	84	45	for	84	171	mbíonn
85	118	spóirt	85	45	maith	85	169	cearta

86	118	ón	86	44	at	86	169	mó
87	116	arís	87	44	cheart	87	167	chuid
88	116	imirt	88	44	den	88	164	faoin
89	116	tosaigh	89	43	chuig	89	164	ombudsman
90	115	ciceáil	90	43	leathanaigh	90	163	roinnt
91	113	tú	91	43	scoil	91	161	baile
92	112	chorp	92	43	siad	92	159	eolas
93	112	má	93	42	agam	93	157	fháil
94	112	á	94	42	á	94	156	ort
95	111	gabháil	95	41	work	95	155	air
96	111	iarraidh	96	41	your	96	155	amháin
97	111	sula	97	40	be	97	153	cuir
98	109	cosa	98	40	cur	98	152	chomh
99	109	fháil	99	40	ghaeilge	99	148	dul
100	105	idir	100	39	amach	100	144	ceart

Appendix 2: Agreements and Required Paperwork

The following agreement template was used when requested by owners/publishers of educational materials.

Mícheál J. Ó Meachair,
Seoladh 1,
Seoladh 2,

Eagraíocht
Seoladh 1,
Seoladh 2,

Conradh Corpais

Gabhaim buíochas mór le__________as an ábhar atá á chur ar fáil acu agus an taighde PhD seo idir lámha agam.

Geallaim, Mícheál J. Ó Meachair, faoi cheangal dlíthe cóipchirt na hÉireann, leis an litir seo go gcloífidh mé lena ndearbhaím thíos.

1. Dearbhaím go gcoimeádfaidh mé an t-ábhar a roinnfidh____________liom i stóras slán, stóras nach mbeidh rochtain ag an bpobal air ná ar a bhfuil ann.
2. Dearbhaím nach bhfoilseoidh mé aon chuid den ábhar seo, ná leagan díorthach den ábhar seo trí mheán ar bith.
3. Dearbhaím nach roinnfear na bunchomhaid, ná leagan ar bith de na comhaid sin, ar an bpobal ná ar thaighdeoir eile.
4. Sa chás go mbeidh ar scrúdaitheoirí na hollscoile breathnú ar chodanna den chorpas, lena mharcáil, cuirfidh mé foireann___ar an eolas cé acu scrúdaitheoirí a d’fhéadfadh breathnú ar an gcuid den chorpas ina bhfuil ábhar__________________. Más fearr le foireann____________nach mbreathnódh aoinne de na scrúdaitheoirí ar a n-ábhar, aimseofar bealach cuid____________ a chur i bhfolach an tráth sin.
5. Dearbhaím nach luafar____________, ná aon dream eile, go sonrach le leibhéal na caighdeán teanga ná teagaisc i dtorthaí an taighde. Cinnteoidh mé féin agus bord eitice na hollscoile seo.

Cuirfidh mé leaganacha d’ábhar___________a chruthófar le linn an taighde ar fáil do___________,

chomh fada agus is féidir agus más mian le foireann_____.

Le gach dea-mhéin,

____

Mícheál J. Ó Meachair, XX / XX / XXXX

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